# Ramsey-Free Composition and One-Stage Capacity Bounds for Linear Shared Decoders

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**Abstract**

We study two-stage compositions $K_2 \circ K_1$ and one-stage capacities of $F$-linear class-uniform shared decoders on $\mathrm{Mat}_{d\times d}(F^b)$, following Section 6 of [10] (which used van der Waerden and Hales–Jewett arithmetic progressions to bound $d$ when $K_2 \circ K_1 = \tau$ with $M_1$ and $M_2$ column classes and to show that $\mathrm{rank}_F(\tau - K_2 \circ K_1)/(bd) \to \infty$ as $d \to \infty$ at fixed $M_1, M_2$ when $\mathrm{char}(F) \nmid d$, i.e. $(d : F) \neq 0$) and Problem 9.1(a)–(b) of [4] (one-stage linear shared-decoder capacity), together with three-term and two-sided interval-mask swap codes ([4, Remark 5.6 and Problem 9.3]). No multitape Turing machine is known to induce a one- or two-stage linear class-uniform decomposition; working within these algebraic models, all numbered results are verified in Lean 4.

For two-stage linear composition, replacing arithmetic progressions by Ramsey-free additive structures (avoiding van der Waerden and Hales–Jewett numbers) yields two bounds. First (Theorem 3.3), over any field $F$ and for $b, M_1, M_2 \geq 1$, replacing progressions by affine Hilbert cubes extracted via iterated Cauchy–Schwarz proves that $K_2 \circ K_1 = \tau$ forces

$$d \;\leq\; \max\bigl(8M_1M_2^2,\; (2M_1)^{2^{M_2}}\bigr).$$

For fixed $M_2 \geq 2$, this gives a cutoff polynomial in $M_1$—ruling out exact two-stage linear factorizations whenever $M_1 = o\bigl(d^{1/2^{M_2}}\bigr)$ (including partitions of $\mathbb{Z}_d$ into $M_1 = 2^{O(\sqrt{\log d})}$ 3-AP-free classes obtained by translate-covering $\mathbb{Z}_d$ with a Behrend set [3]) and whenever $M_2 \leq (1-\varepsilon)\log_2\log_2 d$ at fixed $M_1 \geq 2$ and $\varepsilon \in (0,1)$—whereas at $M_2 = 1$, $M_1 = 1$, and small constant class counts such as $M_1 = M_2 = 2$, Theorem 6.3 and Corollary 6.4 of [10] already give the stronger bounds $d \leq M_1$, $d \leq M_2^2$, and $d \leq 16$. Second (Theorem 4.6), over any field $F$ with $\mathrm{char}(F) \nmid d$, combining shifted $(2M_2)$-fold monochromatic sumsets, arbitrary-subset Vandermonde pullbacks, sparse annihilating polynomials, and iterated Kővári–Sós–Turán extraction proves that for all $b, M_1, M_2 \geq 1$ and all $d \geq D_0(M_1, M_2)$,

$$\mathrm{rank}_F(\tau - K_2 \circ K_1) \;\geq\; c(M_1, M_2)\, b\, d\, (\log d)^{\frac{1}{2M_2-1}},$$

giving $\Omega(b\,d\log d)$ when $M_2 = 1$ (Corollary 4.8) and an explicit log-power rate for $M_1 \geq 2$ and large $d$ (at $M_1 = 1$, [10, Theorem 6.6] already gives $\Omega(bd^2/M_2)$ for $d \geq 4M_2 + 1$, while here $D_0(1,1) = 16^{36} + 2 \approx 2.23 \times 10^{43}$). Finite-dimensional controls at $d = 13$, $d = 4$, and $d = 12$ test the hypotheses of both bounds.

For one-stage linear shared decoders on [4, Problem 9.1(a)], exploiting decoder linearity on maximal monochromatic cyclic runs proves an explicit single-shift intersection bound $B_{\mathrm{run}}(b, p, s)$ when $\mathrm{char}(F) \nmid (d/\gcd(d,s))$ (Theorem 5.2, whose value $\frac{1}{4}d^2 + \frac{3}{2}d$ on the alternating two-coloring for even $d = 2m \geq 2$ at $b = 1, s = 1$ is realized over every field, Theorem 5.3). Averaging over shifts bounds the maximum $b$-lane code dimension $D_b(p)$ by $D_b(p) \leq \frac{19}{32}\,bd^2 + \frac{3}{2}\,bd^{3/2}$ for two classes whenever $\mathrm{char}(F) \nmid d$ (Theorem 5.4, whose leading coefficient $\frac{19}{32}$ lies strictly below both the $\frac{41}{64}$ bound of [10, Theorem 3.4] and the $\frac{5}{8} - o(1)$ nested-component certificate floor of [10, Theorem C.6]), and by $D_b(p) \leq \bigl(1 - \frac{5}{4M} + \frac{1}{M^2} - \frac{1}{4M^3}\bigr)bd^2 + \bigl(\frac{5}{4} - \frac{1}{4M}\bigr)bd$ for $M \geq 2$ at primes

$d \neq \mathrm{char}(F)$ (Theorem 5.5, giving $\frac{19}{32}\,bd^2 + \frac{9}{8}\,bd$ at $M = 2$; its leading coefficient $c_M$ beats the flag bound $1 - \frac{1}{2M}$ for every $M \geq 2$ and is smaller than the optimized difference-cover leading coefficient of [10, Theorem 3.5] if and only if $2 \leq M \leq 40$, whereas [10, Theorem 3.5] has a $1 - \Theta(M^{-1/2})$ leading coefficient that is smaller for every $M \geq 41$ and applies to composite $d$ and arbitrary characteristic). On [4, Problem 9.1(b)], coupled $\mathbb{Z}_6$ Fourier-mode elimination proves $\dim_F(\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})) \leq 14$ and hence $D_1\big(\{0,1,3\},\{2,4,5\}\big) = 25$ over every field with $\mathrm{char}(F) \notin \{2,3\}$ (Theorem 6.2, leaving characteristics 2 and 3 open). For swap codes, we construct three-term families $\sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}$ over every field exceeding $d(d+1)/2$ at $d \in \{6,7,9,12\}$ and over every field at $d = 4$ (an example of [4, Remark 5.6], previously checked over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$), and prove a characteristic-free two-sided bound for two-scale (coarse–fine) families of [4, Problem 9.3] (Theorem 7.4 and Corollary 7.5) giving, on contiguous fine singletons of densities $\alpha = s_A/d, \beta = s_B/d \in (0,1)$ plus one coarse complement, leading coefficient $\frac{1}{2} + \frac{1}{4}\min\big(\beta(1+\alpha), \alpha(1+\beta)\big)$, strictly below the one-sided Theorem TP coefficient $\frac{1}{2} + \frac{1}{2}\min(\alpha,\beta)$ ([4, Theorem 6.4]; and $\frac{11}{16}d^2 + d < \frac{3}{4}d^2$ at $\alpha = \beta = 1/2$ for all even $d \geq 18$). Finally, we prove an Operator Schmidt Rank pass lower bound ($t \geq \log_2 d$) for a restricted shuffle and row/column-uniform pipeline paired with verified $O(L)$-time Turing machines ($L = d^2 \max(1, \lceil \log_2 d \rceil)$) in `lean-machine` for shuffle, unshuffle, and within-row shear showing why $\mathrm{rk}_{\otimes}$ does not answer [4, Problem 9.5], together with cut-causal cone evaluation on frozen `Transpose.TM` fibres, a Lean formalization of the column-class analogue of the $1/(\Delta+1)$ random-order bound at entry read arity $\Delta$ ([5, Proposition 4.14(a)–(b)]), and negative-bracket examples showing why two specific column-class cut reductions fall short of [4, Problem 9.4(b)]. Statement (T) remains open.

**Contributions.** Hans Brende chose the problem, fixed the verification standard, and directed the research. Gemini 4 Carbon (Google DeepMind) developed the mathematics, wrote the Lean 4 formalizations, and wrote the manuscript under his direction. Claude Fable 5.2 (Anthropic) conducted an external review of an earlier draft that identified vacuous statements in a preliminary sweep-extraction section, corrected cross-paper attributions, and recommended refocusing the draft around shared-decoder composition and moving the verified shuffle and shear machines to `lean-machine`. Claude Opus 5.5 (Anthropic) conducted subsequent external reviews that corrected series comparisons, finite-$d$ thresholds, citations, and formalization-scope descriptions, and suggested the present title.

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# 1 Introduction

## 1.1 Context and scope

Let $d \geq 2$, let $b = \lceil \log_2 d \rceil$ be the canonical entry bit-width, and let $L = d^2 b = \Theta(d^2 \log d)$ be the bit-length of a $d \times d$ matrix of $b$-bit entries laid out row by row on the read-only input tape of a deterministic multitape Turing machine with $k$ read-write work tapes and a one-way write-only output tape ([6, Sections 2.1–2.2 and Definition 2.1]). Statement (T) ([6, Definition 2.1], `Transpose.LinearTimeTransposable`) asserts that some such machine transposes all $d \times d$ matrices of $b$-bit entries within $O(L)$ steps; Conjecture 9.1 of [6] (the Tape Coding Conjecture) conjectures, in a robust co-entropy form, that no causal code solves complete exchange at the profile capacities on the block networks of linear-length tape diagrams, and its refutation of Statement (T) is Lean-verified there ([6, Theorem 9.2]). By Harvey and van der Hoeven [16, Corollary 6.9], if Statement (T) is false then no linear-time multitape integer multiplier exists either.

The synthesis paper *Order, Symmetry and the Swap* [4] organized the algebraic obstacles studied across the series into two one-stage mechanisms on $\mathrm{Mat}_{d\times d}(F)$: an order bound for two-term operators $A_1 X B_1^{\mathsf{T}} + A_2 X B_2^{\mathsf{T}} = X^{\mathsf{T}}$ giving dimension at most $d(d+1)/2$ ([4, Theorem 5.1 and Corollary 5.5]; coordinate form of Theorem 5.1 Lean-verified, Corollary 5.5 proved in text), and a translate-pigeonhole bound (Theorem TP, [4, Definition 6.3 and Theorem 6.4], Lean-verified) for class-uniform rules in which column $c$ of the output is computed by applying one of $M$ shared linear decoders to the column-rotated matrix $\mathrm{rot}_c(X)_{i,j} = X_{i,j+c}$. As emphasized in [4, Sections 4.2 and 4.6], abstract storage-code models without shared rules admit high-rate network coding, whereas the class-uniform model captures translation covariance along a single regular sweep.

Multi-stage compositions $K_t \cdots K_1$ of class-uniform hops were first examined for interval classes in [4, Remark 6.14 and Section 7.4] (where $M$ interval classes per hop gave $\dim_F \ker(\mathrm{sw} - K_t \cdots K_1) \leq (d^2 + d + tMd)/2$). In the sequel *Capacity and Composition of Shared Transposition Decoders* [10, Section 6] (where all results cited in this paper are Lean-verified), Brende and GPT-6 Astra studied exact and approximate *two-stage factorizations* $K_2 \circ K_1$ of linear class-uniform operators on $\mathrm{Mat}_{d\times d}(F^b)$ for arbitrary column partitions, where $K_1$ has $M_1$ column classes and $K_2$ has $M_2$ column classes:

- For exact factorizations $K_2 \circ K_1 = \tau$ over any field $F$, Theorem 6.3 of [10] reduced two-stage composition to a common-message subspace inside a cyclic channel module and extracted a monochromatic arithmetic progression in the stage-1 partition, proving $d \leq M_2(\mathsf{W}(M_1, M_2 + 1) - 1)$, where $\mathsf{W}$ denotes a van der Waerden threshold; at $M_1, M_2 \leq 2$, Corollary 6.4 of [10] used $\mathsf{W}(2,3) \leq 9$ to obtain the small-parameter bound $d \leq 16$.

- For approximate factorizations over any field $F$ with $(d : F) \neq 0$, Theorem 6.6 of [10] proved that for all $d \geq N_0$, $r := \mathrm{rank}_F(\tau - K_2 \circ K_1)$ satisfies

$$r \;\geq\; \frac{bk(d - N_0 + 1)}{2}, \qquad \ell = (2M_2 + 1)k - 1, \quad N_0 = \mathsf{W}(M_1, \ell),$$

  showing that $r/(bd) \to \infty$ as $d \to \infty$ for every fixed pair $(M_1, M_2)$, using qualitative finiteness of van der Waerden numbers via the Hales–Jewett theorem [15] without substituting a quantitative upper bound for $\mathsf{W}(M_1, \ell)$.

This paper is primarily a sequel to [10] and [4] in that same linear algebraic model. We state the model boundary up front: *no multitape Turing machine is known to decompose into one- or two-stage linear class-uniform stages*, and none of the results of this paper decides Statement (T)

or bridges general adaptive or oblivious machines to linear shared decoders. Throughout this paper, "Ramsey-free" means avoiding van der Waerden and Hales–Jewett progression numbers by substituting Cauchy–Schwarz density cube extraction [21, 14] and Kővári–Sós–Turán extremal graph bounds [17]. Within the linear class-uniform and swap-code models, exploiting *linearity of the shared decoders* makes progress on three open questions from prior work: in two-stage composition ([10, Section 8]), replacing arithmetic progressions (which translate-covering $\mathbb{Z}_d$ by a Behrend set [3] limits to length $< 3$ at $M_1 = 2^{O(\sqrt{\log d})}$, and whose non-wrapping monochromatic length in $\{0, \dots, d-1\}$ is $O(\log d)$ already under a random 2-coloring [14, 1]) by *affine Hilbert cubes* (over every field) and *shifted monochromatic sumsets* (when $\mathrm{char}(F) \nmid d$, i.e. $(d : F) \neq 0$) eliminates five-fold exponential van der Waerden dependencies [13] for $M_1 \geq 2$; in one-stage linear capacity ([4, Problem 9.1(a)]; [10, Section 3 and Appendix C]), when $\mathrm{char}(F) \nmid d$, run-extension equations at the right endpoints of monochromatic cyclic runs push strictly below the $\frac{41}{64}$ bound of [10, Theorem 3.4] and the $\frac{5}{8} - o(1)$ floor for the nested-component certificates of [10, Theorem C.6] (whose isolation variant [10, Theorem C.7] has floor $\frac{9}{16} - o(1)$; neither is a lower bound on capacity, for any decoders); and on [4, Problem 9.1(b)] (where run-extension alone gives $\leq 26$), a coupled six-mode Fourier analysis determines $D_1(\{0,1,3\},\{2,4,5\}) = 25$ on $\mathbb{Z}_6$ outside characteristics 2 and 3.

### 1.2 Summary of results

1. **Ramsey-free cutoff for exact two-stage linear composition (Section 3).** Over an *arbitrary* field $F$, Lemma 3.1 generalizes [10, Lemma 6.1] from powers of a single endomorphism to ordered products $R_{k-1}^{\varepsilon_{k-1}} \cdots R_0^{\varepsilon_0}$ of $k$ endomorphisms $R_0, \dots, R_{k-1} \in \mathrm{End}_F(A)$: if $W \leq A$ admits an injective linear readout invariant under those $2^k$ products, then

$$(k+1)\dim_F W \;\leq\; \dim_F A \;+\; \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A).$$

Extracting a monochromatic affine $M_2$-cube with small steps $1 \leq s_t \leq \lfloor d/(2M_2^2) \rfloor$ by the classical Cauchy–Schwarz density cube lemma (Lemma 3.2; [21, 14]) yields Theorem 3.3: whenever $b, M_1, M_2 \geq 1$ and two $F$-linear class-uniform stages $K_1, K_2$ on $\mathrm{Mat}_{d\times d}(F^b)$ with $M_1$ and $M_2$ classes satisfy $K_2 \circ K_1 = \tau$, one has

$$d \;\leq\; \max\Big(8M_1M_2^2,\; (2M_1)^{2^{M_2}}\Big).$$

For fixed $M_2 \geq 2$, this cutoff is polynomial in $M_1$—ruling out exact factorizations whenever $M_1(d) = o(d^{1/2^{M_2}})$, including partitions of $\mathbb{Z}_d$ into $M_1(d) = 2^{O(\sqrt{\log d})}$ 3-AP-free classes obtained by translate-covering $\mathbb{Z}_d$ with a Behrend set [3]—whereas every progression-based cutoff from [10, Theorem 6.3] is at least $M_2(\mathsf{W}(M_1, 3) - 1)$, which is superpolynomial in $M_1$. At $M_2 = 1$, $M_1 = 1$, and small constant class counts, however, [10, Theorem 6.3 and Corollary 6.4] are stronger: they give $d \leq M_1$ at $M_2 = 1$, $d \leq M_2^2$ at $M_1 = 1$, and $d \leq 16$ at $M_1 = M_2 = 2$, where Theorem 3.3 gives $\max(8M_1, 4M_1^2)$, $\max(8M_2^2, 2^{2^{M_2}})$, and 256. Two finite controls accompany Theorem 3.3: at $d = 13, q = 2$ on the 3-AP-free 2-cube $J_{13} = \{0,1,4,5\} \subseteq \mathbb{Z}_{13}$, Example 3.5 shows $9 \leq \max \dim_F W \leq 10$ over fields with $(13 : F) \neq 0$ (refuting a naive $2^k$-multiplier bound $\dim_F W \leq 8$); and at $d = 4 = M_1M_2$ on $p_1 = p_2 = [0,0,1,1]$, Example 3.6 constructs exact linear coded factorizations $K_2 \circ K_1 = \tau$ for all lane widths $b \geq 0$ over every field with $(5 : F) \neq 0$ (leaving characteristic 5 open for $b \geq 1$), while showing that entry-copy and bit-lane-copy factorizations fail for $b \geq 1$.

2. **Log-power error-rank lower bound for approximate two-stage composition (Section 4).** Over any field $F$ with $(d : F) \neq 0$, we pass to a splitting field carrying an injective additive character $\chi : \mathbb{Z}_d \to E^\times$ and replace progressions of length $(2M_2+1)k-1$ by $(2M_2)$-fold monochromatic sumsets $\widetilde{J}_0 + \widetilde{B}_0 + \cdots + \widetilde{B}_{2M_2-2}$ with $|\widetilde{J}_0| = k$, $0 \in \widetilde{B}_t$, and $|\widetilde{B}_t| = k+1$. Because $\big(\chi(\nu c)\big)_{c\in\widetilde{J}_0, 0\le\nu<k}$ is a Vandermonde matrix in the distinct scalars $\chi(c)$ $(c \in \widetilde{J}_0)$, stacked evaluation over any $k$-element set $\widetilde{J}_0$ is injective on the frequency-$< k$ band (Lemma 4.2), while each $(k+1)$-element summand $\widetilde{B}_t$ supports a nonzero polynomial $P_t$ of degree at most $\operatorname{diam}_{\mathbb{N}}(\widetilde{B}_t)$ annihilating the band (Lemma 4.3). Combining the resulting rank inequality $2\operatorname{rank}_F(\tau - K_2 \circ K_1) \ge bk(d - \sum_t \operatorname{diam}_{\mathbb{N}}(\widetilde{B}_t))$ (Theorem 4.4) with iterated Kővári–Sós–Turán sumset extraction (Lemma 4.5, with $k = O(\log d)$) yields Theorem 4.6: for all $b, M_1, M_2 \ge 1$ and all $d \ge D_0(M_1, M_2)$ with $(d : F) \neq 0$,

$$\operatorname{rank}_F(\tau - K_2 \circ K_1) \;\ge\; c(M_1, M_2)\, b\, d\, (\log d)^{\frac{1}{2M_2-1}},$$

and in particular $\Omega(b\, d \log d)$ when $M_2 = 1$ (Corollary 4.8). For $M_1 \ge 2$ and sufficiently large $d$, this replaces the iterated-logarithmic growth rates obtained from [10, Theorem 6.6] via Gowers' van der Waerden bound [13] by an explicit log-power rate. By contrast, at $M_1 = 1$, $\mathsf{W}(1, \ell) = \ell$ already gives $\Omega(bd^2/M_2)$ for all $d \ge 4M_2 + 1$ in [10, Theorem 6.6]; at $M_1 = 2, M_2 = 1$, $k = 2$ with $\mathsf{W}(2, 5) = 178$ [19] gives $b(d - 177)$, which Theorem 4.6 ($c(2, 1) = \frac{1}{144 \log 32}$) beats only for $d \gtrsim 10^{216}$ (and the formalized threshold at $(2, 1)$ satisfies $D_0(2, 1) = 32^{36} + 2 \approx 1.53 \times 10^{54}$). Example 4.9 constructs an explicit 24-dimensional shift-covariant 4-channel band witness ($q = 4$; formalized at $b = 2, M_2 = 1$ where $bkd = 48$) over $\mathbb{C}$ at $d = 12, k = 2$: under the text-only parameter reading $(b, M_2) = (1, 2)$ ($24 = bkd$), a single half-density band without any annihilating-polynomial step admits $bkd$-dimensional solutions ($r \ge 0$), showing in that reading that the annihilating-polynomial steps in Theorem 4.4 cannot all be dropped.

3. **Run-extension bounds for one-stage linear shared decoders (Section 5).** In [10, Theorem 3.4], iterating $C \cap \operatorname{rot}_s^{-1}(C) \le H_{\{0,s\}}$ gave $\frac{41}{64}\, bd^2 + \frac{5}{3}\, bd^{3/2}$ for two classes, while the nested-component certificates of [10] cannot go below $(\frac{5}{8} - o(1))bd^2$ on almost every two-class partition ([10, Theorem C.6]; their isolation variant [10, Theorem C.7] has floor $\frac{9}{16} - o(1)$). For $F$-linear decoders (extending the $b = 1$ model of [4, Definition 6.3 and Problem 9.1(a)] to $b$ lanes), evaluating $\operatorname{rot}_s(X) \in \mathcal{X}$ at the right endpoint of each maximal monochromatic run along each non-monochromatic cycle of $c \mapsto c + s$ supplies an extra linear equation per run. Filtering by $\operatorname{rot}_s$-eigenmodes over $\overline{F}$ proves the single-shift bound $B_{\mathrm{run}}(b, p, s)$ (Theorem 5.2) whenever $\operatorname{char}(F) \nmid (d/\gcd(d, s))$, whose value $\frac{1}{4}d^2 + \frac{3}{2}d$ (versus $\dim_F H_{\{0,1\}} = d^2$) on the alternating two-coloring for even $d = 2m \ge 2$ at $b = 1, s = 1$ is realized over *every* field (Theorem 5.3, attaining Theorem 5.2 whenever $\operatorname{char}(F) \nmid 2m$). Averaging over shifts proves $D_b(p) \le \frac{19}{32}\, bd^2 + \frac{3}{2}\, bd^{3/2}$ for two classes whenever $\operatorname{char}(F) \nmid d$ (Theorem 5.4, non-vacuous and improving on [10, Theorem 3.4] for $d \ge 14$, below $\frac{3}{4}\, bd^2$ for $d \ge 93$, and with leading coefficient $\frac{19}{32} < \frac{5}{8} < \frac{41}{64}$; positive characteristics dividing $d$ and the gap from $\frac{19}{32}$ to $\frac{1}{2}$ remain open) and $D_b(p) \le c_M\, bd^2 + (\frac{5}{4} - \frac{1}{4M})bd$ with $c_M = 1 - \frac{5}{4M} + \frac{1}{M^2} - \frac{1}{4M^3}$ for $M \ge 2$ at primes $d \neq \operatorname{char}(F)$ (Theorem 5.5, giving $\frac{19}{32}\, bd^2 + \frac{9}{8}\, bd$ at $M = 2$). Its leading coefficient $c_M$ beats the flag bound $1 - \frac{1}{2M}$ of [4, Theorem 5.1, Corollary 5.9(b), Remark 5.10, and Problem 9.1(a)] (coordinate form of Theorem 5.1 Lean-verified; Corollary 5.9(b) and Remark 5.10 proved in text) for every $M \ge 2$ (whereas the difference-cover bound [10, Theorem 3.5, eq. (6)] beats the flag bound for $M \ge 7$ and its square-root specialization eq. (7) does so for $M \ge 11$). Against the optimized difference-cover bound [10, Theorem 3.5, eq. (6)], $c_M$ has a strictly smaller leading coefficient if and only if $2 \le M \le 40$—improving on [10, Theorem 3.5] at primes $d \neq \operatorname{char}(F)$ past finite

crossovers (from $d \geq 7$ at $M = 2$ up to $d \geq 39{,}163$ at $M = 40$) and, via Theorem 5.4, for composite $d \geq 35$ with $\mathrm{char}(F) \nmid d$ at $M = 2$. Conversely, [10, Theorem 3.5, eq. (6)] has a $1 - \Theta(M^{-1/2})$ leading coefficient that is strictly smaller for every $M \geq 41$, and applies to arbitrary composite $d$ and arbitrary field characteristic.

4. **Exact dimension 25 for $\{0,1,3\} \mid \{2,4,5\}$ on $\mathbb{Z}_6$ outside characteristics 2 and 3 (Section 6).** On [4, Problem 9.1(b)], combining six-mode column-Fourier decomposition on $H_{\{0,1\}}$, an antipodal quotient bound $3 \dim Q + S_K \leq 24$, and kernel-disjointness / $4 \times 4$ row elimination (cleared determinant $2^4 \cdot 3^3 \neq 0$) at $S_K \in \{11, 12\}$ proves $\dim_F(\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})) \leq 14$ (Lemma 6.1) and hence $D_1(\{0,1,3\}, \{2,4,5\}) = 25$ over every field with $\mathrm{char}(F) \notin \{2,3\}$ (Theorem 6.2, paired with the lower bound of [9, Theorem 3.2]); characteristics 2 ($25 \leq D_1 \leq 28$, [4, Theorem 6.4 and Proposition 8.2], [9, Theorem 3.2]) and 3 ($21 \leq D_1 \leq 28$ from symmetric matrices and [4, Theorem 6.4]; whether 25 is attainable in characteristic 3 is [9, Problem 6.1]) remain open.

5. **Three-term swap codes and one- and two-sided interval-mask bounds (Section 7).** In [4, Remark 5.6], examples of three-term swap codes $\{X \in \mathrm{Mat}_{d \times d}(F) : \sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}\}$ exceeding $d(d+1)/2$ were given at $d = 3$ ($8 > 6$, over every field) and $d = 4$ ($11 > 10$, checked numerically over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$), with no example known for $d \geq 5$. In Theorems 7.1 and 7.2, we construct a $3 \times 3$ block family on $d = 3m$ achieving exact dimension $4m(m+1)$ over every field—exceeding $d(d+1)/2$ at $d \in \{3, 6, 9, 12\}$—and pad it to obtain $29 > 28$ at $d = 7$ and $11 > 10$ at $d = 4$ over every field; $d = 5$ and every $d \geq 8$ outside $\{9, 12\}$ remain open. On [4, Problem 9.3], after recording the one-sided endpoint-refinement baseline $(d^2 + d \min(\beta_A, \beta_B))/2$ (Proposition 7.3), we prove a characteristic-free two-sided bound for two-scale (coarse–fine) families (Theorem 7.4 and Corollary 7.5) whose leading coefficient on contiguous fine singletons plus one coarse complement, $\frac{1}{2} + \frac{1}{4} \min(\beta(1+\alpha), \alpha(1+\beta))$, is strictly below one-sided Theorem TP's $\frac{1}{2} + \frac{1}{2} \min(\alpha, \beta)$ ([4, Theorem 6.4]) for all $\alpha, \beta \in (0,1)$ (giving $\frac{11}{16} d^2 + d < \frac{3}{4} d^2$ at $\alpha = \beta = 1/2$ for all even $d \geq 18$), while Proposition 7.7 contrasts interval boundaries ($\leq 2$) with periodic Walsh digit masks ($2^{j+1}$ boundaries, profile 2 per scale, and operator span dimension $\leq (m+1)2^m$).

6. **Restricted shuffle and row/column-uniform operator pipelines and anti-diagonal untwisting (Section 8).** Across the row/column tensor cut $F_{\mathrm{row}}^d \otimes_F (F_{\mathrm{col}}^d \otimes_F F^b)$, Operator Schmidt Rank satisfies $\mathrm{rk}_{\otimes}(\tau) = d^2$ and $\mathrm{rk}_{\otimes}(S) = \mathrm{rk}_{\otimes}(S^{-1}) = 4$ for Stone's perfect shuffle and unshuffle [20, 18, 23] on $d = 2^m \geq 2$ (Theorem 8.2), forcing $t \geq m = \log_2 d$ shuffle/unshuffle passes (attained by $S^m = \tau$) in any pipeline composed solely of $S, S^{-1}$, row-uniform operators $I_d \otimes B$, and column-uniform operators $A \otimes I_{db}$ (Theorem 8.3). As we emphasize at the start of Section 8, this applies only to that restricted operator pipeline and *does not* answer [4, Problem 9.5]: within-row cyclic shear $X_{r,c} \mapsto X_{r,(c+r) \bmod d}$ has $\mathrm{rk}_{\otimes} = d$ (Proposition 8.5; see also [4, Proposition 6.25(a)] for affine row-coset stages) yet runs in $O(L)$ time on a 2-work-tape Turing machine (Proposition 8.4, proved in the shared library `lean-machine` alongside 3-tape shuffle and 2-tape unshuffle). Section 8.2 repackages the shift-covariance subspace of [10, Lemma 3.1] (together with the compatibility equivalence `SharedDecoders.isSharedCode_iff_compatible` in its Lean library) via an explicit `TensorProduct` isomorphism intertwining $\mathrm{rot}_t$ with $\mathrm{id} \otimes \sigma^t$ (where $(\sigma^t v)_a = v_{a+t}$, Theorem 8.7) and records the two-subspace Grassmann relaxation (Proposition 8.8).

7. **Cut-cone evaluation, random-order set capacity, and negative brackets for two column-class cut reductions (Section 9).** On frozen trace/cut fibres of `Transpose`.

TM, Lemma 9.1 extends the causal-cone lemmas of [5, Lemma 4.4 and Corollary 4.5], [7, Lemmas 3.3(b) and 4.7(a)], and [12] by instantiating the $(M.k+1)^D$ cone-size bound on the native step DAG and constructing depth-$D$ symbolic `CausalConeTree`s directly from $M.\delta$ whose syntactic input-cell support has size at most $b(M.k+1)^D$. Theorem 9.3 formalizes in Lean the column-class analogue of the $1/(\Delta+1)$ random-order acyclic-subgraph bound of [5, Proposition 4.14(a)–(b)] (after [2, Theorem 3]). Finally, Remark 9.5 and Proposition 9.6 illustrate that two naive column-class cut reductions—discarding cut-touching outputs, or positing a worst-case full-product class count across per-column cut slots—produce strictly negative deficit brackets along the control spine $u \to u+1$, reinforcing the observation in the commentary on [4, Problem 9.4(b)] regarding per-position fixed-alphabet recovery.

### 1.3 Formalization conventions

Every numbered theorem, lemma, proposition, corollary, and example in this paper has a kernel-verified Lean 4 counterpart (version `v4.34.1` with Mathlib [22]) across the `BoundedArityDecoders` package and the shared `LeanMachine.ShuffleShear` modules in `lean-machine`, depending solely on Lean's three standard axioms (`propext`, `Classical.choice`, and `Quot.sound`), with zero `sorry`, `admit`, `native_decide`, or custom axioms. Each numbered result carries a `Lean`: status line citing proved `theorem` declarations, followed by an outline of the proof; where a companion asymptotic formulation, external-theorem specialization, or parameter re-reading is proved in the text only, the status line marks the result *partially formalized* and states the text-only gap explicitly (summarized in Appendix A).

## 2 Model and preliminaries

### 2.1 Matrix states, class-uniform operators, and bounded-arity set codes

For $d \geq 1$, write $\mathbb{Z}_d = \mathbb{Z}/d\mathbb{Z}$ (formalized as `ZModd`). Given an entry alphabet or module $A$, a $d \times d$ matrix state is a map $X : \mathbb{Z}_d \to \mathbb{Z}_d \to A$, written $X \in \mathrm{Mat}_{d\times d}(A)$ or $A^{d\times d}$ (`SharedDecoders.State`). Its transpose $\tau(X) = X^{\mathsf{T}}$ (`SharedDecoders.swap`, or `SharedDecoders.swapLinear` when $A$ is an $F$-module) and column-cyclic shift $\mathrm{rot}_c(X)$ for $c \in \mathbb{Z}_d$ (`SharedDecoders.rot`) are defined by

$$\tau(X)_{i,j} \;=\; X_{j,i}, \qquad \mathrm{rot}_c(X)_{i,j} \;=\; X_{i,\,j+c} \qquad (i,j \in \mathbb{Z}_d).$$

Thus column 0 of $\mathrm{rot}_c(X)$ is column $c$ of $X$, whereas column $c$ of $\tau(X)$ is the vector $r \mapsto X_{c,r}$ (row $c$ of $X$).

When $A = F^b$ (`Finb->F`) over a field $F$, a linear operator $K : \mathrm{Mat}_{d\times d}(F^b) \to \mathrm{Mat}_{d\times d}(F^b)$ is *class-uniform* for a column partition $p : \mathbb{Z}_d \to \kappa$ (`SharedDecoders.ClassUniform`; cf. [4, Definition 6.3] and [10, Section 2]) if there exist class decoders $f_m : \mathrm{Mat}_{d\times d}(F^b) \to (F^b)^{\mathbb{Z}_d}$ $(m \in \kappa)$ such that column $c$ of $K(X)$ equals $f_{p(c)}(\mathrm{rot}_c(X))$ for every $X$ and every $c \in \mathbb{Z}_d$ (on the full ambient space, linearity of $K$ automatically allows each active $f_m$ to be chosen $F$-linear). On a proper subspace $C \leq \mathrm{Mat}_{d\times d}(A)$, however, decoder linearity is an extra hypothesis: `SharedDecoders.IsSharedCode` ([10, Definition 2.1]) restricts $C$ to an $F$-linear submodule while allowing arbitrary deterministic maps $f_m$.

For the one-stage linear capacity bounds of Sections 5 and 6, we write `laneRot` for the column-rotation map $\mathrm{rot}_s(X)_{i,j} = X_{i,j+s}$ on `Matrix(ZModd)(ZModd)(Finb->F)`. An $F$-linear submodule $\mathcal{X} \leq \mathrm{Mat}_{d\times d}(F^b)$ is a *$b$-lane linear class-uniform code* for $p : \mathbb{Z}_d \to \kappa$ (`IsLinearClassUniformCode`) if there exist $F$-linear decoders $\mathcal{A}_m : \mathrm{Mat}_{d\times d}(F^b) \to (F^b)^{\mathbb{Z}_d}$ $(m \in \kappa)$ such that

$$X_{c,r} \;=\; \mathcal{A}_{p(c)}(\mathrm{rot}_c(X))_r \qquad \text{for all } X \in \mathcal{X} \text{ and all } c,r \in \mathbb{Z}_d$$

(here $c$ indexes row $c$ of $X$, i.e. output column $c$ of $\tau(X)$, while $\mathrm{rot}_c$ shifts the second index of $X$). Its maximum dimension over all such subspaces $\mathcal{X}$ is $D_b(p) := \texttt{linearClassUniformMaxDim}(F, b, p)$ (written $D(p) := D_1(p)$ at $b = 1$). For a single shift $s \in \mathbb{Z}_d$, the *single-shift link-covariance space* (`ShiftLinkCovarianceSpace`, formulated symmetrically in Lean over two-element shift links $\delta, \delta' \in \{0, s\}$ with $p(c) = p(c')$ and $c + \delta = c' + \delta'$) is

$$H_{\{0,s\}} := \Big\{ X \in \mathrm{Mat}_{d\times d}(F^b) \;:\; \forall c \in \mathbb{Z}_d,\ p(c) = p(c+s) \implies \forall j \in \mathbb{Z}_d,\ X_{c+s,j} = X_{c,j+s} \Big\}.$$

We write $\mathrm{char}(F) \nmid n$ for the predicate `CoprimeToChar` $F$ $n$ ($\mathrm{char}(F) = 0$ or $\gcd(n, \mathrm{char}(F)) = 1$).

**Proposition 2.1** (Linear class-uniform bridges and scalar-extension transfer)**.** Lean: `laneRot_apply` [`ExtensionLinearRun.lean`]

Lean: `extension_LinearModelAndCharBridges`, `laneOneMatrixEquiv_laneRot`, `matrixToSharedStateEquiv_laneRot` [`ExtensionLinearRunSharpness.lean`]

Lean: `coprimeToChar_iff_natCast_ne_zero`, `exists_isPrimitiveRoot_of_isAlgClosed`, `exists_isPrimitiveRoot_algebraicClosure`, `linearIndependent_fin_pi_ringHom`, `finrank_le_of_pi_ringHom`, `extendLinearMapPi_comp_map` [`ExtensionFieldTransfer.lean`]

*Let $F$ be a field and let $d \geq 1$ and $b \in \mathbb{N}$.*

*(a) At $b = 1$ and for any class type $\kappa$, pushing forward along* `laneOneMatrixEquiv` *identifies* `IsLinearClassUniformCode` $p\ \mathcal{X}$ *with* `SwapRigidity.IsClassUniformCode` $p\ \mathcal{X}$ *([4, Definition 6.3]), intertwines* `laneRot` $s$ *with* `SwapRigidity.rot` $s$*, and preserves both* $\dim_F \mathcal{X}$ *and* $\dim_F(\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X}))$*. For arbitrary $b$, pushing forward along* `matrixToSharedStateEquiv` *embeds every* `IsLinearClassUniformCode` *into* `SharedDecoders.IsSharedCode` *([10, Definition 2.1]) and intertwines* `laneRot` $s$ *with* `SharedDecoders.rot` $s$*; moreover, $\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X}) \leq H_{\{0,s\}}$ for every $s \in \mathbb{Z}_d$.*

*(b) For every $n \geq 1$,* `CoprimeToChar`$(F, n)$ *holds if and only if $(n : F) \neq 0$; every algebraically closed field $K$ with $(n : K) \neq 0$—and in particular $\overline{F}$ (*`AlgebraicClosureF`*) whenever* `CoprimeToChar`$(F, n)$*—contains a primitive $n$-th root of unity.*

*(c) For every field homomorphism $\varphi : F \to K$, coordinatewise application of $\varphi$ preserves linear independence of finite vector families in $F^\iota$ for any index type $\iota$, satisfies $\dim_F U \leq \dim_K W$ for any finite $\iota$ whenever $U \leq F^\iota$ and $W \leq K^\iota$ obey $\varphi(U) \subseteq W$, and for finite $\iota$ and arbitrary $\kappa'$ extends every $F$-linear map $L : F^\iota \to F^{\kappa'}$ to a $K$-linear map* `extendLinearMapPi`$(\varphi, L) : K^\iota \to K^{\kappa'}$ *commuting with coordinatewise application of $\varphi$.*

*Outline of the formal proof.* In (a), `laneOneMatrixEquiv` drops the unique coordinate $0 \in \mathrm{Fin}\, 1$ and `matrixToSharedStateEquiv` unfolds $\mathrm{Mat}_{d\times d}(F^b)$ to $\mathbb{Z}_d \to \mathbb{Z}_d \to \mathrm{Fin}\, b \to F$; if $X \in \mathcal{X}$ and $\mathrm{rot}_s(X) \in \mathcal{X}$ with $p(c) = p(c+s) = m$, then $X_{c+s,j} = \mathcal{A}_m(\mathrm{rot}_{c+s}(X))_j = \mathcal{A}_m(\mathrm{rot}_c(\mathrm{rot}_s(X)))_j = \mathrm{rot}_s(X)_{c,j} = X_{c,j+s}$. In (b), $(n : F) = 0$ iff $\mathrm{char}(F) \mid n$; when $(n : K) \neq 0$ in an algebraically closed field $K$ (including $K = \overline{F}$ when $(n : F) \neq 0$, via injectivity of $\mathrm{algebraMap}\, F\, \overline{F}$), $Z^n - 1$ has $n$ distinct roots in $K$ ($nZ^{n-1} \neq 0$ at every root), and partitioning those roots by exact multiplicative order across divisors $m \mid n$ with at most $\varphi(m)$ elements per divisor forces $\varphi(n) \geq 1$ primitive $n$-th roots via $\sum_{m|n} \varphi(m) = n$. In (c), preservation of linear independence is proved by induction on the family size via Gaussian pivot elimination over $F$ at a nonzero coordinate, and `extendLinearMapPi` expands $L$ in the standard basis of $F^\iota$. □

For arbitrary subsets $C \subseteq \Sigma^{d\times d}$ over a finite alphabet $\Sigma$ (used in Section 9), we also record syntactic cell footprints:

**Definition 2.2** (Syntactic read footprints and bounded-arity shared set codes)**.** Let $d \geq 1$, let $M, \Delta \in \mathbb{N}$, and let $\Sigma$ be a finite alphabet.

(a) A family of class decoders $f : \operatorname{Fin} M \to \Sigma^{d\times d} \to \Sigma^{\mathbb{Z}_d}$ has *syntactic cell support* $\leq \Delta$ with footprint table $Q : \operatorname{Fin} M \to \mathbb{Z}_d \to \operatorname{Finset}(\mathbb{Z}_d \times \mathbb{Z}_d)$ (`SyntacticCellSupport`) if $|Q_{m,r}| \leq \Delta$ for all $(m, r)$, and whenever $Y_1, Y_2 \in \Sigma^{d\times d}$ agree on every cell $q \in Q_{m,r}$, one has $f_m(Y_1)_r = f_m(Y_2)_r$. (Its whole-column analogue is `SyntacticColumnSupport`.)

(b) Given $p : \mathbb{Z}_d \to \operatorname{Fin} M$, a broken-output set $\mathcal{E} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$ (indexed by output coordinates $(r, c)$ of $\tau(X)_{r,c} = X_{c,r}$), and $C \subseteq \Sigma^{d\times d}$, the data $(\Delta, p, \mathcal{E}, f, Q)$ form a *bounded-arity shared set code on* $C$ (`IsBoundedAritySetCode`) if $(f, Q)$ has syntactic cell support $\leq \Delta$ and

$$f_{p(c)}(\mathrm{rot}_c(X))_r \;=\; X_{c,r} \qquad \text{for all } X \in C \text{ and all } (r, c) \notin \mathcal{E}.$$

## 2.2 Multitape Turing machines, canonical words, and grid bridges

In Sections 8 and 9 we interface with the multitape Turing machine model `Transpose.TM` imported unchanged from [6, Sections 2.1–2.2 and Definition 2.1]: a machine $M$ has state set $\operatorname{Fin}(M.q + 1)$, work alphabet $\operatorname{Fin}(M.g + 1)$, $M.k$ two-way read-write work tapes, a two-way read-only binary input tape of length $L = d^2 b$ where $b =$ `Transpose.entryBits`$(d)$, and a one-way write-only binary output tape. The canonical word alphabet (`CanonicalWordAlphabet`) is $\Sigma_d := \{0, 1\}^b$ (`Fin(Transpose.entryBitsd)->Bool`).

**Lemma 2.3** (Canonical word width and dummy-tape padding)**.** Lean: `entryBits_eq_clog` [`NativeSweepInterfaceProofs.lean`]
Lean: `canonicalWordAlphabet_card_and_ge_two` [`SetCapacityConsequences.lean`]
Lean: `padDummyWorkTape_transposesWithin` [`MachineSweepConsequences.lean`]

*For every $d \geq 0$, $2 \leq |\Sigma_d|$, and for every $d \geq 2$,* `Transpose.entryBits`$(d) = \lceil \log_2 d \rceil$ *and* $|\Sigma_d| = 2^{\lceil \log_2 d \rceil}$*. Moreover, for every time bound $T : \mathbb{N} \to \mathbb{N}$ and every machine $M$ satisfying* `Transpose.TransposesWithin` $M$ $T$*, there exists a machine $M^+$ with $1 \leq M^+.k$ that also satisfies* `Transpose.TransposesWithin` $M^+$ $T$*.*

*Outline of the formal proof.* Since $1 \leq$ `Transpose.entryBits`$(d)$ for all $d$, $|\Sigma_d| = 2^b \geq 2$, and for $d \geq 2$ the definition $\lfloor \log_2(d-1) \rfloor + 1$ agrees with `Nat.clog`$(2, d)$. When $M.k = 0$, adjoining a single blank work tape whose head never moves preserves states, input-head positions, and emitted output bits step for step. □

To connect Boolean tape matrices $A :$ `Transpose.Matrix` (read on $d \times d$ entries at canonical bit-width $b =$ `Transpose.entryBits`$(d)$) with the linear operators of Section 8, `encodeTMMatrixToGrid` maps each bit $A(r, c, \ell) \in \{0, 1\}$ to $0, 1 \in F$ in `GridState`$(F, d, b) = (\operatorname{Fin} d \to \operatorname{Fin} d \to \operatorname{Fin} b \to F)$.

**Proposition 2.4** (Equivariance of the matrix-to-grid bridge)**.** Lean: `tmMatrix_gridLinear_bridges` [`ShuffleShearBridge.lean`]

*For every ring $F$,* `encodeTMMatrixToGrid` *intertwines* `stoneShuffleTMMatrix` *with* `stoneShuffleGridLinear` *(for $d = 2^m, m \geq 0$),* `stoneUnshuffleTMMatrix` *with* `stoneUnshuffleGridLinear` *(for $d = 2^m, m \geq 0$),* `rowShearTMMatrix` *with* `rowShearLinear` *(for $d \geq 1$), and* `Transpose.transpose` *with* `gridSwapLinear`*.*

*Outline of the formal proof.* Verified directly by unfolding `encodeTMMatrixToGrid` on each coordinate permutation. □

# 3 Affine Hilbert cubes and Ramsey-free two-stage cutoffs

## 3.1 Common-message growth on ordered operator cubes

Suppose two $F$-linear class-uniform operators $K_1, K_2 : \mathrm{Mat}_{d\times d}(F^b) \to \mathrm{Mat}_{d\times d}(F^b)$ with class partitions $p_1 : \mathbb{Z}_d \to \mathrm{Fin}\, M_1$ and $p_2 : \mathbb{Z}_d \to \mathrm{Fin}\, M_2$ satisfy $K_2 \circ K_1 = \tau$. By the common-message reduction of [10, Lemma 6.1 and Theorem 6.3], any such factorization produces a subspace $W \le A$ of dimension $\dim_F W = bd$ inside a $q$-channel cyclic shift module $A = (\mathbb{Z}_d \to F^q)$ with $q \le bM_2$ ($\dim_F A = qd \le bM_2 d$), together with a linear readout $L : A \to Z$ and a reference shift $c_{\mathrm{ref}} \in J$ in a single $p_1$-class $J = p_1^{-1}(m_1)$ such that $w \mapsto L(T_{c_{\mathrm{ref}}} w)$ is injective on $W$ and $L(T_c w) = L(T_{c_{\mathrm{ref}}} w)$ for all $w \in W$ and all $c \in J$ (where $(T_s w)_u = w_{u+s}$, formalized as `cyclicChannelShift`; passing from $W$ to $T_{c_{\mathrm{ref}}}(W)$ centers the invariance at shift 0). In [10, Theorem 6.3], choosing a monochromatic arithmetic progression $\{c_0 + js : 0 \le j < \ell\} \subseteq J$ gave $\ell \dim_F W \le \dim_F A + (\ell - 1)\dim_F \ker(T_s - \mathrm{id})$, yielding $d \le M_2(\mathsf{W}(M_1, M_2 + 1) - 1)$.

To dispense with arithmetic progressions over an arbitrary field $F$, consider any $k$ linear endomorphisms $R_0, \dots, R_{k-1} \in \mathrm{End}_F(A)$ (not required to commute) and define for each $\varepsilon \in \{0,1\}^k$ the ordered operator cube product (`orderedCubeMap`)

$$R^{\varepsilon} \;:=\; R_{k-1}^{\varepsilon_{k-1}} \cdots R_0^{\varepsilon_0} \in \mathrm{End}_F(A).$$

When $R_0 = \cdots = R_{k-1} = R$, the lemma below specializes to [10, Lemma 6.1] with the same nested-span argument.

**Lemma 3.1** (Affine $k$-cube common-message growth; cf. [10, Lemma 6.1])**.** Lean: `lemma_AffineCubeCommonMessageGrowth` [`AffineCubeCompositionProofs.lean`]

*Let $F$ be any field, $A$ a finite-dimensional $F$-vector space, $Z$ an $F$-vector space, $R_0, \dots, R_{k-1} \in \mathrm{End}_F(A)$, $W \le A$ a subspace, and $L : A \to Z$ an $F$-linear map injective on $W$. If $L(R^{\varepsilon} x) = L(x)$ for all $x \in W$ and all $\varepsilon \in \{0,1\}^k$, then*

$$(k+1)\dim_F W \;\le\; \dim_F A \;+\; \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A).$$

*Outline of the formal proof.* Define the nested cube-orbit subspaces $S_0 := W$ and $S_{t+1} := S_t + R_t(S_t)$ for $0 \le t < k$. By induction on $t$, every $u \in S_t$ is a linear combination of elements $R^{\varepsilon} x$ with $x \in W$ and $\varepsilon_s = 0$ for $s \ge t$. Comparing $\varepsilon_t = 1$ with $\varepsilon_t = 0$ in the cube invariance $L(R^{\varepsilon} x) = L(x)$ shows that $P_t := R_t - \mathrm{id}_A$ maps $S_t$ into $S_{t+1} \cap \ker L$. Since $W \subseteq S_t \subseteq S_{t+1}$ and $L$ is injective on $W$, we have $W \cap P_t(S_t) \subseteq W \cap \ker L = \{0\}$, so

$$\dim_F S_{t+1} \;\ge\; \dim_F W + \dim_F P_t(S_t) \;\ge\; \dim_F W + \dim_F S_t - \dim_F \ker(R_t - \mathrm{id}_A).$$

Summing from $t = 0$ to $k - 1$ gives $(k+1)\dim_F W \le \dim_F S_k + \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A) \le \dim_F A + \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A)$. □

## 3.2 Iterated Cauchy–Schwarz cube extraction and Theorem 2A

Unlike arithmetic progressions, affine $k$-cubes exist in every dense subset $J \subseteq \mathbb{Z}_d$ (including 3-AP-free sets) by the classical Szemerédi (density Hilbert) cube lemma [21, 14], implemented here via bucket pigeonholing and Cauchy–Schwarz with explicit step bounds $1 \le s_t \le \lfloor d/(2k^2) \rfloor$ keeping $\dim_F \ker(T_{s_t} - \mathrm{id})$ small.

**Lemma 3.2** (Cauchy–Schwarz affine $k$-cube extraction and dense-class bound; cf. [21, 14])**.** Lean: `cauchySchwarzAffineCubeExtraction` [`AffineCubeCompositionProofs.lean`]
Lean: `singleSubset_CauchySchwarzAffineCubeExtraction`, `denseSubset_AffineCubeCommonMessageBound` [`AffineCubeConsequences.lean`]

*Let $d, M_{\text{eff}}, k \geq 1$ satisfy $\max\bigl(8M_{\text{eff}}k^2,\ (2M_{\text{eff}})^{2^k}\bigr) < d$.*

*(a)* Single-subset form*: for every subset $J \subseteq \mathbb{Z}_d$ of size $|J| \geq d/M_{\text{eff}}$, there exist $c_0 \in J$ and steps $s_0, \dots, s_{k-1} \in \{1, \dots, \lfloor d/(2k^2)\rfloor\}$ such that $c_0 + \sum_{t:\varepsilon_t=1} s_t \in J$ for all $\varepsilon \in \{0,1\}^k$. Over any field $F$, any subspace $W \leq (\mathbb{Z}_d \to F^q)$ admitting an $F$-linear readout $L : (\mathbb{Z}_d \to F^q) \to Z$ into an $F$-vector space $Z$ and a reference shift $c_{\text{ref}} \in J$ such that $w \mapsto L(T_{c_{\text{ref}}} w)$ is injective on $W$ and $L(T_c w) = L(T_{c_{\text{ref}}} w)$ for all $c \in J$ and all $w \in W$ satisfies $(k+1)\dim_F W \leq qd + qk\lfloor d/(2k^2)\rfloor$, ruling out exact two-stage linear factorizations (for any stage-1 class type $\kappa_1$) whenever one stage-1 class has size $\geq d/M_{\text{eff}}$ at $k = M_2$ and $b \geq 1$.*

*(b)* Coloring form*: setting $M_{\text{eff}} = M_1$ and $k = M_2$, every $M_1$-coloring $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ admits a monochromatic affine $M_2$-cube with steps in $\{1, \dots, \lfloor d/(2M_2^2)\rfloor\}$.*

*Outline of the formal proof.* Lift $J$ to $S_0 \subseteq \{0, \dots, d-1\}$. Partitioning $\{0, \dots, d-1\}$ into $Q = 2k^2$ intervals of length $\leq \lfloor d/Q\rfloor + 1$ and applying Cauchy–Schwarz to the bucket sizes shows that for any $S \subseteq \{0, \dots, d-1\}$ with $|S| > Q$, some step $s \in \{1, \dots, \lfloor d/Q\rfloor\}$ satisfies $|S|(|S| - Q) \leq 2d\,|\{x \in S : x + s \in S \text{ in } \mathbb{N}\}|$. Iterating $k$ times with $D_0 = M_{\text{eff}}$ and $D_{t+1} = 2D_t^2$ ($D_k = \frac{1}{2}(2M_{\text{eff}})^{2^k}$) maintains $|S_t| \geq d/D_t - Q/2$ for $1 \leq t < k$. At $k = 1$, $d > 8M_{\text{eff}}$ gives $|S_0| \geq d/M_{\text{eff}} > Q = 2$; for $k \geq 2$, $|S_{k-1}| > Q$ follows from $d/D_{k-1} > 3Q/2$, i.e. $d > \frac{3}{2}k^2(2M_{\text{eff}})^{2^{k-1}}$, which holds by $d > (2M_{\text{eff}})^{2^k}$ whenever $(M_{\text{eff}}, k) \neq (1,2)$ and by $d > 8M_{\text{eff}}k^2 = 32 > 24$ at $(M_{\text{eff}}, k) = (1,2)$, giving $|S_k| \geq 1$. Because each step filters $x + s_t \in S_t$ in $\mathbb{N}$ inside $S_0 \subseteq \{0, \dots, d-1\}$, every cube sum $x_0 + \sum_{t:\varepsilon_t=1} s_t$ ($x_0 \in S_k$) lies in $S_0$ with natural sum $< d$, preventing wraparound modulo $d$. Feeding the extracted cube into Lemma 3.1 with $\dim_F \ker(T_{s_t} - \mathrm{id}) \leq q s_t \leq q\lfloor d/(2k^2)\rfloor$ gives the dimension bound. □

**Theorem 3.3** (Theorem 2A: Ramsey-free two-stage exact factorization cutoff)**.** Lean: `theorem2A_TwoStageRamseyFreeCutoff` [`AffineCubeCompositionProofs.lean`]
Lean (partial): `theorem2A_GrowingClassesNonFactorization` [`AffineCubeConsequences.lean`]
Not formalized: the asymptotic $o\bigl(d^{1/2^{M_2}}\bigr)$ and $(1-\varepsilon)\log_2\log_2 d$ specializations of the pointwise inequality and the translate-covering step turning Behrend's 3-AP-free set [3] into a $2^{O(\sqrt{\log d})}$-class partition of $\mathbb{Z}_d$.

*Let $F$ be any field, let $d, b, M_1, M_2 \geq 1$, and let $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ and $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$. If two $F$-linear class-uniform maps $K_1, K_2$ on $\mathrm{Mat}_{d\times d}(F^b)$ satisfy $K_2 \circ K_1 = \tau$, then*

$$d \ \leq\ \max\Bigl(8M_1M_2^2,\ (2M_1)^{2^{M_2}}\Bigr).$$

*Consequently, for any $b \geq 1$ and any class schedules $M_1(d), M_2(d) \geq 1$ eventually satisfying both $8M_1(d)M_2(d)^2 < d$ and $(2M_1(d))^{2^{M_2(d)}} < d$—in particular whenever $M_2 \geq 2$ is fixed and $M_1(d) = o\bigl(d^{1/2^{M_2}}\bigr)$ (covering partitions of $\mathbb{Z}_d$ into $M_1(d) = 2^{O(\sqrt{\log d})}$ 3-AP-free classes via Behrend [3]) or whenever $M_1 \geq 2$ and $\varepsilon \in (0,1)$ are fixed and $M_2(d) \leq (1-\varepsilon)\log_2\log_2 d$—for all sufficiently large $d$ no exact two-stage linear factorization $K_2 \circ K_1 = \tau$ on $\mathrm{Mat}_{d\times d}(F^b)$ exists over any field $F$.*

*Outline of the formal proof and proof of the remaining claims.* Assume for contradiction that $\max(8M_1M_2^2, (2M_1)^{2^{M_2}}) < d$ and $K_2 \circ K_1 = \tau$. By Lemma 3.2(b), some $p_1$-class contains an affine $M_2$-cube with steps $1 \leq s_t \leq \lfloor d/(2M_2^2)\rfloor$. On the $q$-channel common-message module $A = (\mathbb{Z}_d \to F^q)$ ($q \leq bM_2$, $\dim_F A = qd \leq bM_2 d$), [10, Lemma 6.1 and Theorem 6.3] supplies a

subspace $W \le A$ of dimension $\dim_F W = bd$ while $\dim_F \ker(T_{s_t} - \mathrm{id}) \le qs_t \le bM_2\lfloor d/(2M_2^2)\rfloor$. Applying Lemma 3.1 with $k = M_2$ gives

$$(M_2+1)bd \;\le\; bM_2 d + \sum_{t=0}^{M_2-1} bM_2\lfloor d/(2M_2^2)\rfloor \;\le\; bM_2 d + \frac{1}{2}bd,$$

contradicting $bd > 0$. For the asymptotic clauses (proved in text), if $M_2 \ge 2$ is fixed and $M_1(d) = o(d^{1/2^{M_2}})$, then $(2M_1(d))^{2^{M_2}} = o(d)$ and $8M_1(d)M_2^2 = o(d^{1/4}) = o(d)$; if $M_1 \ge 2$ is fixed and $M_2(d) \le (1-\varepsilon)\log_2\log_2 d$, then $2^{M_2(d)} \le (\log_2 d)^{1-\varepsilon}$, so $\log_2((2M_1)^{2^{M_2(d)}}) \le (\log_2(2M_1))(\log_2 d)^{1-\varepsilon} = o(\log_2 d)$ while $8M_1M_2(d)^2 = O((\log\log d)^2) = o(d)$. Finally, for $d \ge 4$ and $N = \lfloor d/2\rfloor$, Behrend [3] constructs an integer 3-AP-free set $B \subseteq \{0,\dots,N-1\}$ of size $|B| \ge N\exp(-c\sqrt{\log N})$. Because $0 \le x+z, 2y < d$ for all $x,y,z \in B$, $B$ (and hence every cyclic translate $t+B \subseteq \mathbb{Z}_d$) is 3-AP-free modulo $d$; greedily choosing translates $t_i + B$ covering at least a $|B|/d$ fraction of the remaining uncovered residues covers $\mathbb{Z}_d$ in at most $\lceil (d/|B|)\log d\rceil + 1 = 2^{O(\sqrt{\log d})}$ translates, and taking disjoint remainders $(t_i+B)\setminus\bigcup_{j<i}(t_j+B)$ partitions $\mathbb{Z}_d$ into $2^{O(\sqrt{\log d})}$ 3-AP-free classes. □

**Remark 3.4** (Comparison with van der Waerden cutoffs across parameter regimes)**.** Comparing Theorem 3.3 with [10, Theorem 6.3] across parameter regimes shows that Theorem 3.3 never improves on [10, Theorem 6.3] at $M_1 = 1$ or $M_2 = 1$, loses at $M_1 = M_2 = 2$, and wins for every fixed $M_2 \ge 2$ once $M_1$ is sufficiently large, whereas for fixed $M_1 \ge 2$ and growing $M_2$ the comparison depends on the unknown growth rate of $\mathsf{W}(M_1, M_2+1)$:

- At $M_2 = 1$, two distinct columns in a single stage-1 class already contradict $\ell d \le M_2 d + (\ell-1)M_2\gcd(d,s)$ at $\ell = 2$ in [10, Theorem 6.3], giving the exact pigeonhole bound $d \le M_1$ ($\mathsf{W}(M_1,2) = M_1+1$), whereas Theorem 3.3 gives $d \le \max(8M_1, 4M_1^2)$.
- At $M_1 = 1$, $\mathsf{W}(1, M_2+1) = M_2+1$ gives $d \le M_2^2$ in [10, Theorem 6.3], whereas Theorem 3.3 gives $d \le \max(8M_2^2, 2^{2^{M_2}})$.
- At $M_1 = M_2 = 2$, [10, Corollary 6.4] uses $\mathsf{W}(2,3) \le 9$ to obtain $d \le 2(9-1) = 16$, whereas Theorem 3.3 gives $d \le \max(64, 4^4) = 256$.
- Conversely, for every fixed $M_2 \ge 2$, any progression-based cutoff from [10, Theorem 6.3] is at least $M_2(\mathsf{W}(M_1, M_2+1) - 1) \ge M_2(\mathsf{W}(M_1,3) - 1)$, which is superpolynomial in $M_1$ by Behrend [3], whereas Theorem 3.3 gives the polynomial-in-$M_1$ bound $(2M_1)^{2^{M_2}}$ for fixed $M_2 \ge 2$. For fixed $M_1 \ge 2$ and growing $M_2$, Theorem 3.3 gives the doubly exponential bound $(2M_1)^{2^{M_2}}$, while explicit van der Waerden upper bounds [13] show only that $M_2(\mathsf{W}(M_1, M_2+1) - 1)$ is at most five-fold exponential in $M_2$ (and known lower bounds on $\mathsf{W}(M_1, M_2+1)$ are exponential in $M_2$).

### 3.3 Finite controls at $d = 13$ and $d = 4$

Two finite-dimensional examples test the linear-growth multiplier $(k+1)$ in Lemma 3.1 and the coded-versus-copy distinction at $d = 4 = M_1M_2$.

**Example 3.5** (3-AP-free 2-cube witness at $d = 13$ ($9 \le \max \dim_F W \le 10$))**.** Lean: `control_D13TwoCubeAPFreeWitness` [`AffineCubeCompositionProofs.lean`]
Lean: `control_D13_GeneralFieldAndProgressionBridge` [`AffineCubeConsequences.lean`]

Let $d = 13$, $q = 2$ ($\dim_F A = 26$), and consider the 2-cube $J_{13} = \{0, 1, 4, 5\} = \{0, 1\} + \{0, 4\} \subseteq \mathbb{Z}_{13}$.

(a) $J_{13}$ is 3-AP-free modulo 13 (`ThreeAPFreeMod`), so every non-constant progression in $J_{13}$ has length $\ell \le 2$, where the single-step progression bound gives only $\dim_F W \le 14$ (attained on $\{0, 1\}$ over every field).

(b) Over every field $F$, Lemma 3.1 with $k = 2$ and $\dim_F \ker(T_s - \mathrm{id}) = 2$ ($s \in \{1, 4\}$) gives $3 \dim_F W \le 26 + 2 + 2 = 30$, hence $\dim_F W \le 10$.

(c) Over $\mathbb{R}$, and over every field $F$ with $(13 : F) \neq 0$, there exists an explicit $(W, L)$ shift-invariant on $J_{13}$ of dimension $\dim_F W = 9$. Thus the maximum invariant dimension on $J_{13}$ over such fields lies in $\{9, 10\}$ (whether 10 is attainable is not decided here); by contrast, on every non-constant 4-term arithmetic progression in $\mathbb{Z}_{13}$ the progression bound forces $4 \dim_F W \le 26 + 3 \cdot 2 = 32$ ($\dim_F W \le 8 < 9$), and a naive $2^k$-multiplier inequality $2^k \dim_F W \le \dim_F A + (2^k - 1)q$ fails at $k = 2$.

*Outline of the formal proof.* Residue inspection verifies that $J_{13}$ has no non-constant 3-term progression modulo 13, and cyclic shift periodicity gives $\dim_F \ker(T_s - \mathrm{id}) = 2$ for every $s \neq 0$ in $\mathbb{Z}_{13}$, so [10, Lemma 6.1] (`SharedDecoders.commonMessage_finrank_le`) yields $2 \dim_F W \le 28$ on 2-term progressions and $4 \dim_F W \le 26 + 3 \cdot 2 = 32$ ($\dim_F W \le 8$) on non-constant 4-term progressions. Over every field $F$ with $(13 : F) \neq 0$, explicit integer matrices $E \in \mathbb{Z}^{26 \times 9}$ and $L_{13} \in \mathbb{Z}^{9 \times 26}$ satisfy $L_{13}(T_c E) = 13 I_9$ for all $c \in J_{13}$, directly certifying both $\dim_F W = 9$ and shift-invariant injectivity of $L_{13}$. □

**Example 3.6** (Same-interval coded factorizations at $d = 4 = M_1 M_2$ when $(5 : F) \neq 0$)**.** Lean: `control_D4SameIntervalCodedFactorization` [`AffineCubeCompositionProofs.lean`]
Lean: `control_D4_AllWidthsAndBinaryFields` [`AffineCubeConsequences.lean`]

Let $d = 4$ and let both stages use the same two-interval partition $p_1 = p_2 = [0, 0, 1, 1]$ (`halfIntervalPartition4`, $M_1 = M_2 = 2$).

(a) *Uncoded impossibility*: for every lane width $b \ge 1$ and finite alphabet $|\Sigma| \ge 2$, neither entry-copy rules nor bit-lane-copy rules $K_1, K_2$ with $p_1 = p_2 = [0, 0, 1, 1]$ can satisfy $K_2 \circ K_1 = \tau$.

(b) *Coded factorization for* $(5 : F) \neq 0$: over every field $F$ with $(5 : F) \neq 0$ (combining an explicit rational table valid whenever $(20 : F) \neq 0$ with a separate binary 0/1 table valid whenever $(2 : F) = 0$, including $\mathbb{F}_2$ at $b = 1$), and for every lane width $b \ge 0$, there exist $F$-linear class-uniform maps $K_1, K_2$ on $\mathrm{Mat}_{4\times 4}(F^b)$ with $p_1 = p_2 = [0, 0, 1, 1]$ satisfying $K_2 \circ K_1 = \tau$. (Whether such a factorization exists in characteristic 5 for $b \ge 1$ is not decided here.)

*Outline of the formal proof.* In (a), because stage 2 can route across distinct bit lanes, the proof argues on all $4b$ wires $\mathbb{Z}_4 \times \mathrm{Fin}\, b$ simultaneously: comparing columns $c = 0$ and $c = 1$ ($p_2(0) = p_2(1) = 0$) on two-symbol test inputs shows that for every wire $q = (i, \ell)$, stage 2's shift $u_2(0, i, \ell)$ lies in $\{1, 3\}$ and stage 1's source row on $\varphi_0(q) := (r_2(0, i, \ell), \ell_2(0, i, \ell))$ lies in $\{0, 1\}$, and moreover $\varphi_0$ is injective and hence surjective on $\mathbb{Z}_4 \times \mathrm{Fin}\, b$; thus stage 1 can never output input row 2, contradicting column $c = 2$. In (b), separate explicit linear tables verify $K_2 \circ K_1 = \tau$ lane-by-lane when $(20 : F) \neq 0$ and when $(2 : F) = 0$; every field with $(5 : F) \neq 0$ satisfies one of the two conditions. □

# 4 Shifted sumsets and logarithmic two-stage rank deficits

## 4.1 Minimum-shifted monochromatic sumsets

For approximate two-stage compositions at fixed class counts $M_1, M_2 \geq 1$, write

$$r \;:=\; \mathrm{rank}_F(\tau - K_2 \circ K_1).$$

In [10, Theorem 6.6], extracting a monochromatic progression of length $\ell = (2M_2+1)k - 1$ inside $p_1$ gave $r \geq bk(d - N_0 + 1)/2$ for all $d \geq N_0$ with $N_0 = \mathsf{W}(M_1, \ell)$ over every field $F$ with $\mathrm{char}(F) \nmid d$ ($(d : F) \neq 0$), proving $r/(bd) \to \infty$ as $d \to \infty$. At $M_1 = 1$, $\mathsf{W}(1, \ell) = \ell$ with $k \approx d/(2(2M_2+1))$ already yields a quadratic rank deficit $\Omega(bd^2/M_2)$ for all $d \geq 4M_2 + 1$ ([10, Theorem 6.6], `SharedDecoders.twoHop_rank_interval_any_field`). For $M_1 \geq 2$, however, [10, Theorem 6.6] is stated with qualitative $\mathsf{W}(M_1, \ell)$ via the Hales–Jewett theorem [15], and substituting Gowers' van der Waerden bound [13] gives only an iterated-logarithmic growth rate for $r/(bd)$ (moreover, non-wrapping monochromatic progressions in $\{0, \ldots, d-1\}$ under random 2-colorings have length $O(\log d)$ by a standard first-moment calculation [14, 1], and translate-covering $\mathbb{Z}_d$ by a Behrend set [3] limits progressions modulo $d$ to length $< 3$ at $M_1 = 2^{O(\sqrt{\log d})}$). Throughout this section, log denotes the natural logarithm (`Real.log`).

Over any field $F$ in which $d$ is invertible ($(d : F) \neq 0$), we pass to a splitting field $E/F$ carrying an injective additive character $\chi : \mathbb{Z}_d \to E^\times$ and replace progressions by $(n+1)$-fold monochromatic sumsets at $n = 2M_2 - 1$ (still at frequency width $k = O(\log d)$, but replacing five-fold exponential van der Waerden thresholds for $M_1 \geq 2$ by Kővári–Sós–Turán density bounds): an *unshifted* $(n+1)$-fold monochromatic sumset in class $m_1$ consists of a base set $J_0 \subseteq \mathbb{Z}_d$ ($|J_0| = k$) and $n$ natural-number shift sets $B_0, \ldots, B_{n-1} \subseteq \mathbb{N}$ ($|B_t| = k+1$) whose cast sumset in $\mathbb{Z}_d$ satisfies

$$J_0 + B_0 + \cdots + B_{n-1} \;\subseteq\; p_1^{-1}(m_1).$$

For any finite $S \subseteq \mathbb{N}$, write $\mathrm{diam}_{\mathbb{N}}(S) := \max S - \min S$ (0 if $S = \emptyset$; `finsetNatDiam`). Shifting each $B_t$ down by $\min B_t$ places $0 \in \widetilde{B}_t$ while preserving $\mathrm{diam}_{\mathbb{N}}(\widetilde{B}_t) = \mathrm{diam}_{\mathbb{N}}(B_t)$, producing a *shifted* $(n+1)$-fold monochromatic sumset $(\widetilde{J}_0, \widetilde{B}, \mathrm{diamB})$ ($|\widetilde{J}_0| = k$, $0 \in \widetilde{B}_t \subseteq [0, \mathrm{diamB}_t] \subseteq \mathbb{N}$, $|\widetilde{B}_t| = k+1$, and $\widetilde{J}_0 + \sum_{t=0}^{n-1} \widetilde{B}_t \subseteq p_1^{-1}(m_1)$) so that all partial subset sums remain in $p_1^{-1}(m_1)$ simultaneously.

**Lemma 4.1** (Minimum-shifting monochromatic sumsets)**.** Lean: `unshiftedToShiftedSumset` [`SumsetRankDeficitProofs.lean`]

Lean: `finsetNatDiam_shift_min`, `unshiftedToShiftedSumset_explicit`, `unshiftedToShiftedSumset_intrinsicDiam` [`SumsetRankConsequences.lean`]

*Let $d, M_1 \geq 1$ and $p_1 : \mathbb{Z}_d \to \mathrm{Fin}\, M_1$. Given any unshifted monochromatic sumset $J_0 + B_0 + \cdots + B_{n-1} \subseteq p_1^{-1}(m_1)$ ($J_0 \subseteq \mathbb{Z}_d$, $B_t \subseteq \mathbb{N}$) with $|J_0| = k$, $|B_t| = k+1$, and $B_t \subseteq [0, L_0]$, setting $\widetilde{B}_t := B_t - \min B_t$ and $\widetilde{J}_0 := J_0 + \sum_{t=0}^{n-1} \min B_t$ satisfies $0 \in \widetilde{B}_t$, $|\widetilde{B}_t| = k+1$, $\mathrm{diam}_{\mathbb{N}}(\widetilde{B}_t) = \mathrm{diam}_{\mathbb{N}}(B_t) \leq L_0$, $|\widetilde{J}_0| = k$, and*

$$\widetilde{J}_0 + \sum_{t \in S} \widetilde{B}_t \;\subseteq\; p_1^{-1}(m_1) \qquad \text{for every subset } S \subseteq \{0, \ldots, n-1\}$$

*(proved directly for the explicit definitions $\widetilde{B}_t, \widetilde{J}_0$ in `unshiftedToShiftedSumset_explicit`, and packaged with $\mathrm{diamB}_t = \mathrm{diam}_{\mathbb{N}}(B_t)$ at $n = 2M_2 - 1$ for $M_2 \geq 1$ in `unshiftedToShiftedSumset_intrinsicDiam`).*

*Outline of the formal proof.* Choosing $b_t \in \widetilde{B}_t$ for $t \in S$ and $0 \in \widetilde{B}_t$ for $t \notin S$ rewrites $c_0 + \sum_{t=0}^{n-1} \min B_t + \sum_{t \in S} b_t = c_0 + \sum_{t=0}^{n-1} (b_t + \min B_t)$ with $b_t + \min B_t \in B_t$. Translation by $-\min B_t$ on $\mathbb{N}_{\geq \min B_t}$ preserves both cardinality and max − min. □

### 4.2 Arbitrary-subset Vandermonde pullback and sparse annihilating polynomials

On an arbitrary $k$-element subset $J_0 \subseteq \mathbb{Z}_d$, the character-evaluation matrix $\big(\chi(\nu \cdot c)\big)_{c\in J_0,\, 0\le\nu<k}$ is a $k \times k$ Vandermonde matrix in the $k$ distinct scalars $z_c = \chi(c)$ ($c \in J_0$), since $\chi(\nu \cdot c) = \chi(c)^\nu$. Thus $J_0$ need not be an interval or progression for stacked evaluation on the frequency-$< k$ band to be injective. Following [10, Section 6], for a partition $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$ and $k \le d$, let $H := \texttt{SharedDecoders.bandRowSpace}\ (\chi, p_2, k) \le (E^b)^{\mathbb{Z}_d}$ be the $E$-span of the $bM_2k$ class-modulated character vectors $i \mapsto \mathbf{1}_{p_2(i)=m_2}\, \chi(\nu i)\, e_\ell$ ($m_2 \in \operatorname{Fin} M_2$, $0 \le \nu < k$, $\ell \in \operatorname{Fin} b$, so $\dim_E H \le bM_2k$), and let $\texttt{SharedDecoders.ColumnAmbient}(H) = (\mathrm{Unit} \to \mathbb{Z}_d \to H) \cong (\mathbb{Z}_d \to H)$ embed into $\mathrm{Mat}_{d\times d}(E^b)$ via $(\iota_H x)_{i,c} = (x_{(),c})_i$ (`SharedDecoders.columnAmbientEmbed`), with column shift $(T_{\mathrm{band}}x)_{(),c} = x_{(),c+1}$ (`SharedDecoders.channelRotation` at shift 1) satisfying $\iota_H(T_{\mathrm{band}}x) = \mathrm{rot}_1(\iota_H x)$.

**Lemma 4.2** (Arbitrary-subset Vandermonde injectivity and stacked band pullback)**.** `Lean`: `vandermondeArbitrarySubsetInjective`, `stackedBandPullbackInjective` [`SumsetRankDeficitProofs.lean`]

*Let $d \ge 1$, let $E$ be a field with an injective additive character $\chi : \mathbb{Z}_d \to E^\times$, and let $J_0 \subseteq \mathbb{Z}_d$ satisfy $|J_0| = k \le d$.*

*(a) For any $E$-vector space $V$, degree-$< k$ character evaluation $a \mapsto \big(\sum_{\nu=0}^{k-1} \chi(\nu c)\, a_\nu\big)_{c\in J_0}$ (`charBandEval`) is injective on $V^k$.*

*(b) Let $M_1, M_2 \ge 1$, $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$, and $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$. For any two $E$-linear class-uniform stages $K_1, K_2$ on $\mathrm{Mat}_{d\times d}(E^b)$ with $J_0 \subseteq p_1^{-1}(m_1)$ and stage-1 class-$m_1$ decoder $f_{1,m_1}$, the frequency-$< k$ band space $H$ (`SharedDecoders.bandRowSpace` at $(\chi, p_2, k)$, with $\dim_E H \le bM_2k$) carries a pullback subspace $W \le \texttt{SharedDecoders.ColumnAmbient}(H)$ satisfying*

$$\dim_E W + \mathrm{rank}_E(\tau - K_2 \circ K_1) \;\ge\; bkd$$

*such that the stacked evaluation $x \mapsto \big(f_{1,m_1}(\mathrm{rot}_c(\iota_H x))\big)_{c\in J_0}$ is injective on $W$.*

*Outline of the formal proof.* In (a), $\chi(\nu c) = \chi(c)^\nu$, and injectivity of $\chi$ makes $\{\chi(c) : c \in J_0\}$ $k$ distinct scalars in $E$, so a degree-$< k$ vector-valued polynomial vanishing on $J_0$ is zero. For (b), parameterize intermediate frequency-$< k$ band states by $E_{\mathrm{band}}(u)_{i,c} := \sum_{\nu=0}^{k-1} \chi(\nu c)\, u_{\nu,i}$ ($bkd$ dimensions, injective since $k \le d$). Since $x \mapsto \tau K_2 x$ restricts to a bijection $\ker(\mathrm{id} - K_1\circ\tau\circ K_2) \xrightarrow{\sim} \ker(\mathrm{id} - \tau\circ K_2\circ K_1)$ and $\mathrm{id} - \tau\circ K_2\circ K_1 = \tau(\tau - K_2\circ K_1)$ ($\tau^2 = \mathrm{id}$), one has $\mathrm{rank}_E(\mathrm{id} - K_1\circ\tau\circ K_2) = \mathrm{rank}_E(\tau - K_2\circ K_1)$. Pulling back $\mathrm{id} - K_1\circ\tau\circ K_2$ along $E_{\mathrm{band}}$ therefore cuts out $Q := \ker\big((\mathrm{id} - K_1\circ\tau\circ K_2)\circ E_{\mathrm{band}}\big)$ of dimension $\dim_E Q \ge bkd - \mathrm{rank}_E(\tau - K_2\circ K_1)$, and $u \mapsto \tau K_2(E_{\mathrm{band}}(u))$ embeds $Q$ injectively into $\iota_H\big(\texttt{SharedDecoders.ColumnAmbient}(H)\big)$ as $W$ ($\dim_E W = \dim_E Q$). For $x \in W$ coming from $u \in Q$, $X := \iota_H(x) = \tau K_2(E_{\mathrm{band}}(u))$ has intermediate state $K_1 X = E_{\mathrm{band}}(u)$ whose column $c$ is $\sum_{\nu=0}^{k-1} \chi(\nu c)\, u_\nu$; if $f_{1,m_1}(\mathrm{rot}_c(X)) = (K_1X)_{:,c} = 0$ for all $c \in J_0 \subseteq p_1^{-1}(m_1)$, part (a) gives $u = 0$, hence $K_1X = 0$ and $X = \tau K_2 K_1 X = 0$ ($x = 0$). □

Each summand $\widetilde{B}_t$ of size $k+1$ provides $k+1$ monomials against $k$ frequencies $\nu \in \{0, \dots, k-1\}$, guaranteeing a nonzero sparse polynomial supported in $\widetilde{B}_t$ that annihilates the frequency-$< k$ band and has degree at most $\mathrm{diam}_{\mathbb{N}}(\widetilde{B}_t)$.

**Lemma 4.3** (Sparse sumset annihilating polynomial)**.** Lean: `sumsetAnnihilatingPolynomial`, `cyclicChannelShift_aeval_ker_finrank_le` [`SumsetRankDeficitProofs.lean`]
Lean: `sumsetAnnihilatingPolynomial_bandRowSpace` [`SumsetRankConsequences.lean`]

*Let $d \geq 1$, let $E$ be a field with an additive character $\chi : \mathbb{Z}_d \to E^\times$, and let $\widetilde{B} \subseteq \mathbb{N}$ satisfy $0 \in \widetilde{B}$, $|\widetilde{B}| = k+1$, and $\max \widetilde{B} \leq L_0$. Then there exists a nonzero polynomial $P \in E[Z]$ supported in $\widetilde{B}$ with $\deg P \leq \operatorname{diam}_{\mathbb{N}}(\widetilde{B}) \leq L_0$ such that $P(\chi(\nu)) = \sum_{x \in \widetilde{B}} P_x \chi(\nu x) = 0$ for all $0 \leq \nu < k$. Moreover, for any $q \in \mathbb{N}$, on the $q$-channel cyclic shift module $(\mathbb{Z}_d \to E^q)$ with unit shift $(T_1 w)_c = w_{c+1}$ one has*

$$\dim_E \ker P(T_1) \;\leq\; q \deg P \;\leq\; q\, L_0,$$

*and on* `SharedDecoders.ColumnAmbient` *over* `SharedDecoders.bandRowSpace` *for $(\chi, p_2, k)$ (where $\dim_E H \leq bM_2k$) one has $\dim_E \ker P(T_{\mathrm{band}}) \leq bM_2kL_0$.*

*Outline of the formal proof.* The homogeneous linear system $\sum_{x \in \widetilde{B}} P_x \chi(\nu x) = 0$ $(0 \leq \nu < k)$ has $k$ equations in $k+1$ unknowns, hence a nonzero solution $P$ supported in $\widetilde{B}$ of degree $\leq \max \widetilde{B} = \operatorname{diam}_{\mathbb{N}}(\widetilde{B}) \leq L_0$. On $(\mathbb{Z}_d \to E^q)$, setting $m := \deg P$ $(P_m \neq 0)$, restricting $\ker P(T_1)$ to the $m$ residues $(0 \bmod d, \ldots, (m-1) \bmod d)$ ($qm$ coordinates) is injective for every $m$: if $P(T_1)w = 0$ and $w_{(j \bmod d)} = 0$ for all $0 \leq j < m$, strong induction on $n \in \mathbb{N}$ shows $w_{(n \bmod d)} = 0$ for all $n \geq m$ by evaluating $(P(T_1)w)_{((n-m) \bmod d)} = 0$ and dividing by $P_m \neq 0$. □

## 4.3 Sumset rank bound, Kővári–Sós–Turán extraction, and Theorem 2B

**Theorem 4.4** (Two-stage sumset recurrence rank bound)**.** Lean: `sumsetTwoHopRankBound` [`SumsetRankDeficitProofs.lean`]

*Let $F$ be a field with $(d : F) \neq 0$, let $M_1, M_2, b \geq 1$, and let $1 \leq k \leq d$. Suppose $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ admits a shifted $(2M_2)$-fold monochromatic sumset $(\widetilde{J}_0, \widetilde{B}, \mathrm{diamB})$ in class $m_1$ with $\sum_{t=0}^{2M_2-2} \mathrm{diamB}_t \leq d$. Then for every $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$ and every pair of $F$-linear class-uniform maps $K_1, K_2$ on $\mathrm{Mat}_{d\times d}(F^b)$, writing $r := \operatorname{rank}_F(\tau - K_2 \circ K_1)$,*

$$2r \;\geq\; bk\Big(d - \sum_{t=0}^{2M_2-2} \mathrm{diamB}_t\Big), \qquad \text{and if } \mathrm{diamB}_t \leq \lfloor d/(4M_2) \rfloor \text{ for all } t, \quad 4r \;\geq\; bkd.$$

*Outline of the formal proof.* Since $(d : F) \neq 0$, $Z^d - 1$ is separable over $F$; passing to a cyclotomic splitting field $E/F$ satisfies $\operatorname{rank}_E(\tau - K_2 \circ K_1) \leq \operatorname{rank}_F(\tau - K_2 \circ K_1)$ (`SharedDecoders.stateExtension_rank_le`, so any lower bound over $E$ transfers to $F$) and furnishes an injective character $\chi : \mathbb{Z}_d \to E^\times$. Let $W \leq A_{\mathrm{band}}$ $(\dim_E A_{\mathrm{band}} \leq bM_2kd)$ be the pullback subspace from Lemma 4.2(b) on $\widetilde{J}_0$, with $\dim_E W \geq bkd - r$, and let $P_t$ be the sparse annihilating polynomial on $\widetilde{B}_t$ from Lemma 4.3. Because $P_t$ annihilates the frequency-$< k$ band while every shift in $\widetilde{J}_0 + \sum_{s \leq t} \widetilde{B}_s$ lies in $p_1^{-1}(m_1)$, $P_t(T_{\mathrm{band}})$ maps the $t$-th partial sumset span $U_t$ into the kernel of stacked $\widetilde{J}_0$-evaluation, which meets $W$ trivially. Each of the $2M_2 - 1$ steps therefore adds at least $\dim_E W - bM_2k\,\mathrm{diamB}_t$ dimensions inside $A_{\mathrm{band}}$:

$$2M_2 \dim_E W - bM_2k \sum_{t=0}^{2M_2-2} \mathrm{diamB}_t \;\leq\; \dim_E A_{\mathrm{band}} \;\leq\; bM_2kd.$$

Substituting $\dim_E W \geq bkd - r$ and dividing by $M_2$ yields $2r \geq bk(d - \sum_t \mathrm{diamB}_t)$, and $\mathrm{diamB}_t \leq \lfloor d/(4M_2) \rfloor$ gives $\sum_t \mathrm{diamB}_t \leq d/2$, hence $4r \geq bkd$. □

**Lemma 4.5** (Iterated Kővári–Sós–Turán monochromatic sumset extraction)**.** Lean: `iteratedKovariSosTuranSumset` [`SumsetRankDeficitProofs.lean`]

*For every $M_1, M_2 \geq 1$, there exist $0 < c_{\mathrm{lo}} \leq C_{\mathrm{hi}}$ and $D_0 \in \mathbb{N}$ such that for all $d \geq D_0$, every $M_1$-coloring $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ contains an unshifted (and hence shifted) $(2M_2)$-fold monochromatic sumset in $\mathbb{Z}_d$ whose $2M_2 - 1$ shift summands $B_t$ (and $\widetilde{B}_t$) lie in $[0, \lfloor d/(4M_2) \rfloor] \subseteq \mathbb{N}$ at width scale $1 \leq k \leq d$ satisfying*

$$c_{\mathrm{lo}} \, (\log d)^{\frac{1}{2M_2-1}} \;\leq\; k \;\leq\; C_{\mathrm{hi}} \, (\log d)^{\frac{1}{2M_2-1}}.$$

*Outline of the formal proof.* Start from the densest color class $A_0 \subseteq \{0, \ldots, d-1\}$ ($|A_0| \geq \lceil d/M_1 \rceil$). At each of the $2M_2 - 1$ steps, partition $\{0, \ldots, d-1\}$ into $4M_2$ intervals of width at most $\lfloor d/(4M_2) \rfloor + 1$, pick the densest bucket $I \cap A$ of size $m \geq \lceil |A|/(4M_2) \rceil$, split $I \cap A$ into its lower $\lceil m/2 \rceil$ and upper $\lfloor m/2 \rfloor$ elements, and apply the Kővári–Sós–Turán double count [17] to $(k+1)$-tuples of differences $a_{\mathrm{hi}} - a_{\mathrm{lo}} \in [0, \lfloor d/(4M_2) \rfloor]$ (so $a_{\mathrm{lo}} + b = a_{\mathrm{hi}} \in A \subseteq [0, d)$ never wraps modulo $d$). Iterating $2M_2 - 1$ times extracts $k$ base elements and $2M_2 - 1$ shift sets of size $k+1$ at density cost exponential in $k^{2M_2-1}$, balancing at $k = \Theta_{M_1,M_2}\big((\log d)^{1/(2M_2-1)}\big)$. □

**Theorem 4.6** (Theorem 2B: log-power two-stage rank deficit over admissible fields)**.** Lean: `theorem2B_TwoStageLogPowerRankDeficit` [`SumsetRankDeficitProofs.lean`]

Lean: `theorem2B_ExplicitConstant`, `theorem2BExplicitConst_pos`, `theorem2BExplicitThreshold_eq_pow` [`SumsetRankConsequences.lean`]

*For all integers $M_1, M_2 \geq 1$, define the explicit constant and dimension threshold (*`theorem2BExplicitConst`*,* `theorem2BExplicitThreshold`*, with natural logarithm* $\log =$ `Real.log`*)*

$$\begin{aligned} c(M_1, M_2) \;&:=\; \frac{1}{144 \, \big(\log(16 M_1 M_2)\big)^{\frac{1}{2M_2-1}}} \;>\; 0, \\ D_0(M_1, M_2) \;&:=\; \max\Big(4M_2,\; \big\lceil \exp\big(36^{2M_2-1} \log(16 M_1 M_2)\big) \big\rceil + 2\Big) \\ &=\; \max\Big(4M_2,\; (16 M_1 M_2)^{36^{2M_2-1}} + 2\Big). \end{aligned}$$

*Then for every field $F$, every $d \geq D_0(M_1, M_2)$ with $(d : F) \neq 0$, every lane width $b \geq 1$, and every pair of $F$-linear class-uniform stages $K_1, K_2$ on $\mathrm{Mat}_{d \times d}(F^b)$ with $M_1$ and $M_2$ classes,*

$$\operatorname{rank}_F(\tau - K_2 \circ K_1) \;\geq\; c(M_1, M_2) \, b \, d \, (\log d)^{\frac{1}{2M_2-1}}.$$

*Outline of the formal proof.* Write $n := 2M_2 - 1 \geq 1$ and $M_0 := 16 M_1 M_2 > 1$. Starting from a $p_1$-class $A_0 \subseteq \{0, \ldots, d-1\}$ of size $|A_0| \geq d/M_1$ and setting $R_0 := M_1$ and $R_{t+1} := (16 M_2 R_t)^{k+2}$, the $n$-step Kővári–Sós–Turán recursion of Lemma 4.5 succeeds whenever $d \geq 4M_2$ and $16M_2(k+1)R_n \leq d$, leaving $|A_n| \geq d/R_n$ and shift summands in $[0, \lfloor d/(4M_2) \rfloor]$. For $k \geq 2$, induction gives $R_n \leq M_0^{3^n (k+2)^n} \leq M_0^{9^n k^n}$ (since $k + 2 \leq 3k$), and combining $16M_2 \leq M_0$, $k + 1 \leq 2^k \leq M_0^{k^n}$, and $1 + k^n + 9^n k^n \leq (2 + 9^n) k^n \leq 18^n k^n$ yields $R_n k < 16 M_2 (k+1) R_n \leq M_0^{18^n k^n}$. Setting $x := \big(\frac{\log d}{18^n \log M_0}\big)^{1/n}$ and $k := \lfloor x \rfloor$, the hypothesis $d \geq D_0(M_1, M_2) \geq M_0^{36^n} = \exp(2^n \cdot 18^n \log M_0)$ implies $x \geq 2$, so $k \geq \max(2, x/2)$, $M_0^{18^n k^n} \leq \exp(18^n x^n \log M_0) = d$, and $|A_n| \geq k$. Converting to a shifted sumset (Lemma 4.1) and applying $4r \geq bkd$ (Theorem 4.4) yields $r \geq \frac{1}{4} bkd \geq \frac{1}{8} bdx = \frac{bd \, (\log d)^{1/n}}{144 \, (\log M_0)^{1/n}} = c(M_1, M_2) \, b \, d \, (\log d)^{1/(2M_2-1)}$. □

**Remark 4.7** (Comparison with [10, Theorem 6.6], asymptotic threshold, and characteristic restriction)**.** Three qualifications on Theorem 4.6 should be noted explicitly:

(a) *Parameter comparison*: Theorem 4.6 improves on [10, Theorem 6.6] only when $M_1 \geq 2$ and $d$ is sufficiently large. At $M_1 = 1$, $\mathsf{W}(1, \ell) = \ell$ already gives a quadratic rank deficit $\Omega(bd^2/M_2)$ for all $d \geq 4M_2 + 1$ ([10, Theorem 6.6], `SharedDecoders.twoHop_rank_interval_any_field`). At $M_1 = 2$ and $M_2 = 1$, [10, Theorem 6.6] at $k = 2$ with $\mathsf{W}(2, 5) = 178$ [19] gives $r \geq b(d-177)$, which Theorem 4.6 ($c(2, 1) = \frac{1}{144 \log 32}$) beats only for $d \gtrsim 10^{216}$. For fixed $M_1 \geq 2$ as $d \to \infty$, however, Theorem 4.6 replaces the iterated-logarithmic growth rates obtained from Gowers' van der Waerden bound [13] by $(\log d)^{1/(2M_2-1)}$.

(b) *Threshold size*: the formalized threshold $D_0(M_1, M_2)$ is purely asymptotic: already at $M_1 = M_2 = 1$ one has $D_0(1, 1) = 16^{36} + 2 = 2^{144} + 2 \approx 2.23 \times 10^{43}$, and the exponent $36^{2M_2-1}$ grows exponentially in $M_2$.

(c) *Characteristic*: like [10, Theorem 6.6] (and unlike Theorem 3.3), Theorem 4.6 requires $(d : F) \neq 0$ (excluding $\mathbb{F}_2$ at $d = 2^m$) so that $\mathbb{Z}_d$ admits $d$ distinct character values.

**Corollary 4.8** (Single-class second stage: $\Omega(b\, d \log d)$ rank deficit)**.** Lean: `theorem2B_OneSecondStage_OmegaLog` [`SumsetRankConsequences.lean`]

*When $M_2 = 1$, for every $M_1 \geq 1$ there exist $c > 0$ and $D_0 \geq 2$ such that for every field $F$, every $d \geq D_0$ with $(d : F) \neq 0$, every $b \geq 1$, and every $M_1$-class $K_1$ and 1-class $K_2$ on $\mathrm{Mat}_{d\times d}(F^b)$,*

$$\mathrm{rank}_F(\tau - K_2 \circ K_1) \;\geq\; c\, b\, d \log d.$$

*Outline of the formal proof.* Specialize Theorem 4.6 to $M_2 = 1$, where $\frac{1}{2M_2-1} = 1$. □

### 4.4 Single-band witness without sumset iteration at $d = 12$

Theorem 4.4 iterates the sparse annihilating-polynomial step across all $2M_2 - 1$ summands of a $(2M_2)$-fold monochromatic sumset rather than stopping at a single half-density subset $J \subseteq \mathbb{Z}_d$ and a single $k = 2$ band (with one fixed shift-covariant linear readout $L$ applied to $T_c w$ for $c \in J$). The following 4-channel witness at $d = 12$ ($q = bM_2k = 4$) shows that under the text-only parameter reading $(b, M_2) = (1, 2)$ ($bkd = 24$, whereas in the formalized $(b, M_2) = (2, 1)$ bridge one has $bkd = 48$ and $24 = \frac{1}{2}bkd$), a single half-density band without any annihilating-polynomial step admits $bkd = 24$-dimensional shift-covariant solutions ($r \geq 0$), so the annihilating-polynomial steps cannot all be dropped.

**Example 4.9** (Shift-covariant 4-channel band witness at $d = 12$ over $\mathbb{C}$)**.** Lean: `control_D12SingleClassAbstractBand` [`SumsetRankDeficitProofs.lean`]

Lean (partial): `exists_primitive_twelfth_root_complex`, `d12AddChar_injective_and_primitive`, `control_D12_UnconditionalExists`, `control_D12_BandRowSpaceChannelEquiv` [`SumsetRankConsequences.lean`]

Not formalized: the alternative $(b = 1, M_2 = 2)$ parameter reading of $q = bM_2k = 4$ (Lean proves the shift-equivariant `SharedDecoders.bandRowSpace` bridge for $b = 2, M_2 = 1, k = 2$, where $24 = kd$ is half of $bkd = 48$).

At $d = 12$ and $k = 2$ on the 4-channel cyclic module $(\mathbb{Z}_{12} \to \mathbb{C}^4)$ ($q = 4$, identified shift-equivariantly via `control_D12_BandRowSpaceChannelEquiv` with `SharedDecoders.ColumnAmbient` over `SharedDecoders.bandRowSpace` for $(\chi, p_0, 2)$ on the single-class partition $p_0 : \mathbb{Z}_{12} \to \mathrm{Fin}\, 1$ at $b = 2, M_2 = 1, k = 2$, where $L$ reads 12 scalar coordinates per shift and $24 = kd = \frac{1}{2}bkd$; in text, $q = 4$ also matches the dimension bound $\dim H \leq bM_2k$ at $b = 1, M_2 = 2, k = 2$ where $24 = bkd$), let $J_{12} = \{0, 1, 3, 4, 7, 9\} \subseteq \mathbb{Z}_{12}$ ($|J_{12}| = 6$). With $\zeta_{12} = \exp(2\pi i/12) \in \mathbb{C}$, there exist a fixed linear readout $L : (\mathbb{Z}_{12} \to \mathbb{C}^4) \to \mathbb{C}^{12}$ and a 24-dimensional subspace $W \leq (\mathbb{Z}_{12} \to \mathbb{C}^4)$ on which stacked

shift-covariant evaluation $w \mapsto (L(T_c w))_{c \in J_{12}}$ is injective and realizes every pair $(a_0, a_1) \in (\mathbb{C}^{12})^2$ as $L(T_c w) = a_0 + \zeta_{12}^c a_1$ for all $c \in J_{12}$.

*Outline of the formal proof.* Verify that $\zeta_{12} = \exp(2\pi i/12)$ has exact multiplicative order 12 in $\mathbb{C}$, construct the explicit 24-dimensional 4-channel encoding whose shifts on $J_{12}$ satisfy $L(T_c w) = a_0 + \zeta_{12}^c a_1$ for one fixed linear map $L$, and identify $(\mathbb{Z}_{12} \to \mathbb{C}^4)$ shift-equivariantly with `SharedDecoders.ColumnAmbient` over `SharedDecoders.bandRowSpace` for $(\chi, p_0, 2)$ at $b = 2, M_2 = 1, k = 2$. Under the text-only parameter reading $(b, M_2) = (1, 2)$ ($24 = bkd$, so the band alone gives only $r \geq 0$), a single half-density shift set $J_{12}$ without any annihilating-polynomial step admits $bkd = 24$-dimensional shift-covariant solutions, showing in that reading that the annihilating-polynomial steps in Theorem 4.4 cannot all be dropped (note that because $q = 4$ corresponds to $bM_2 = 2$ rather than $b = M_2 = 1$, this example does not provide a $b = 1, M_1 = M_2 = 1$ matrix factorization). □

# 5 Run-extension bounds for one-stage linear shared decoders

## 5.1 Certificate floors at 5/8 and 9/16, and run-extension for linear decoders

We now turn from two-stage compositions to the one-stage linear shared-decoder capacity problem (posed at $b = 1$ in [4, Definition 6.3 and Problem 9.1(a)] and studied for $b$ lanes in [10]): for a partition $p : \mathbb{Z}_d \to \kappa$ into $M$ column classes, bound the maximum dimension $D_b(p) =$ `linearClassUniformMaxDim`$(F, b, p)$ of an $F$-linear class-uniform code $\mathcal{X} \leq \mathrm{Mat}_{d \times d}(F^b)$ (`IsLinearClassUniformCode`). When each class is a cyclic interval, Theorem TP ([4, Theorem 6.4], Lean-verified at $b = 1$) gives $D_1(p) \leq (d^2 + Md)/2$, whereas scattered classes can exceed that interval value ([4, Proposition 8.2], Lean-verified, and [9, Theorem 4.9] at $d \in \{8, 12, 16\}$, Lean-verified; [4, Proposition 6.17 and Corollary 6.19] and [9, Theorem 4.8 and Corollary 4.10], proved in text).

Theorem 3.4 of [10] (Lean-verified, as are Theorems 3.2, 3.5, C.6, and C.7 cited below) proved the universal two-class ceiling $D_b(p) \leq \frac{41}{64}\, bd^2 + \frac{5}{3}\, bd^{3/2}$ over every field by iterating the single-shift inclusion $\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X}) \leq H_{\{0,s\}}$ along doubled shift sets $E_0 = \{0\}, E_j = E_{j-1} \cup (E_{j-1} + t_j)$ ([10, Theorem 3.2]). For general $M$, Theorem 3.5 of [10] proved the difference-cover bound

$$D_b(p) \;\leq\; bd^2 - \frac{bd(d-U)}{u+v}, \qquad R = \min(uv, d-1), \quad U = \min\bigl(d, M\lfloor d/(R+1)\rfloor\bigr)$$

for all $u, v \geq 1$ (`SharedDecoders.IsSharedCode.difference_cover_real_bound`, with no $M \leq d/2$ restriction and optimized leading coefficient $1 - \frac{1+o(1)}{3\sqrt{3M}}$), together with its square-root specialization $D_b(p) \leq (1 - 1/(4\lceil\sqrt{2M}\,\rceil))bd^2$ for $M \leq d/2$ (`SharedDecoders.IsSharedCode.square_cover_bound`). Against the flag bound $(1 - \frac{1}{2M})bd^2 + O(bd)$ ([4, Theorem 5.1, Corollary 5.9(b), Remark 5.10, and Problem 9.1(a)]; coordinate form of Theorem 5.1 Lean-verified, Corollary 5.9(b) and Remark 5.10 proved in text), the optimized difference-cover bound improves from $M \geq 7$ ($u = 4, v = 5$ gives $\frac{25}{27} < \frac{13}{14}$) and its square-root specialization improves from $M \geq 11$. In comparison, the leading coefficient $c_M = 1 - \frac{5}{4M} + \frac{1}{M^2} - \frac{1}{4M^3}$ of Theorem 5.5 below beats the flag bound $1 - \frac{1}{2M}$ for every $M \geq 2$ and is strictly smaller than the optimized leading coefficient of [10, Theorem 3.5] *if and only if* $2 \leq M \leq 40$: at primes $d \neq \mathrm{char}(F)$, Theorem 5.5(b) improves on [10, Theorem 3.5] for $2 \leq M \leq 40$ once $d$ exceeds a finite crossover (from $d \geq 7$ at $M = 2$, where $\frac{19}{32}\, bd^2 + \frac{9}{8}\, bd$ is also $< bd^2$ for all primes $d \geq 3$ and $< \frac{3}{4}\, bd^2$ for all primes $d \geq 11$, up to $d \geq 39{,}163$ at $M = 40$), while at $M = 2$ for general (possibly composite) $d$ with $\mathrm{char}(F) \nmid d$, Theorem 5.4(b) lies strictly below [10, Theorem 3.4] for all $d \geq 1$, is non-vacuous ($< bd^2$) for $d \geq 14$, improves on [10, Theorem 3.5] for $d \geq 35$, and lies below $\frac{3}{4}\, bd^2$ for $d \geq 93$ (all exact rational threshold and $(u, v)$-minimization checks are recorded in `tools/verify_linear_capacity_crossovers.py`). Conversely, for every $M \geq 41$

the $1-\Theta(M^{-1/2})$ leading coefficient of [10, Theorem 3.5] is strictly smaller than $c_M$ (at $M=41$, $u=v=11$ gives $\frac{2603}{2684}\approx 0.96982 < c_{41}\approx 0.97010$), and [10, Theorem 3.5] applies to arbitrary composite $d$ and arbitrary field characteristic.

Regarding two-class certificate limits, Theorem C.6 of [10] proved that the nested-component certificates of [10] cannot go below $(\frac{5}{8}-o(1))bd^2$ ($\frac{5}{8}=\frac{40}{64}=0.625$) on almost every two-class partition, while their full-ambient isolation variant cannot go below $(\frac{9}{16}-o(1))bd^2$ ([10, Theorem C.7]; neither is a lower bound on capacity, for any decoders). Separately, the inclusion $C\cap \mathrm{rot}_s^{-1}(C)\le H_{\{0,s\}}$ uses only equality of decoder outputs across monochromatic edges $p(c)=p(c+s)$, a property shared by arbitrary deterministic decoders $f_m:\mathrm{Mat}_{d\times d}(F^b)\to(F^b)^{\mathbb{Z}_d}$ (`SharedDecoders.IsSharedCode`).

When the class decoders $\mathcal{A}_m:\mathrm{Mat}_{d\times d}(F^b)\to(F^b)^{\mathbb{Z}_d}$ are *$F$-linear* (the $b$-lane form of [4, Definition 6.3]), the single-shift intersection $U:=\mathcal{X}\cap\mathrm{rot}_s^{-1}(\mathcal{X})$ satisfies additional linear constraints beyond $H_{\{0,s\}}$ whenever an orbit of $c\mapsto c+s$ is not monochromatic. Fix $s\in\mathbb{Z}_d$ (identified with its canonical representative in $\{0,\dots,d-1\}$), let $g_s:=\gcd(d,s)$ (`shiftGcd`) and $n:=d/g_s$ (`shiftOrbitLen`), and partition $\mathbb{Z}_d$ into the $g_s$ cyclic orbits $O_a$ ($a\in\mathrm{Fin}\,g_s$, `shiftOrbit`) of $c\mapsto c+s$, each of length $n$. Along $O_a$, consider any maximal contiguous monochromatic run $I_r=\{c_r,c_r+s,\dots,c_r+(\ell_r-1)s\}\subseteq J_m\cap O_a$ of class $m$, where $J_m=p^{-1}(m)$:

- Since $U\le H_{\{0,s\}}$, along $I_r$ the rows $X_{c_r+ts,\cdot}=\mathrm{rot}_s^t(x_r^{(a)})$ ($0\le t\le\ell_r-1$) of $X$ are cyclic shifts of the single vector $x_r^{(a)}:=X_{c_r,\cdot}\in(F^b)^{\mathbb{Z}_d}$ (writing $(\mathrm{rot}_s\,v)_j:=v_{j+s}$ on $(F^b)^{\mathbb{Z}_d}$), and $X\in\mathcal{X}$ gives $\mathcal{A}_m(\mathrm{rot}_{c_r+ts}(X))=\mathrm{rot}_s^t(x_r^{(a)})$ for $0\le t\le\ell_r-1$.
- Because $\mathrm{rot}_s(X)$ *also* lies in $\mathcal{X}$ and the right endpoint $c_r+(\ell_r-1)s$ still belongs to $J_m$, evaluating the class-$m$ rule for $\mathrm{rot}_s(X)$ at output column (row index) $c_r+(\ell_r-1)s$ yields *one additional linear equation* at $t=\ell_r$ whenever $c_r+\ell_r s\notin J_m$:

$$\mathcal{A}_m(\mathrm{rot}_{c_r+\ell_r s}(X)) \;=\; \mathcal{A}_m(\mathrm{rot}_{c_r+(\ell_r-1)s}(\mathrm{rot}_s(X))) \;=\; \mathrm{rot}_s(X)_{c_r+(\ell_r-1)s,\cdot} \;=\; \mathrm{rot}_s^{\ell_r}(x_r^{(a)}),$$

  even though $c_r+\ell_r s\notin J_m$. Thus $\mathcal{A}_m$ is pinned down at $\ell_r+1$ points along $I_r$ rather than $\ell_r$.

When $\mathrm{char}(F)\nmid n$, passing to $\overline{F}$ with a primitive $n$-th root of unity $\omega_n$ (Proposition 2.1(b)–(c)) expands $\mathrm{rot}_c(X)$ in $\mathrm{rot}_s$-eigenmodes and turns those run-extended equations into root-vanishing conditions on polynomials of bounded degree:

**Lemma 5.1** (Polynomial root-vanishing bound)**.** Lean: `finrank_vanishCoeffs_le` [`ExtensionLinearRunSingleShift.lean`]

*Let $K$ be any field, let $D\in\mathbb{N}$, and let $S\subseteq K$ be a finite subset. The subspace* `vanishCoeffs` $(K,D,S)\le K^{\mathrm{Fin}\,D}$ *of coefficient tuples $(c_0,\dots,c_{D-1})$ satisfying $\sum_{j=0}^{D-1}c_jz^j=0$ for all $z\in S$ has dimension at most $D-|S|$ (with truncated subtraction in $\mathbb{N}$, hence $0$ whenever $|S|\ge D$).*

*Outline of the formal proof.* Induct on $D$. At $D=0$ the ambient space $K^{\mathrm{Fin}\,0}$ is zero-dimensional. For $D+1$ and nonempty $S$, choose $z_0\in S$; every polynomial $P(Z)$ of degree $\le D$ vanishing at $z_0$ factors uniquely as $(Z-z_0)Q(Z)$ via a $K$-linear quotient map on coefficient tuples, injectively mapping `vanishCoeffs`$(K,D+1,S)$ into `vanishCoeffs`$(K,D,S\setminus\{z_0\})$. □

For each orbit $a\in\mathrm{Fin}\,g_s$ and class $m\in\mathrm{Fin}\,M$, denote the orbit class size and right-endpoint run count (`orbitClassCard`, `orbitRunCount`) by

$$d_{m,a}\;:=\;|J_m\cap O_a|,\qquad T_{m,a}\;:=\;|\{c\in J_m\cap O_a\;:\;c+s\notin J_m\}|.$$

Write the single-shift quadratic bound (`theorem1ASingleShiftBound`) as

$$B_{\mathrm{run}}(b,p,s)\;:=\;bg_s\sum_{a=0}^{g_s-1}\sum_{m=0}^{M-1}\Big(T_{m,a}(n-d_{m,a})\;-\;\tfrac12\,T_{m,a}^2\;+\;\tfrac12\,T_{m,a}\Big)\;+\;bg_sd.$$

**Theorem 5.2** (Single-shift run-extension bound for linear shared decoders)**.** Lean: `extension_Theorem1A_SingleShift`, `linearClassUniform_singleShift_finrank_le` [`ExtensionLinearRunSingleShift.lean`]

*Let $F$ be a field, let $d, M, b \geq 1$, let $p : \mathbb{Z}_d \to \operatorname{Fin} M$, and let $s \in \mathbb{Z}_d$ with $g_s = \gcd(d, s)$ and $n = d/g_s$.*

*(a) For every $a \in \operatorname{Fin} g_s$ and $m \in \operatorname{Fin} M$, $T_{m,a} \leq n - d_{m,a}$.*

*(b) Whenever $\operatorname{char}(F) \nmid n$ (`CoprimeToChar F n`), every subspace $\mathcal{X} \leq \operatorname{Mat}_{d\times d}(F^b)$ satisfying `IsLinearClassUniformCode p` $\mathcal{X}$ obeys*

$$\dim_F(\mathcal{X} \cap \operatorname{rot}_s^{-1}(\mathcal{X})) \;\leq\; B_{\mathrm{run}}(b, p, s).$$

*Outline of the formal proof.* Part (a) holds because the map $c \mapsto c+s$ injects $\{c \in J_m \cap O_a : c+s \notin J_m\}$ into $O_a \setminus J_m$. For (b), extend scalars to $K = \overline{F}$ via Proposition 2.1(b)–(c) and pick a primitive $n$-th root of unity $\omega \in K^\times$. Decompose $K$-matrices into $\operatorname{rot}_s$-eigenmodes $E_\eta = \{Y : \operatorname{rot}_s(Y) = \omega^\eta Y\}$ $(0 \leq \eta < n)$ via `fourierCoeffs` and `modeLift`, and apply `LeanMachine.SequentialDimension.finrank_le_sum_of_prefix` in descending degree order $D = n - 1 - k$ $(0 \leq k < n)$. When modes $> D$ of $X \in \mathcal{X}_K \cap \operatorname{rot}_s^{-1}(\mathcal{X}_K)$ vanish, on each row-orbit $O_a$, class $m$, and column-orbit $a' \in \operatorname{Fin} g_s$, evaluating $\mathcal{A}_{m,K}$ on both $X$ and $\operatorname{rot}_s(X)$ expresses the mode-$D$ coefficients along $c = a_0 + ts \in J_m \cap O_a$ via a $(K^b)$-valued polynomial $P(Z) = \sum_{\eta=0}^{D} v_\eta Z^\eta$ obeying both $P(\omega^t) = \alpha_D(a_0 + ts, a')$ and $P(\omega^{t+1}) = \omega^D \alpha_D(a_0 + ts, a')$. Hence the twisted values $\omega^{-tD} P(\omega^t)$ are constant across all $\ell_r + 1$ points $t_r, \ldots, t_r + \ell_r$ of each run of $J_m \cap O_a$, and $P(\omega Z) - \omega^D P(Z) = \sum_{\eta=0}^{D-1} (\omega^\eta - \omega^D) v_\eta Z^\eta$ vanishes at all $d_{m,a}$ roots $\omega^t$ $(a_0 + ts \in J_m \cap O_a)$. Since $\omega^\eta \neq \omega^D$ for $\eta < D$, Lemma 5.1 bounds each $(a, m, a')$ block at step $D$ by $b \cdot$ `stepRunBound`$(d_{m,a}, T_{m,a}, D)$, where `stepRunBound` equals 0 if $d_{m,a} = 0$, 1 if $d_{m,a} = n$ ($T_{m,a} = 0$, summing over $0 \leq D < n$ to $n = d_{m,a}$), and $\min(T_{m,a},\, 1 + \max(0, D - d_{m,a}))$ when $1 \leq d_{m,a} < n$ ($T_{m,a} \geq 1$). In the latter case, using $T_{m,a} \leq n - d_{m,a}$, the $d_{m,a}$ steps $0 \leq D < d_{m,a}$ contribute 1 each $(+d_{m,a})$ and the remaining steps $d_{m,a} \leq D < n$ contribute $\sum_{u=0}^{n-d_{m,a}-1} \min(T_{m,a}, 1+u) = T_{m,a}(n - d_{m,a} - T_{m,a}) + \binom{T_{m,a}+1}{2} = T_{m,a}(n - d_{m,a}) - \frac{1}{2}T_{m,a}^2 + \frac{1}{2}T_{m,a}$. Summing over $a' \in \operatorname{Fin} g_s$, $a \in \operatorname{Fin} g_s$, and $m \in \operatorname{Fin} M$ with $\sum_{a,m} d_{m,a} = d$ gives $B_{\mathrm{run}}(b, p, s)$. □

Unlike $H_{\{0,1\}}$, which has full dimension $d^2$ whenever no two adjacent columns share a class, Theorem 5.2 cuts the single-shift intersection down to $\frac{1}{4}d^2 + \frac{3}{2}d$ on the alternating two-coloring when $\operatorname{char}(F) \nmid 2m$, and the value $\frac{1}{4}d^2 + \frac{3}{2}d$ itself is realized over *every* field:

**Theorem 5.3** (Realization of $B_{\mathrm{run}}(1, p, 1)$ over every field and sharpness when $\operatorname{char}(F) \nmid 2m$)**.** Lean: `extension_Theorem1A_AlternatingSharpness`, `alternatingSharpCode_finrank_and_shift_inf` [`ExtensionLinearRunSharpness.lean`]

*Let $F$ be* any *field (of arbitrary characteristic) and let $m \geq 1$ and $d = 2m$. For the alternating two-coloring $p(c) = c \bmod 2$ (`altTwoColoring`, for $c \in \{0, \ldots, 2m-1\}$) at $b = 1$ and shift $s = 1$:*

*(a) With $g_1 = 1$, $n = 2m$, $d_{0,0} = d_{1,0} = m$, and $T_{0,0} = T_{1,0} = m$, the covariance space $H_{\{0,1\}}$ has $\dim_F H_{\{0,1\}} = (2m)^2 = d^2$, and $B_{\mathrm{run}}(1, p, 1) = m^2 + 3m = \frac{1}{4}d^2 + \frac{3}{2}d$.*

*(b) The explicit position-space symmetric decoder pair `alternatingSharpDecoder` defines a linear class-uniform code $\mathcal{X} :=$ `alternatingSharpCode`$(F, m)$ (`SwapRigidity.IsClassUniformCode`) satisfying*

$$\dim_F \mathcal{X} \;=\; 2m(m+1) \;=\; \frac{1}{2}d^2 + d, \qquad \dim_F(\mathcal{X} \cap \operatorname{rot}_1^{-1}(\mathcal{X})) \;=\; m^2 + 3m \;=\; \frac{1}{4}d^2 + \frac{3}{2}d,$$

*matching $B_{\mathrm{run}}(1, p, 1)$ over every field $F$—hence attaining Theorem 5.2(b) with equality whenever $\operatorname{char}(F) \nmid 2m$, and showing even when $\operatorname{char}(F) \mid 2m$ (including characteristic 2 for every $m \geq 1$) that $m^2 + 3m$ cannot be lowered.*

*Outline of the formal proof.* Because $p(c) \neq p(c+1)$ for all $c \in \mathbb{Z}_{2m}$, the implication in $H_{\{0,1\}}$ is vacuous, giving $\dim_F H_{\{0,1\}} = (2m)^2$. Decomposing $X \in \mathrm{Mat}_{2m\times 2m}(F)$ into four $m \times m$ parity blocks $(B_{i,j}, A_{i,j}, C_{i,j}, D_{i,j}) = (X_{2i,2j}, X_{2i,2j+1}, X_{2i+1,2j}, X_{2i+1,2j+1})$ $(i,j \in \mathbb{Z}_m)$, `alternatingShar pDecoder` reads only even rows at class 0 and only odd rows at class 1, so $X \in \mathcal{X}$ if and only if $B$ and $C$ are symmetric $(M_{i,j} = M_{j,i})$ while $A$ and $D$ are shifted-symmetric $(M_{i,j} = M_{j+1,i-1})$. Over every field $F$, each of the four blocks is parameterized by unordered pairs in $\mathrm{Sym}^2(\mathbb{Z}_m)$ of dimension $\frac{m(m+1)}{2}$, giving $\dim_F \mathcal{X} = 4 \cdot \frac{m(m+1)}{2} = 2m(m+1)$. Intersecting with $\mathrm{rot}_1^{-1}(\mathcal{X})$ keeps $B$ and $C$ symmetric ($\frac{m(m+1)}{2}$ each) and forces $A$ and $D$ to be both symmetric and shifted-symmetric, hence constant along cyclic anti-diagonals $i + j \in \mathbb{Z}_m$ (dimension $m$ each), leaving exact dimension $2 \cdot \frac{m(m+1)}{2} + 2m = m^2 + 3m$. □

### 5.2 Two-class 19/32 bound when $\mathrm{char}(F) \nmid d$ and prime-$d$ multi-class bounds

Averaging Theorem 5.2 across shifts $s \in \mathbb{Z}_d$ via the Grassmann inequality $2\dim_F \mathcal{X} \leq bd^2 + \dim_F(\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X}))$ yields leading coefficient $\frac{19}{32}$ for two classes whenever $\mathrm{char}(F) \nmid d$ (with Theorem 5.4(b) non-vacuous for $d \geq 14$, improving on [10, Theorem 3.5] for $d \geq 35$, and lying below $\frac{3}{4}bd^2$ for $d \geq 93$)—strictly below both the $\frac{41}{64}$ bound of [10, Theorem 3.4] and the $\frac{5}{8} - o(1)$ nested-component certificate floor of [10, Theorem C.6]—and leading coefficient $c_M < 1 - \frac{1}{2M}$ for every $M \geq 2$ at prime $d \neq \mathrm{char}(F)$ (at $M = 2$, $\frac{19}{32}bd^2 + \frac{9}{8}bd$ is $< bd^2$ for primes $d \geq 3$, beats [10, Theorem 3.5] for primes $d \geq 7$, and is $< \frac{3}{4}bd^2$ for primes $d \geq 11$; see `tools/verify_linear_capacity_crossovers.py`):

**Theorem 5.4** (Two-class 19/32 capacity ceiling when $\mathrm{char}(F) \nmid d$)**.** Lean: `extension_Theorem1B_TwoClass`, `linearClassUniform_twoClass_finrank_le` [`ExtensionLinearRunProofs.lean`]

*Let $F$ be a field and let $d, b \geq 1$. For any finite set $\kappa$, any $p : \mathbb{Z}_d \to \kappa$, and any $s \in \mathbb{Z}_d$, write $Q_s := |\{c \in \mathbb{Z}_d : p(c) \neq p(c+s)\}|$ (`shiftTransitionCount`).*

*(a)* Per-shift orbit-aggregated bound*: for every $p : \mathbb{Z}_d \to \mathrm{Fin}\,2$ and every $s \in \mathbb{Z}_d$ with $\mathrm{char}(F) \nmid (d/g_s)$, every $\mathcal{X}$ satisfying* `IsLinearClassUniformCode` *$p$ $\mathcal{X}$ obeys*

$$\dim_F(\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X})) \;\leq\; b\Big(\frac{1}{2}\,Q_s d \;-\; \frac{1}{4}\,Q_s^2 \;+\; \frac{1}{2}\,g_s Q_s \;+\; g_s d\Big).$$

*(b)* Two-class bound for $\mathrm{char}(F) \nmid d$*: whenever $\mathrm{char}(F) \nmid d$, for every finite set $\kappa$ with $|\kappa| \leq 2$ and every $p : \mathbb{Z}_d \to \kappa$, every $\mathcal{X}$ satisfying* `IsLinearClassUniformCode` *$p$ $\mathcal{X}$—and hence $D_b(p) =$* `linearClassUniformMaxDim`*$(F, b, p)$—satisfies*

$$\dim_F \mathcal{X} \;\leq\; \frac{19}{32}\,bd^2 \;+\; \frac{3}{2}\,bd^{3/2}.$$

*Here $\frac{19}{32} = \frac{38}{64} = 0.59375 < \frac{5}{8} = \frac{40}{64} < \frac{41}{64}$.*

*Outline of the formal proof.* Re-index $\kappa$ into $\mathrm{Fin}\,2$. On each orbit $O_a$ of $c \mapsto c+s$, a two-coloring has equal transition counts $T_{0,a} = T_{1,a} =: t_a$ with $(n - d_{0,a}) + (n - d_{1,a}) = n$ and $2\sum_{a=0}^{g_s-1} t_a = Q_s$. Thus Theorem 5.2(b) gives $B_{\mathrm{run}}(b,p,s) = bg_s \sum_a (t_a n - t_a^2 + t_a) + bg_s d$, and Cauchy–Schwarz across the $g_s$ orbits ($g_s \sum_a t_a^2 \geq (Q_s/2)^2$) cancels $g_s$ in the quadratic term to prove (a). For (b), since $n_s = d/g_s \mid d$, $\mathrm{char}(F) \nmid d$ implies $\mathrm{char}(F) \nmid n_s$ for all $s \in \mathbb{Z}_d$. Averaging $2\dim_F \mathcal{X} \leq bd^2 + \dim_F(\mathcal{X} \cap \mathrm{rot}_s^{-1}(\mathcal{X}))$ uniformly over all $d$ shifts $s \in \mathbb{Z}_d$ uses $\frac{1}{d}\sum_s Q_s = 2\alpha(1-\alpha)d \leq d/2$ (where $\alpha d$ is a class size), $\frac{1}{d}\sum_s Q_s^2 \geq \left(\frac{1}{d}\sum_s Q_s\right)^2$ by Cauchy–Schwarz, $\max_{q\in[0,1/2]}(\frac{1}{2} + \frac{1}{4}q - \frac{1}{8}q^2) = \frac{19}{32}$ at $q = \frac{1}{2}$, $Q_s \leq d$, and divisor counting $\sum_{s\in\mathbb{Z}_d} g_s = \sum_{g\mid d} g\,\varphi(d/g) \leq d \cdot \#\{g : g \mid d\} \leq 2d^{3/2}$, so the subleading term $\frac{1}{2}b(\frac{1}{2}g_s Q_s + g_s d) \leq \frac{3}{4}bg_s d$ averages over $s \in \mathbb{Z}_d$ to at most $\frac{3}{4}b \cdot 2d^{3/2} = \frac{3}{2}bd^{3/2}$. □

**Theorem 5.5** (Multi-class linear capacity ceiling at prime dimensions)**.** Lean: `extension_Theorem1B_MultiClassPrime`, `multiClass_simplex_max_iff`, `linearClassUniform_multiClass_prime_finrank_le_sharp`, `linearClassUniform_multiClass_prime_maxDim_le_sharp`, `linearClassUniform_twoClass_prime_finrank_le`, `linearClassUniform_multiClass_prime_finrank_le`, `linearClassUniform_multiClass_prime_maxDim_le`, `multiClassLeadingCoeff_sub_flag` [ExtensionLinearRunProofs.lean]

*For $M \geq 2$, define the rational leading coefficient (*`multiClassLeadingCoeff`*)*

$$c_M \;:=\; 1 \;-\; \frac{5}{4M} \;+\; \frac{1}{M^2} \;-\; \frac{1}{4M^3}.$$

*(a)* Unique simplex maximum*: for every $M \geq 2$ and every probability vector $\alpha : \operatorname{Fin} M \to \mathbb{R}$ ($\alpha_m \geq 0$, $\sum_{m=0}^{M-1} \alpha_m = 1$),*

$$\frac{1}{2} \;+\; \frac{1}{2}\sum_{m=0}^{M-1} \alpha_m(1-\alpha_m)^2\Big(1-\frac{\alpha_m}{2}\Big) \;\leq\; c_M,$$

*with equality if and only if $\alpha_m = 1/M$ for all $m \in \operatorname{Fin} M$.*

*(b)* Prime-$d$ bound*: for every prime $d$, every field $F$ with $\operatorname{char}(F) \neq d$, every $M \geq 2$, every $b \geq 1$, and every $p : \mathbb{Z}_d \to \operatorname{Fin} M$, every subspace $\mathcal{X}$ satisfying* `IsLinearClassUniformCode` *$p$ $\mathcal{X}$—and hence $D_b(p)$—obeys*

$$\dim_F \mathcal{X} \;\leq\; c_M\, bd^2 \;+\; \big(\frac{5}{4} - \frac{1}{4M}\big)bd$$

*(giving $\dim_F \mathcal{X} \leq \frac{19}{32}\, bd^2 + \frac{9}{8}\, bd$ for any finite $|\kappa| \leq 2$ at $M = 2$, and implying the simpler form $\dim_F \mathcal{X} \leq c_M\, bd^2 + bMd$ for $M \geq 3$).*

*(c)* Strict win over the flag bound for all $M \geq 2$*: for every $M \geq 2$,*

$$\Big(1 - \frac{1}{2M}\Big) - c_M \;=\; \frac{(M-1)(3M-1)}{4M^3} \;>\; 0,$$

*so $c_M < 1 - \frac{1}{2M}$ (in particular $c_3 = \frac{37}{54} \approx 0.6852 < \frac{5}{6}$ and $c_4 = \frac{191}{256} \approx 0.7461 < \frac{7}{8}$).*

*Outline of the formal proof.* For (a), let $f(x) := x(1-x)^2(1-x/2)$. When $M = 2$, $\alpha_1 = 1 - \alpha_0$ and the explicit sum-of-squares identity $2f(1/2) - (f(x) + f(1-x)) = (x - 1/2)^2(\frac{1}{2} + (x-1/2)^2) \geq 0$ vanishes if and only if $x = 1/2$. When $M \geq 3$, set $c := 1/M \leq 1/3$; for every $x \in [0,1]$, the global tangent-line sum-of-squares identity

$$f(c) + f'(c)(x-c) - f(x) \;=\; (x-c)^2\Big(\tfrac{1}{2}(1-x)(3-x-2c) + \tfrac{3}{2}\big(\tfrac{1}{3}-c\big)^2 + 2\big(\tfrac{1}{3}-c\big) + \tfrac{1}{6}\Big) \;\geq\; \tfrac{1}{6}(x-c)^2$$

sums over $m \in \operatorname{Fin} M$ using $\sum_m (\alpha_m - c) = 0$ to give $\sum_m f(\alpha_m) \leq M f(1/M)$, with equality if and only if every $\alpha_m = 1/M$. For (b), when $d$ is prime, every $s \in \mathbb{Z}_d \setminus \{0\}$ has $g_s = 1$ and $n_s = d$; writing $d_m := |J_m|$, summing $2\dim_F \mathcal{X} \leq bd^2 + \dim_F(\mathcal{X} \cap \operatorname{rot}_s^{-1}(\mathcal{X}))$ at $s = 0$ ($\dim_F \mathcal{X} \leq bd^2$) and Theorem 5.2(b) over all $s \neq 0$ uses $\sum_{s\neq 0} T_{m,0}(s) = d_m(d - d_m)$, Cauchy–Schwarz over the $d-1$ nonzero shifts ($d\sum_{s\neq 0} T_{m,0}(s)^2 \geq (d-1)\sum_{s\neq 0} T_{m,0}(s)^2 \geq d_m^2(d-d_m)^2$), and (a) at $\alpha_m = d_m/d$, while the remaining subleading terms contribute $\frac{1}{2d}b(d^2 + (d-1)d + \frac{1}{2}\sum_m d_m(d-d_m)) \leq (\frac{5}{4} - \frac{1}{4M})bd$ since $\sum_m d_m^2 \geq d^2/M$ by Cauchy–Schwarz. Part (c) is proved directly in `multiClassLeadingCoeff_sub_flag`. □

**Remark 5.6** (Scope in positive characteristic dividing $d$, and remaining gap to $1/2$ (not formalized))**.** Because Theorem 5.2(b) relies on distinct $n$-th roots of unity in $\overline{F}$, Theorem 5.4(b) requires $\mathrm{char}(F) \nmid d$ and does not apply, for example, at $d = 2^m$ ($m \geq 2$) in characteristic 2 (where the Mahler binomial polynomial $\binom{c}{j} \bmod 2$ can vanish on more than $j$ elements of $\{0, \ldots, 2^m - 1\}$: already at $j = 1$, $\binom{c}{1} \equiv 0 \pmod 2$ vanishes on all $2^{m-1} > 1$ even integers in $\{0, \ldots, 2^m - 1\}$). Even when $\mathrm{char}(F) \nmid d$, the constant $\frac{19}{32} = 0.59375$ narrows the gap toward $\frac{1}{2} + o(1)$ in [4, Problem 9.1(a)] (dropping below the $\frac{41}{64}$ bound of [10, Theorem 3.4] and the $\frac{5}{8} - o(1)$ nested-component certificate floor of [10, Theorem C.6], while remaining above the $\frac{9}{16} - o(1)$ full-ambient isolation certificate floor of [10, Theorem C.7]), leaving $D_b(p) \leq (\frac{1}{2} + o(1))bd^2$ open.

# 6 Exact dimension 25 for $\{0,1,3\} \mid \{2,4,5\}$ on $\mathbb{Z}_6$ outside characteristics 2 and 3

Problem 9.1(b) of [4] asks for the exact maximum scalar ($b = 1$) dimension $D_1(\{0,1,3\}, \{2,4,5\})$ (written $D$ in [4]) of a linear class-uniform code on $\mathbb{Z}_6$ for the two-class partition $J_0 = \{0,1,3\}$, $J_1 = \{2,4,5\}$ (`SwapRigidity.aperiodicCls`, the partition of [4, Proposition 8.2], which has no shift almost-period by [4, Remark 8.1(c)] and whose codes beat the interval value $(d^2 + Md)/2 = 24$). Prior to this work, the known bounds were $25 \leq D_1 \leq 28$ over every field in which $3 \neq 0$ and $21 \leq D_1 \leq 28$ in characteristic 3: Theorem TP ([4, Theorem 6.4], Lean-verified) gives $D_1 \leq 28$ over every field; symmetric matrices $X = X^{\mathsf{T}}$ (with single decoder $A(Y)_r = Y_{r,0}$, giving $X_{c,r} = X_{r,c}$) achieve dimension $6 \cdot 7/2 = 21$ in every characteristic; and a 25-dimensional code was constructed over $\mathbb{F}_2$ in [4, Proposition 8.2] (Lean-verified) and extended to every field with $3 \neq 0$ in [9, Theorem 3.2] (`ClassUniformMemo.exists_code_of_three_ne_zero`, Lean-verified, pushing forward the $\mathbb{F}_2$ certificate in characteristic 2 and supplying two integer echelon certificates for all characteristics outside $\{2,3\}$; whether 25 is attainable in characteristic 3 is [9, Problem 6.1]). Note that when $\mathrm{char}(F) \nmid 6$, Theorem 5.2(b) alone at $d = 6, b = 1, s = 1$ ($d_{0,0} = d_{1,0} = 3$, $T_{0,0} = T_{1,0} = 2$) gives $\dim_F(\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})) \leq 2(2 \cdot 3 - \frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 2) + 6 = 16$, yielding $2 \dim_F \mathcal{X} \leq 36 + 16 = 52$, i.e. $\dim_F \mathcal{X} \leq 26$, still one dimension above 25.

By analyzing the coupled cross-mode constraints across all six column-Fourier modes of $H_{\{0,1\}}$ simultaneously, we sharpen the single-shift intersection bound to 14 and establish $D_1(\{0,1,3\}, \{2,4,5\}) = 25$ in [4, Problem 9.1(b)] over every field of characteristic outside $\{2,3\}$.

**Lemma 6.1** (Single-shift intersection bound 14 for $\{0,1,3\} \mid \{2,4,5\}$ on $\mathbb{Z}_6$)**.** Lean: `extension_Theorem2_Preliminaries`, `aperiodicSix_shift_inf_finrank_le_14` [`ExtensionAperiodicSixProofs.lean`]

*Let $F$ be a field.*

*(a) The Boolean partition* `SwapRigidity.aperiodicCls` $: \mathbb{Z}_6 \to \mathrm{Bool}$ *has* ***`true`****-fiber* $\{0,1,3\}$ *and* ***`false`****-fiber* $\{2,4,5\}$*, and for every field $F$ the three conditions* $(\mathrm{char}(F) \neq 2 \wedge \mathrm{char}(F) \neq 3)$*,* $((2 : F) \neq 0 \wedge (3 : F) \neq 0)$*, and* `CoprimeToChar`$(F, 6)$ *are equivalent.*

*(b) Whenever* $\mathrm{char}(F) \notin \{2,3\}$*, every subspace* $\mathcal{X} \leq \mathrm{Mat}_{6\times 6}(F)$ *satisfying* `SwapRigidity.IsClassUniformCode SwapRigidity.aperiodicCls` $\mathcal{X}$ *obeys*

$$\dim_F(\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})) \;\leq\; 14.$$

*Outline of the formal proof.* Part (a) is direct verification on $\mathbb{Z}_6$ and prime factors of 6. For (b), extend the two $F$-linear class decoders $\mathcal{A}_0, \mathcal{A}_1$ by scalars to $K = \overline{F}$ via Proposition 2.1(b)–(c),

let $\mathcal{X}_K := \{X \in \mathrm{Mat}_{6\times 6}(K) : \forall c \in \mathbb{Z}_6,\ X_{c,:} = \mathcal{A}_{\mathrm{cls}(c),K}(\mathrm{rot}_c(X))\}$ be the *maximal* $K$-linear class-uniform code cut out by $\mathcal{A}_{0,K}, \mathcal{A}_{1,K}$ (containing the scalar-extended image of $\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})$ inside $W := \mathcal{X}_K \cap \mathrm{rot}_1^{-1}(\mathcal{X}_K)$, so $\dim_F(\mathcal{X} \cap \mathrm{rot}_1^{-1}(\mathcal{X})) \le \dim_K W$), and choose a primitive 6-th root of unity $\zeta \in K^\times$ ($\zeta^2 = \zeta - 1$, with $(2:K) \neq 0$ and $(3:K) \neq 0$). Because $W$ lies in $H_{\{0,1\}}$ ($X_{1,j} = X_{0,j+1}$ and $X_{5,j} = X_{4,j+1}$), $H_{\{0,1\}}$ has dimension $4\cdot 6 = 24$ and decomposes into six 4-dimensional column-Fourier eigenspaces $V_\eta = \{Y \in H_{\{0,1\}} : \mathrm{rot}_1(Y) = \zeta^\eta Y\}$ ($\eta \in \mathbb{Z}_6$). Let $K_\eta := W \cap V_\eta$, $W_{\mathrm{pure}} := \bigoplus_{\eta\in\mathbb{Z}_6} K_\eta$, $S_K := \sum_{\eta\in\mathbb{Z}_6} \dim_K K_\eta$, and $Q := W/W_{\mathrm{pure}}$, so $\dim_K W = S_K + \dim_K Q$.

1. *Pure-mode cap $S_K \le 12$:* on a single mode $Y \in V_\eta$, $\mathrm{rot}_c(Y) = \zeta^{c\eta} Y$, so the linear decoder $\mathcal{A}_{m,K}$ outputs $\zeta^{c\eta}\mathcal{A}_{m,K}(Y)$ across every $c \in J_m$ ($m \in \{0,1\}$), forcing $Y$ to be class-shift covariant on all of $J_0$ and all of $J_1$; this cuts each $K_\eta$ down to a 2-dimensional subspace $L_\eta \le V_\eta$ and gives $\dim_K K_\eta \le 2$ for every $\eta \in \mathbb{Z}_6$, hence $S_K \le 12$.

2. *Antipodal quotient inequality $3\dim_K Q + S_K \le 24$:* for each $a \in \{0,1,2\}$, let $S_a := \mathbb{Z}_6 \setminus \{a, a+3\}$ and $W_{S_a} := W \cap \bigoplus_{\eta\in S_a} V_\eta$. Because both classes have size 3 with antipodal cancellation and $(Z^6-1)/(Z-\zeta^{c_0})$ has all six coefficients nonzero, any $Y \in W_{S_a}$ has each of its four mode components $\pi_\eta(Y)$ ($\eta \in S_a$) individually satisfy the decoder equations defining $\mathcal{X}_K$ and the shifted-membership equations for $\mathrm{rot}_1^{-1}(\mathcal{X}_K)$, hence lie in $W$ by maximality of $\mathcal{X}_K$, i.e. $W_{S_a} = \bigoplus_{\eta\in S_a} K_\eta$. Thus the induced map $Q \to (V_a/K_a) \oplus (V_{a+3}/K_{a+3})$ is injective for each $a \in \{0,1,2\}$. Summing $\dim_K Q \le 8 - \dim_K K_a - \dim_K K_{a+3}$ over $a \in \{0,1,2\}$ yields $3\dim_K Q + S_K \le 24$. Whenever $S_K \le 10$, this immediately gives $\dim_K W = S_K + \dim_K Q \le S_K + \lfloor (24 - S_K)/3 \rfloor \le 14$.

3. *Closing $S_K = 11$ and $S_K = 12$:* using the exact anti-diagonal zeros of the $\mathbb{Z}_6$ DFT interpolation matrices at shift $\eta + 3 \leftrightarrow \eta$ to eliminate the single unconstrained column ($c = 5$ for $J_0$, $c = 1$ for $J_1$), pairwise disjointness of the antipodal projection kernels eliminates $S_K = 11, \dim_K Q \ge 4$ (where a $2+2$ kernel decomposition forces two coupled blocks to vanish and forces two kernels to coincide) and reduces $S_K = 12, \dim_K Q \ge 3$ to a parameter-free $4\times 4$ linear system over $\mathbb{Z}[\zeta, \frac{1}{6}]$ (determinant $\frac{16}{3}$, or $3^4 \cdot \frac{16}{3} = 432 = 2^4 \cdot 3^3$ with entries $\pm 3, \pm(2\zeta - 1)$ after clearing thirds) proved nonsingular by direct row elimination using $(2:K) \neq 0$, $(3:K) \neq 0$, and $(2\zeta-1)^2 = -3 \neq 0$. Hence $\dim_K W \le 14$ in all cases.

□

**Theorem 6.2** (Exact dimension 25 for $\{0,1,3\} \mid \{2,4,5\}$ outside characteristics 2 and 3)**.** `Lean`: `aperiodicSix_finrank_le_25`, `aperiodicSix_maxDim_eq_25`, `extension_Theorem2_AperiodicSixExact` [`ExtensionAperiodicSixProofs.lean`]

*Let $F$ be any field with $\mathrm{char}(F) \neq 2$ and $\mathrm{char}(F) \neq 3$.*

*(a) Every subspace $\mathcal{X} \le \mathrm{Mat}_{6\times 6}(F)$ satisfying* `SwapRigidity.IsClassUniformCode SwapRigidity.aperiodicCls` $\mathcal{X}$ *obeys $2\dim_F \mathcal{X} \le 36 + 14 = 50$ and hence $\dim_F \mathcal{X} \le 25$.*

*(b) Combining (a) with [9, Theorem 3.2] (*`ClassUniformMemo.exists_code_of_three_ne_zero`*), the set of achievable dimensions $\{\dim_F \mathcal{X}\}$ over all $\mathcal{X}$ satisfying* `SwapRigidity.IsClassUniformCode SwapRigidity.aperiodicCls` $\mathcal{X}$ *has greatest element 25, and*

$$D_1(\{0,1,3\},\ \{2,4,5\}) \ =\ 25$$

*(formalized as* `linearClassUniformMaxDim`$(F, 1, \mathrm{cls}) = 25$ *at* cls = `SwapRigidity.aperiodicCls`*). The same bounds and exact maximum 25 hold for every coloring $\mathrm{cls} : \mathbb{Z}_6 \to \kappa$ with fibers $\{0,1,3\}$ and $\{2,4,5\}$ (*`IsAperiodicSixPartition`*).*

*Outline of the formal proof.* Since $\mathrm{rot}_1$ is an $F$-linear automorphism of $\mathrm{Mat}_{6\times 6}(F)$ of ambient dimension 36, Grassmann's formula $\dim_F(\mathcal{X}+\mathrm{rot}_1^{-1}(\mathcal{X}))+\dim_F(\mathcal{X}\cap\mathrm{rot}_1^{-1}(\mathcal{X}))=2\dim_F\mathcal{X}$ together with Lemma 6.1(b) yields $2\dim_F\mathcal{X}\le 36+14=50$, so $\dim_F\mathcal{X}\le 25$. Since $\mathrm{char}(F)\neq 3$ implies $(3:F)\neq 0$, `ClassUniformMemo.exists_code_of_three_ne_zero` supplies a 25-dimensional code over $F$, proving `IsGreatest` and `linearClassUniformMaxDim` $= 25$; re-indexing the two fibers transfers the result to any cls satisfying `IsAperiodicSixPartition`. □

**Remark 6.3** (Remaining characteristics 2 and 3)**.** Over $\mathrm{char}(F)=2$ ($25\le D_1\le 28$, by [4, Theorem 6.4 and Proposition 8.2] and [9, Theorem 3.2]) and $\mathrm{char}(F)=3$ ($21\le D_1\le 28$, from symmetric matrices and [4, Theorem 6.4], with attainability of 25 posed in [9, Problem 6.1]), $Z^6-1$ factors with repeated roots as $(Z^3-1)^2$ and $(Z^2-1)^3$. Thus the group algebra $\overline{F}[\mathbb{Z}_6]$ is not semisimple, explaining why the six-mode Fourier decomposition above is unavailable there; determining the exact value of $D_1$ in characteristics 2 and 3 remains open.

# 7 Three-term swap codes and one- and two-sided interval-mask bounds

## 7.1 Three-term swap codes exceeding $d(d+1)/2$ at $d\in\{6,7,9,12\}$ and over all fields at $d=4$

For finite matrix families $A,B:\alpha\to\mathrm{Mat}_{d\times d}(F)$ over a field $F$, the swap code (`SwapRigidity.swapCode`) is the solution space

$$\texttt{SwapRigidity.swapCode}(A,B)\;:=\;\Big\{X\in\mathrm{Mat}_{d\times d}(F)\;:\;\sum_{a\in\alpha}A_aXB_a^{\mathsf T}=X^{\mathsf T}\Big\}.$$

In [4, Theorem 5.1 and Corollary 5.5] (where the coordinate form of Theorem 5.1 is Lean-verified and Corollary 5.5 is proved in text), any two-term swap code ($|\alpha|=2$) has dimension at most the symmetric pairing value $d(d+1)/2$, proved by triangularizing a matrix pencil over an algebraic closure. Immediately afterward, Remark 5.6 of [4] noted that three terms can exceed $d(d+1)/2$, giving an explicit three-term family of dimension $8>6$ at $d=3$ over every field and an example of dimension $11>10$ at $d=4$ (checked numerically there over $\mathbb{F}_2,\mathbb{F}_3,\mathbb{F}_5,\mathbb{F}_7$), and remarked: *"We have no three-term example for $d\ge 5$."*

Here we give explicit three-term examples above $d(d+1)/2$ at $d\in\{6,7,9,12\}$ over every field $F$, together with a proof over every field at $d=4$ (for a padded family matching the dimension $11>10$ of [4, Remark 5.6], previously checked over $\mathbb{F}_2,\mathbb{F}_3,\mathbb{F}_5,\mathbb{F}_7$), by lifting the $3\times 3$ skew-pair construction into $m\times m$ blocks (`threeTermBlockA`, `threeTermBlockB`) and adjoining a diagonal tail (`threeTermPaddedA`, `threeTermPaddedB`). Write the three off-diagonal pairs of $\{0,1,2\}$ as $(i_0,j_0)=(0,1)$, $(i_1,j_1)=(0,2)$, and $(i_2,j_2)=(1,2)$, and set

$$A_t\;:=\;(E_{i_t,j_t}-E_{j_t,i_t})\otimes I_m,\qquad B_t\;:=\;-A_t\;=\;(E_{j_t,i_t}-E_{i_t,j_t})\otimes I_m\qquad(t\in\{0,1,2\}).$$

Because $E_{j_t,i_t}^{\mathsf T}-E_{i_t,j_t}^{\mathsf T}=E_{i_t,j_t}-E_{j_t,i_t}$, we have $B_t^{\mathsf T}=A_t$, so the swap-code equation $\sum_{t=0}^{2}A_tXB_t^{\mathsf T}=X^{\mathsf T}$ is

$$\sum_{t=0}^{2}\big((E_{i_t,j_t}-E_{j_t,i_t})\otimes I_m\big)\,X\,\big((E_{i_t,j_t}-E_{j_t,i_t})\otimes I_m\big)\;=\;X^{\mathsf T}.$$

**Theorem 7.1** (Dimension $4m(m+1)$ of the three-term block family on $d = 3m$)**.** Lean: `bonus1_ThreeTermBlockSwapCodeDim` [`SwapCodeAnswersProofs.lean`]

*Let $F$ be any field and let $m \geq 1$. Partition $X \in \mathrm{Mat}_{3m\times 3m}(F)$ into a $3 \times 3$ grid of $m \times m$ blocks $(X_{i,j})_{0\leq i,j<3}$.*

*(a) $X$ lies in* `SwapRigidity.swapCode`$(A, B) = \{X : \sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}\}$ *if and only if every block $X_{i,j}$ is symmetric ($X_{i,j}^{\mathsf{T}} = X_{i,j}$ for all $0 \leq i, j < 3$) and the three diagonal blocks sum to zero ($X_{0,0} + X_{1,1} + X_{2,2} = 0$;* `IsThreeByThreeSymmetricZeroTraceBlocks`*).*

*(b) Consequently, over every field $F$,*

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;=\; 8 \cdot \frac{m(m+1)}{2} \;=\; 4m(m+1).$$

*Since $4m(m+1) - \frac{3m(3m+1)}{2} = \frac{m(5-m)}{2}$, this exceeds the pairing value $d(d+1)/2$ ($d = 3m$) for $m \in \{1, 2, 3, 4\}$ ($d = 3 : 8 > 6$, matching [4, Remark 5.6]; $d = 6 : 24 > 21$; $d = 9 : 48 > 45$; and $d = 12 : 80 > 78$).*

*Outline of the formal proof.* For each $t \in \{0, 1, 2\}$, multiplying on both sides by $(E_{i_t,j_t} - E_{j_t,i_t}) \otimes I_m$ places $+X_{j_t,i_t}$ in block $(i_t, j_t)$, $+X_{i_t,j_t}$ in block $(j_t, i_t)$, $-X_{j_t,j_t}$ in block $(i_t, i_t)$, and $-X_{i_t,i_t}$ in block $(j_t, j_t)$. Summing over the three pairs sends each off-diagonal block $(i, j)$ $(i \neq j)$ to $+X_{j,i}$ and each diagonal block $(i, i)$ to $-\sum_{j\neq i} X_{j,j}$. Equating with $(X_{j,i}^{\mathsf{T}})_{i,j}$ gives $X_{j,i} = X_{j,i}^{\mathsf{T}}$ for $i \neq j$ together with $-X_{1,1} - X_{2,2} = X_{0,0}^{\mathsf{T}}$, $-X_{0,0} - X_{2,2} = X_{1,1}^{\mathsf{T}}$, and $-X_{0,0} - X_{1,1} = X_{2,2}^{\mathsf{T}}$. Over every ring (including characteristic 2), subtracting pairs of diagonal equations yields $X_{0,0}^{\mathsf{T}} - X_{1,1}^{\mathsf{T}} = X_{0,0} - X_{1,1}$ and cyclic analogues, which together with $X_{2,2} = -X_{0,0}^{\mathsf{T}} - X_{1,1}$ is equivalent to $X_{i,i}^{\mathsf{T}} = X_{i,i}$ for $i \in \{0, 1, 2\}$ and $X_{0,0} + X_{1,1} + X_{2,2} = 0$. Eight independent symmetric $m \times m$ blocks (6 off-diagonal and 2 free diagonal blocks) each have dimension $m(m+1)/2$, giving $4m(m+1)$. □

Embedding the $3m \times 3m$ family in the top-left corner of $\mathrm{Mat}_{d\times d}(F)$ ($d = 3m + s$) and setting both $A_2$ and $B_2$ equal to $I_s$ on the trailing $s \times s$ diagonal block extends the construction to $d = 4$ and $d = 7$.

**Theorem 7.2** (Padded three-term swap codes across $d \in \{3, 4, 6, 7, 9, 12\}$)**.** Lean: `bonus1_ThreeTermPaddedSevenAndFive` [`SwapCodeAnswersProofs.lean`]

Lean: `bonus1_UnifiedThreeTermExceedsPairingBound` [`DualityAndSwapConsequences.lean`]

*Over every field $F$:*

*(a) Padding $m = 2$ ($3m = 6$) to $d = 7$ achieves dimension $29 > 28 = \frac{7\cdot 8}{2}$; padding $m = 1$ ($3m = 3$) to $d = 4$ achieves $11 > 10 = \frac{4\cdot 5}{2}$; and padding $m = 1$ to $d = 5$ achieves $15 = \frac{5\cdot 6}{2}$.*

*(b) Consequently, for every field $F$ and every $d \in \{3, 4, 6, 7, 9, 12\}$, there exist $A, B : \mathrm{Fin}\, 3 \to \mathrm{Mat}_{d\times d}(F)$ with*

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;>\; \frac{d(d+1)}{2}.$$

*Outline of the formal proof.* On the trailing $s = d - 3m$ coordinates, $A_2$ and $B_2^{\mathsf{T}}$ both act as $+I_s$, while $A_0, B_0, A_1, B_1$ vanish outside the top-left $3m \times 3m$ block. Inspecting the coupled cross-blocks between the active $(1, 2)$ pair and the $s$-coordinate tail yields $24+5 = 29$ dimensions at $(m, d) = (2, 7)$, $8 + 3 = 11$ at $(1, 4)$, and $8 + 7 = 15$ at $(1, 5)$. Combining $(1, 4)$ and $(2, 7)$ with $m \in \{1, 2, 3, 4\}$ in Theorem 7.1 covers $d \in \{3, 4, 6, 7, 9, 12\}$. □

More generally (proved in text; not formalized for general $m, s$), for $d = 3m + s$, partitioning $X \in \mathrm{Mat}_{d\times d}(F)$ into three $m$-coordinate blocks $0, 1, 2$ and a trailing $s$-coordinate block 3, the top-left $3m\times 3m$ blocks still contribute $4m(m+1)$ dimensions by Theorem 7.1, while equating $\sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}$ on row/column 3 gives $X_{0,3} = 0$, $X_{3,0} = 0$, $X_{2,3} = X_{3,1}^{\mathsf{T}}$ ($ms$ free entries), $-X_{1,3} = X_{3,2}^{\mathsf{T}}$ ($ms$ free entries), and $X_{3,3} = X_{3,3}^{\mathsf{T}}$ ($s(s+1)/2$ free entries). Thus for $m = \lfloor d/3 \rfloor$ and $s = d - 3m \in \{0, 1, 2\}$, this block-and-tail construction achieves dimension $4m(m+1) + \frac{s(s+1)}{2} + 2ms = \frac{d(d+1)}{2} + \frac{m(5-m-2s)}{2}$, which is at most $d(d+1)/2$ for every $(m \geq 1, s \geq 0)$ with $d \geq 5$ outside $\{6, 7, 9, 12\}$ (giving $15 = 15$ at $d = 5$, $35 < 36$ at $d = 8$, $55 = 55$ at $d = 10$, $63 < 66$ at $d = 11$, $120 = 120$ at $d = 15$, and strictly less than $d(d+1)/2$ once $m \geq 6$). Whether a three-term swap code can exceed $d(d+1)/2$ remains open for $d = 5$ and for every $d \geq 8$ outside $\{9, 12\}$.

### 7.2 One-sided multi-scale and two-sided two-scale (coarse–fine) bounds (Problem 9.3)

Problem 9.3 of [4] asks for upper bounds on $\dim_F \texttt{SwapRigidity.swapCode}(A, B)$ when $d = 2^m$ and both $A_a$ and $B_a$ lie in the span of dyadic interval masks across all scales composed with cyclic rotations ([4, Corollary 6.13], proved in text, covers one-sided tensor-factor rules commuting with a single non-derogatory matrix on one address digit). We first record the one-sided endpoint-refinement baseline across all scales, then prove a characteristic-free two-sided bound for two-scale (coarse–fine) interval families that improves the leading coefficient over the one-sided Theorem TP baseline $(d^2 + \min(s_A + c_A, s_B + c_B)d)/2$ implied by the decomposition counts alone (where each element of $S_A, S_B$ may carry its own singleton mask, as in Corollary 7.5) when $s_A/d$ and $s_B/d$ are bounded away from 0 and 1 in $(0, 1)$ and the run and coarse counts are $o(d)$, and finally contrast interval masks with periodic Walsh digit masks.

For contiguous cyclic or aligned dyadic intervals (`IsCyclicInterval`, `IsAlignedDyadicInterval`; spans `InMultiscaleIntervalRotSpan`, `InAlignedDyadicIntervalRotSpan`), each interval mask introduces at most two boundary transitions around $\mathbb{Z}_d$, so applying Theorem TP ([4, Theorem 6.4], Lean-verified) to the common refinement of the interval endpoints on one side at a time gives a nontrivial upper bound whenever at least one side uses $o(d)$ intervals in total:

**Proposition 7.3** (One-sided endpoint-refinement bound for multi-scale interval masks)**.** Lean: `bonus2_MultiscaleIntervalBothSides`, `bonus2_MultiscaleFullProfileAttainsFullSwap` [`SwapCodeAnswersProofs.lean`]
Lean: `bonus2_SublinearProfileAndTendstoHalf`, `bonus2_SublinearProfileExplicitThreshold` [`DualityAndSwapConsequences.lean`]

*Let $F$ be any field, $m \geq 0$, and $d = 2^m$.*

*(a) Every* `InAlignedDyadicIntervalRotSpan` *family lies in* `InMultiscaleIntervalRotSpan`*, and whenever finite families $A, B : \alpha \to \mathrm{Mat}_{d\times d}(F)$ lie in* `InMultiscaleIntervalRotSpan` *with per-scale interval counts $M_A, M_B : \mathrm{Fin}\, m \to \mathbb{N}$, setting $\beta_A := \max(1, 2\sum_j M_A(j))$ and $\beta_B := \max(1, 2\sum_j M_B(j))$,*

$$2 \dim_F \texttt{SwapRigidity.swapCode}(A, B) \;\leq\; d^2 + d \min(\beta_A, \beta_B).$$

*(b) Along any sequence of fields $(F_m)$ and families $(A_m, B_m)$ whose per-scale interval sums $S_{A,m} := \sum_j M_{A,m}(j)$ and $S_{B,m} := \sum_j M_{B,m}(j)$ satisfy $\min(S_{A,m}, S_{B,m})/2^m \to 0$ as $m \to \infty$, for every $\varepsilon > 0$ one eventually has $\dim_{F_m} \texttt{SwapRigidity.swapCode}(A_m, B_m) \leq (\frac{1}{2} + \varepsilon)d^2$ (uniformly for all $m \geq \lfloor 1/\varepsilon \rfloor + 1$ whenever $\min(S_{A,m}, S_{B,m}) \leq (\varepsilon/2)\, 2^m$).*

*(c) At the full aligned dyadic profile* $M_{\mathrm{full}}(j) = 2^{j+1}$ *(satisfying* $\sum_{j=0}^{m-1} 2^{j+1} \geq d$ *for* $m \geq 1$*), there exist aligned dyadic interval-rotation families attaining* $\dim_F$ `SwapRigidity.swapCode`$(A, B) = d^2$.

*Outline of the formal proof.* In (a), the intervals used in $A$ create at most $2\sum_j M_A(j)$ endpoint cuts around $\mathbb{Z}_d$, partitioning $\mathrm{Fin}\, d$ into at most $\beta_A$ contiguous classes on each of which every $A_a$ acts as a fixed linear combination of rotations. Hence $\sum_a A_a X B_a^{\mathsf{T}} = X^{\mathsf{T}}$ defines a $\beta_A$-class-uniform code, giving $2\dim_F \leq d^2 + d\beta_A$ by [4, Theorem 6.4]. Passing to $Y = X^{\mathsf{T}}$ exchanges $A$ and $B$ via $Y^{\mathsf{T}} = \sum_a B_a Y A_a^{\mathsf{T}}$, giving $2\dim_F \leq d^2 + d\beta_B$ as well. Part (b) divides by $2d^2$. In (c), $\sum_{j=0}^{m-1} 2^{j+1} = 2d - 2 \geq d$ for $m \geq 1$, and aligned dyadic intervals down to length $2^0 = 1$ span all diagonal projectors $E_{r,r}$ and hence all matrix units $E_{r,c} = E_{r,r}\sigma^{c-r}$ (where $(\sigma v)_a = v_{a+1}$). □

Note the scope of Proposition 7.3: the bound uses only the sparser of the two sides rather than any two-sided interaction between $A$ and $B$, carries no matching sparse lower bound, and becomes vacuous ($\geq d^2$) as soon as $\min(\beta_A, \beta_B) \geq d$—in particular at the full profile $M_{\mathrm{full}}(j) = 2^{j+1}$ for $m \geq 2$, where $\beta = 4d - 4 > d$.

To exploit sparsity on *both* sides simultaneously in a two-scale (coarse–fine) setting over an arbitrary field $F$, suppose the intervals used on side $A$ separate into a "fine" coordinate set $S_A \subseteq \mathrm{Fin}\, d$ of cardinality $s_A = |S_A|$ and $c_A$ pairwise-disjoint nonempty coarse cyclic intervals $\mathrm{coarse}_A : \mathrm{Fin}\, c_A \to \mathrm{Finset}(\mathrm{Fin}\, d)$ partitioning the complement $L_A := \mathrm{Fin}\, d \setminus S_A$, so that every $A_a$ lies in the $F$-span of $\{$`intervalMaskRotMatrix`$(F, I, e)\}$ with $I \subseteq S_A$, $I = \mathrm{coarse}_A(i)$, or $I = \mathrm{Fin}\, d$ (`HasCoarseFineIntervalDecomposition`), and similarly $(S_B, c_B)$ on side $B$ ($s_B = |S_B|$, $L_B = \mathrm{Fin}\, d \setminus S_B$). Note that `HasCoarseFineIntervalDecomposition` allows only masks supported in $S$, single coarse intervals partitioning $\mathrm{Fin}\, d \setminus S$, or the full set $\mathrm{Fin}\, d$: nested dyadic families with three or more active scales force intermediate-scale intervals into $S$, so Theorem 7.4 below is a bound for two-scale (coarse–fine) families rather than a general multi-scale profile bound for [4, Problem 9.3]. For any $S \subseteq \mathrm{Fin}\, d$, write $\dot{m}(S) :=$ `cyclicRunEndpointCount`$(S) = |\{r \in S : (r+1) \bmod d \notin S\}|$ and $m(S) :=$ `cyclicRunCount`$(S)$ ($= 1$ if $S = \mathrm{Fin}\, d$ and $\dot{m}(S)$ otherwise, so $\dot{m}(S) \leq m(S)$), and set (`theorem3UnsymmetrizedBound`)

$$B_{\mathrm{cf}}(d;\, s_A, m_A, c_A;\, s_B, m_B, c_B) \;:=\; \frac{1}{2}\,d^2 + \frac{1}{4}\,s_B(d - s_A) + \frac{1}{2}\,s_A s_B + \frac{1}{2}\,c_B d + \frac{1}{4}\,c_A s_B + \frac{1}{2}\,s_B m_A + \frac{1}{4}\,m_B(d - s_A).$$

**Theorem 7.4** (Characteristic-free two-sided coarse–fine bound for interval-rotation swap codes)**.**
Lean: `extension_Theorem3_TwoSidedCoarseFine`, `swapCode_finrank_le_twoSidedCoarseFine` [`ExtensionTwoSidedProfileProofs.lean`]

*Let* $F$ *be* any *field (of arbitrary characteristic), let* $d \geq 1$*, let* $\alpha$ *be a finite index type, and let* $A, B : \alpha \to \mathrm{Mat}_{d\times d}(F)$ *satisfy* `HasCoarseFineIntervalDecomposition` $(S_A, c_A, A)$ *and* `HasCoarseFineIntervalDecomposition` $(S_B, c_B, B)$ *with* $s_A = |S_A|$ *and* $s_B = |S_B|$*. Then with either* $(m_A, m_B) = (\dot{m}(S_A), \dot{m}(S_B))$ *or* $(m_A, m_B) = (m(S_A), m(S_B))$*,*

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;\leq\; \min\Big(B_{\mathrm{cf}}(d;\, s_A, m_A, c_A;\, s_B, m_B, c_B),\; B_{\mathrm{cf}}(d;\, s_B, m_B, c_B;\, s_A, m_A, c_A)\Big).$$

*Outline of the formal proof.* Identify $\mathrm{Fin}\, d$ with $\mathbb{Z}_d$ and let $K(X) := \sum_{a\in\alpha} A_a X B_a^{\mathsf{T}}$, so $\mathcal{X} :=$ `SwapRigidity.swapCode`$(A, B) = \{X : X^{\mathsf{T}} = K(X)\}$.

1. *Corner perturbation*: subtracting the fine-by-fine corner $X \mapsto P_{S_A} K(X) P_{S_B}$ (where $(P_S v)_a = \mathbf{1}_{a \in S}\, v_a$, image dimension $\le s_A s_B$) gives $\dim_F \mathcal{X} \le s_A s_B + \dim_F \mathcal{Y}$, where $\mathcal{Y} := \{X : X^{\mathsf{T}} = K(X) - P_{S_A} K(X) P_{S_B}\}$. Every $X \in \mathcal{Y}$ has $(X^{\mathsf{T}})_{r,c} = 0$ on $S_A \times S_B$, i.e. $X_{b,a} = 0$ for all $(b,a) \in S_B \times S_A$.

2. *Column-shift Grassmann step*: for $\sigma_{\mathrm{col}}(X)_{b,a} := X_{b,a+1}$, the sum $\mathcal{Y} + \sigma_{\mathrm{col}}^{-1}(\mathcal{Y})$ vanishes on $S_B \times \{a \in S_A : a - 1 \in S_A\}$ of cardinality $s_B(s_A - \dot{m}(S_A))$, so $2\dim_F \mathcal{Y} + s_B(s_A - \dot{m}(S_A)) \le d^2 + \dim_F \mathcal{Y}_1$ where $\mathcal{Y}_1 := \mathcal{Y} \cap \sigma_{\mathrm{col}}^{-1}(\mathcal{Y})$.

3. *$B$-side coarse anchors*: along each $B$-coarse interval $\mathrm{coarse}_B(j)$ with start $\mathrm{start}_B(j)$, shift covariance of the $B_a$ forces $X_{b+1,a} = X_{b,a+1}$ across all $a \in \mathbb{Z}_d$ for every $X \in \mathcal{Y}_1$ and every interior step of $\mathrm{coarse}_B(j)$. Setting the $c_B d$ anchor entries $X_{\mathrm{start}_B(j),a} = 0$ defines $\mathcal{Z}_0 \le \mathcal{Y}_1$ with $\dim_F \mathcal{Y}_1 \le c_B d + \dim_F \mathcal{Z}_0$, and every $X \in \mathcal{Z}_0$ vanishes outside $S_B \times L_A$.

4. *Row-shift Grassmann step*: for $\sigma_{\mathrm{row}}(X)_{b,a} := X_{b+1,a}$, the sum $\mathcal{Z}_0 + \sigma_{\mathrm{row}}^{-1}(\mathcal{Z}_0)$ is supported inside $(S_B \cup (S_B + 1)) \times L_A$ of size at most $(s_B + \dot{m}(S_B))(d - s_A)$, while on $\mathcal{Z}_1 := \mathcal{Z}_0 \cap \sigma_{\mathrm{row}}^{-1}(\mathcal{Z}_0)$ shift covariance of the $A_a$ along each $A$-coarse interval forces $X_{b,\mathrm{start}_A(i)+u} = X_{b+u,\mathrm{start}_A(i)}$, injecting $\mathcal{Z}_1$ into $F^{\mathrm{Fin}\, c_A \times S_B}$ of dimension $c_A s_B$. Hence $2\dim_F \mathcal{Z}_0 \le (s_B + \dot{m}(S_B))(d - s_A) + c_A s_B$.

Combining steps 1–4 gives $\dim_F \mathcal{X} \le B_{\mathrm{cf}}(d;\, s_A, \dot{m}(S_A), c_A;\, s_B, \dot{m}(S_B), c_B)$; transposing $Y = X^{\mathsf{T}}$ swaps $(A, S_A, c_A)$ with $(B, S_B, c_B)$, and monotonicity in $(m_A, m_B)$ gives the $m(S)$ form since $d - s_A \ge 0$. □

On the concrete family `InContiguousCoarseFineSpan` generated on side $A$ by $s_A$ contiguous singleton masks $\{0\}, \dots, \{s_A - 1\}$ (`contiguousFineSet`, $m_A = 1$) together with one complementary coarse interval $\{s_A, \dots, d-1\}$ (`contiguousCoarseComplement`, $c_A = 1$) and full rotations—and analogously $s_B$ contiguous singletons plus one coarse complement on side $B$—applying Theorem TP's interval clause ([4, Theorem 6.4], $(d^2 + Md)/2$) to one side at a time partitions $\mathrm{Fin}\, d$ into $M = s_A + 1$ cyclic intervals on side $A$ (so $\sum_a A_a X B_a^{\mathsf{T}} = X^{\mathsf{T}}$ makes $X^{\mathsf{T}}$ an $(s_A+1)$-class-uniform code) and $s_B + 1$ cyclic intervals on side $B$ (passing to $Y = X^{\mathsf{T}}$). That one-sided Theorem TP baseline gives $(d^2 + (\min(s_A, s_B) + 1)d)/2$, with leading coefficient $\frac{1}{2} + \frac{1}{2}\min(\alpha, \beta)$ ($\alpha = s_A/d$, $\beta = s_B/d$), whereas Theorem 7.4 achieves a strictly smaller leading coefficient for all $\alpha, \beta \in (0,1)$ (note that at small $d$ or $s = 1$, Theorem TP can be tighter: on $s_A = s_B = s$, $\frac{1}{2}(d^2 + (s+1)d) - B_{\mathrm{cf}}(d; s, 1, 1; s, 1, 1) = \frac{1}{4}(s(d-s) - 2s - d)$, and if $S$ is scattered with $\dot{m}(S) = \Theta(d)$, such as the even residues, $B_{\mathrm{cf}}$ can exceed $d^2$):

**Corollary 7.5** (Leading-coefficient win over one-sided Theorem TP and $\frac{11}{16}d^2 + d$ benchmark)**.**
Lean (partial): `extension_Theorem3_ConcreteWinOverTP`, `swapCode_finrank_le_contiguousCoarseFine` [`ExtensionTwoSidedProfileProofs.lean`]
Not formalized: applying [4, Theorem 6.4] to `InContiguousCoarseFineSpan` to obtain the one-sided baseline $(d^2 + (\min(s_A, s_B) + 1)d)/2$ (part (a) in Lean is the real-coefficient inequality).

*(a) For all real $\alpha, \beta \in (0,1)$,*

$$\frac{1}{2} \;+\; \frac{1}{4}\min\big(\beta(1+\alpha),\ \alpha(1+\beta)\big) \;<\; \frac{1}{2} \;+\; \frac{1}{2}\min(\alpha, \beta).$$

*(b) For every field $F$, every finite index type $\alpha_0$, every $d \ge 2$, every $1 \le s_A, s_B < d$ ($\alpha := s_A/d$, $\beta := s_B/d$), and all $A, B : \alpha_0 \to \mathrm{Mat}_{d\times d}(F)$ satisfying* `InContiguousCoarseFineSpan` $(s_A, A)$ *and* `InContiguousCoarseFineSpan` $(s_B, B)$*, both sides satisfy* `HasCoarseFineIntervalDecomposition` *with $c_A = c_B = m_A = m_B = 1$, and*

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;\le\; \left(\frac{1}{2} + \frac{1}{4}\min\big(\beta(1+\alpha),\ \alpha(1+\beta)\big)\right) d^2 \;+\; \frac{5}{4}\, d.$$

*(c) At the half-fine benchmark $d = 2k$ ($k \geq 1$) with $s_A = s_B = k$ ($\alpha = \beta = 1/2$), one has $B_{\mathrm{cf}}(2k;\, k,1,1;\, k,1,1) = \frac{11}{16}\, d^2 + d$ ($\frac{11}{16} = 0.6875 < \frac{3}{4}$), so every pair $A, B$ in* `InContiguousCoarseFineSpan` $(k)$ *satisfies*

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;\leq\; \frac{11}{16}\, d^2 + d,$$

*which is strictly less than $\frac{3}{4}\, d^2$ for all even $d = 2k \geq 18$ ($k \geq 9$).*

*Outline of the formal proof and proof of the one-sided baseline.* For the one-sided baseline (proved in text), on any $A$ in $\texttt{InContiguousCoarseFineSpan}(s_A)$, every $A_a$ acts on each of the $s_A + 1$ cyclic intervals $\{0\}, \ldots, \{s_A - 1\}, \{s_A, \ldots, d-1\}$ as a fixed linear combination of rotations, so $\sum_a A_a X B_a^{\mathsf{T}} = X^{\mathsf{T}}$ places $X^{\mathsf{T}}$ in an $(s_A+1)$-interval class-uniform code of dimension at most $(d^2 + (s_A+1)d)/2$ by [4, Theorem 6.4] (and $(d^2 + (s_B+1)d)/2$ on $X$ by symmetry). In (a), assuming $\alpha \leq \beta$ by symmetry gives $\min(\beta(1+\alpha), \alpha(1+\beta)) \leq \alpha(1+\beta) < 2\alpha = 2\min(\alpha, \beta)$ since $\alpha > 0$ and $1 + \beta < 2$. In (b), $\texttt{contiguousFineSet}(s)$ has cardinality $s$ and a single cyclic run ($m = 1$) while $\texttt{contiguousCoarseComplement}(s)$ is a single cyclic interval of length $d - s$ ($c = 1$), and substituting $c_A = c_B = m_A = m_B = 1$ into Theorem 7.4 yields $\frac{1}{2}d^2 + \frac{1}{4}s_B d + \frac{1}{4}s_A s_B + \frac{1}{2}d + \frac{3}{4}s_B + \frac{1}{4}(d - s_A) \leq (\frac{1}{2} + \frac{1}{4}\beta(1+\alpha))d^2 + \frac{5}{4}d$ when $s_B \leq s_A$ (and symmetrically when $s_A \leq s_B$). Part (c) evaluates $B_{\mathrm{cf}}(2k;\, k,1,1;\, k,1,1) = 2k^2 + \frac{1}{4}k^2 + \frac{1}{2}k^2 + k + \frac{1}{4}k + \frac{1}{2}k + \frac{1}{4}k = \frac{11}{4}k^2 + 2k = \frac{11}{16}d^2 + d$, which satisfies $\frac{11}{16}d^2 + d < \frac{3}{4}d^2 \iff d > 16 \iff k \geq 9$. □

**Remark 7.6** (Numerical: rank-minimization search on small two-sided interval partitions)**.** A generic random operator $K \in \mathcal{U}(\mathcal{I}) \otimes \mathcal{U}(\mathcal{J})$ in the two-sided interval-rotation span has $\dim \ker(\mathrm{sw} - K) = 0$. To probe maximum swap-code dimensions over $\mathbb{R}$ at $d \in \{5, 6, 7, 8\}$ (reproducible via `tools/verify_two_sided_svd.py`; not formalized, and no proof in this paper uses this computation), we ran Nesterov-accelerated alternating projections with adaptive restart between $\mathcal{U}(\mathcal{I}) \otimes \mathcal{U}(\mathcal{J})$ and bounded-rank manifolds $\{K : \mathrm{rank}(\mathrm{sw} - K) \leq d^2 - D\}$ via truncated SVD across random restarts. Writing $D_\diamond(\mathcal{I}, \mathcal{J}) := \lfloor \frac{1}{2}(d^2 + \sum_{i,j} \min(|I_i|, |J_j|)) \rfloor$, the search found no witness at $D = D_\diamond + 1$ on any tested partition. Bounded-norm operators converged to machine precision ($\leq 10^{-11}$) at $D = D_\diamond$ for the contiguous fine-singleton + coarse-complement partitions $(d, s) \in \{(5,2), (6,3), (7,4), (8,4)\}$ (including the half-density cases $s = d/2$, where $D_\diamond = \frac{5}{8}d^2 + O(d)$), for $d = 6$ with $s_A = 3$ and $s_B = 4$, and for the $d = 6$ two-run partition $\{0\}, \{1,2\}, \{3\}, \{4,5\}$; by contrast, $D_\diamond$ was not reached by the budgeted search for $(d, s) = (7, 3)$ ($D_\diamond = 34$), $(d, s) = (8, 3)$ ($D_\diamond = 42$), and the $d = 7$ partition $1 + 1 + 5$ against $2 + 2 + 3$ ($D_\diamond = 31$), where the witnesses found have dimensions 32, 40, and 27. Whether $D_\diamond(\mathcal{I}, \mathcal{J})$ is an upper bound for all two-sided interval partitions remains open.

By contrast, if "scale-$j$ dyadic mask" means the periodic Walsh/Rademacher bit projector $P_{W_j,\mathrm{bit}}$ onto $\{r \in \mathrm{Fin}(2^m) : \mathrm{bit}_j(r) = \mathrm{bit}\}$ (bit $\in \{0, 1\}$; `walshBitSet`, `walshDigitMaskRotMatrix`, `IsPeriodicScaleDyadicMask`), then a single mask per scale has exponentially many boundaries at fine scales, so one-sided endpoint refinement becomes vacuous even though only two periodic masks per scale are used:

**Proposition 7.7** (Cyclic boundaries and operator-span dimension for periodic Walsh masks)**.** Lean: `walshDigitMask_in_periodicDyadicSpan` [`SwapCodeAnswersProofs.lean`]

Lean: `walshBitSet_cyclicBoundaryCount`, `isCyclicInterval_cyclicBoundaryCount_le_two`, `isPeriodicScaleDyadicMask_cyclicBoundaryCount`, `walshBitSet_joint_separates_points`, `walshDigitMaskRotSpan_finrank_le`, `bonus2_WalshDichotomyStructuralProperties` [`DualityAndSwapConsequences.lean`]

*Let $m \geq 0$ and $d = 2^m$.*

*(a) Every cyclic interval $I \subseteq \operatorname{Fin} d$ ($d \geq 1$) satisfies* `cyclicBoundaryCount`$(I) \leq 2$*, whereas for each $j \in \operatorname{Fin} m$, both $W_j =$* `walshBitSet`$(m, j)$ *and every periodic scale-$j$ mask (*`IsPeriodicScaleDyadicMask`*) have* `cyclicBoundaryCount` $= 2^{j+1}$ *(at least 4 once $j \geq 1$).*

*(b) Across $j \in \operatorname{Fin} m$, the $m$ Walsh bit sets $W_0, \ldots, W_{m-1}$ jointly separate $\operatorname{Fin}(2^m)$ into $2^m$ singleton atoms.*

*(c) Over every field $F$, every family in the span of $\{$*`walshDigitMaskRotMatrix`$(F, j, \mathrm{bit}, s) : j \in \operatorname{Fin} m,\ \mathrm{bit} \in \{0,1\},\ s \in \operatorname{Fin}(2^m)\}$ *belongs to* `InMultiscalePeriodicDyadicRotSpan` *with constant profile $M(j) = 2$ per scale ($\sum_{j=0}^{m-1} M(j) = 2m = 2\log_2 d$), and the entire Walsh-masked rotation span has dimension at most $(m+1)2^m$ over $F$.*

*Outline of the formal proof.* In (a), $W_j$ alternates blocks of $2^{m-1-j}$ zeros and $2^{m-1-j}$ ones $2^j$ times around $\operatorname{Fin}(2^m)$, producing $2^{j+1}$ boundary transitions, and any periodic scale-$j$ mask is a cyclic translate of $W_j$ or its complement. Part (b) recovers $r \in \operatorname{Fin}(2^m)$ from its $m$ binary digits. In (c), the two bit values bit $\in \{0,1\}$ at scale $j$ give two periodic scale-$j$ masks ($M(j) = 2$), while $P_{W_j,0}\sigma^s + P_{W_j,1}\sigma^s = \sigma^s$ shows that the span is generated by the $2^m$ pure rotations together with the $m \cdot 2^m$ operators $P_{W_j,1}\sigma^s$. □

Whether $\dim_F$ `SwapRigidity.swapCode`$(A, B)$ is bounded away from $d^2$ when both $A$ and $B$ lie in the at most $(m+1)2^m$-dimensional Walsh-masked rotation span (a periodic variant of [4, Problem 9.3]) remains open.

# 8 Restricted shuffle and row/column-uniform operator pipelines and anti-diagonal untwisting

We state the scope of this section up front. In Section 8.1 we analyze Operator Schmidt Rank ($\mathrm{rk}_\otimes$) across the row/column tensor cut $F^d_{\mathrm{row}} \otimes_F (F^d_{\mathrm{col}} \otimes_F F^b)$ for linear pipelines built from Stone's shuffle and unshuffle [20] together with row-uniform operators $I_d \otimes B$ and column-uniform operators $A \otimes I_{db}$. This invariant applies *only* to that restricted operator pipeline: it *does not* answer [4, Problem 9.5] and *does not* yield a lower bound for Turing machines. Indeed, Problem 9.5 of [4] asks for a functional that is subadditive over machine steps, large for the swap at every scale separately, and *"not raised at all scales by a single linear-time operation"*, whereas within-row cyclic shear has $\mathrm{rk}_\otimes = d$ (Proposition 8.5(a)) yet is computed in $O(L)$ steps on a 2-work-tape Turing machine (Proposition 8.4). Within the specified operator class, the pass bound $t \geq m$ of Theorem 8.3(a) is attained with zero row/column-uniform stages by Stone's identity $S^m = \tau$ [20] (`stoneShuffleGridLinear_iterate_eq_swap`).

## 8.1 Operator Schmidt Rank on restricted shuffle pipelines and linear-time machines

Write $\texttt{GridState}(F, d, b) := (\mathrm{Fin}\, d \to \mathrm{Fin}\, d \to \mathrm{Fin}\, b \to F)$, separating the row index $r \in \mathrm{Fin}\, d$ from the within-row column-and-lane index $(c, \ell) \in \mathrm{Fin}\, d \times \mathrm{Fin}\, b$. An $F$-linear operator $K$ on $\texttt{GridState}(F, d, b)$ has *Operator Schmidt Rank at most* $R$, written $\mathrm{rk}_{\otimes}(K) \le R$ (`HasOperatorSchmidtRankLe`; see [18, 23]), if there exist $R$ row matrices $A_i \in \mathrm{Mat}_{d\times d}(F)$ and $R$ within-row linear maps $B_i \in \mathrm{End}_F(F^d \otimes_F F^b)$ such that

$$(KX)_{r,c,\ell} \;=\; \sum_{i=0}^{R-1} \sum_{r'=0}^{d-1} (A_i)_{r,r'}\, \big(B_i(X_{r'})\big)_{c,\ell},$$

and we write $\mathrm{rk}_{\otimes}(K) = R$ (`HasExactOperatorSchmidtRank`) when $\mathrm{rk}_{\otimes}(K) \le R$ and no $R' < R$ suffices.

**Lemma 8.1** (Submultiplicativity and unit-rank operators; cf. [18, 23])**.** Lean: `operatorSchmidtRankSubmultiplicative`, `hasOperatorSchmidtRankLe_comp` [`OperatorDualityProofs.lean`]
Lean: `isRowOrColUniform_rankLe_one` [`DualityAndSwapConsequences.lean`]

*Over any commutative semiring $F$ (and in particular any field):*

*(a) If* $\mathrm{rk}_{\otimes}(K_1) \le R_1$ *and* $\mathrm{rk}_{\otimes}(K_2) \le R_2$*, then* $\mathrm{rk}_{\otimes}(K_2 \circ K_1) \le R_2 R_1$*.*

*(b) Every row-uniform column operator* $I_d \otimes B$ *(*`IsRowUniformColOp`*,* $(KX)_r = B(X_r)$*) and every column-uniform row operator* $A \otimes I_{db}$ *(*`IsColUniformRowOp`*,* $(KX)_{r,c,\ell} = \sum_{r'} A_{r,r'} X_{r',c,\ell}$*) satisfies* $\mathrm{rk}_{\otimes}(K) \le 1$*.*

*Outline of the formal proof.* Expanding $K_2 \circ K_1$ over $(i_2, i_1) \in \mathrm{Fin}\, R_2 \times \mathrm{Fin}\, R_1$ multiplies the row matrices $A_{2,i_2} A_{1,i_1}$ and composes the within-row maps $B_{2,i_2} \circ B_{1,i_1}$ using $R_2 R_1$ terms; $I_d \otimes B$ and $A \otimes I_{db}$ are single elementary tensors. □

**Theorem 8.2** (Operator Schmidt Rank of transpose and Stone's shuffle; cf. [18, 23])**.** Lean: `operatorSchmidtRankSwap`, `operatorSchmidtRankStoneShuffle` [`OperatorDualityProofs.lean`]

*Let $F$ be any field and let $b \ge 1$.*

*(a) For every $d \ge 1$, the matrix transpose $\tau = \texttt{gridSwapLinear}(F, d, b)$ satisfies $\mathrm{rk}_{\otimes}(\tau) = d^2$.*

*(b) On $d = 2^m$, Stone's 1-bit cyclic shuffle $S = \texttt{stoneShuffleGridLinear}(F, m, b)$ and unshuffle $S^{-1} = \texttt{stoneUnshuffleGridLinear}(F, m, b)$ are mutual inverses with $\mathrm{rk}_{\otimes}(S) \le 4$ and $\mathrm{rk}_{\otimes}(S^{-1}) \le 4$ for all $m \ge 0$, and for every $m \ge 1$ ($d = 2^m \ge 2$),*

$$\mathrm{rk}_{\otimes}(S) \;=\; \mathrm{rk}_{\otimes}(S^{-1}) \;=\; 4.$$

*Outline of the formal proof.* In (a), decomposing $\tau$ by source/target row pairs uses $d^2$ terms, and evaluating any $R$-term decomposition on row-localized lane-0 basis inputs produces $R$ matrices spanning $\mathrm{Mat}_{d\times d}(F)$, forcing $R \ge d^2$. In (b), for $m \ge 1$, Stone's shuffle sends $(r, c)$ to $(r', c')$ by moving the top bit of $r$ into the low bit of $c'$ and the top bit of $c$ into the low bit of $r'$; splitting on (low bit of $c'$, low bit of $r'$) $\in \{0,1\}^2$ gives a 4-term decomposition whose four nonzero quadrant blocks are linearly independent once $2^m \ge 2$ (and symmetrically for $S^{-1}$). □

**Theorem 8.3** (Pass lower bounds and attainment for restricted shuffle and row/column-uniform pipelines)**.** Lean: `stoneShufflePipelinePassLowerBound` [`OperatorDualityProofs.lean`]
Lean: `boundedOSRPipelinePassLowerBound`, `stoneShuffleGridLinear_iterate_eq_swap` [`DualityAndSwapConsequences.lean`]

*Let $F$ be any field and let $b \geq 1$.*

*(a) On $d = 2^m$ ($m \geq 0$), if an operator list $K_1, \ldots, K_n$ has composition* `composeGridOps` *equal to $\tau$, where $t$ indices are equal to $S$ or $S^{-1}$ and every remaining operator satisfies* `IsRowUniformColOp` *or* `IsColUniformRowOp` *(of the form $I_d \otimes B$ or $A \otimes I_{db}$), then $(2^m)^2 \leq 4^t$ and $t \geq m$. Conversely, iterating $S$ $m$ times satisfies $S^m = \tau$ (*`stoneShuffleGridLinear_iterate_eq_swap`*, over any semiring and for any $m, b \geq 0$), attaining $t = m$.*

*(b) More generally, for any $d \geq 1$ and $R \geq 1$, if a composed pipeline equals $\tau$ with $t$ operators satisfying $\mathrm{rk}_\otimes \leq R$ and all remaining operators satisfying $\mathrm{rk}_\otimes \leq 1$, then $d^2 \leq R^t$, and for $R \geq 2$ one has $t \geq \dfrac{2 \log d}{\log R}$.*

*Outline of the formal proof.* By Lemma 8.1 and Theorem 8.2(b), the composed pipeline has $\mathrm{rk}_\otimes \leq 4^t$ (respectively $\mathrm{rk}_\otimes \leq R^t$), whereas $\mathrm{rk}_\otimes(\tau) = d^2$ by Theorem 8.2(a); taking real logarithms when $R \geq 2$ gives $2 \log d \leq t \log R$. For attainment in (a), induction on $0 \leq k \leq m$ shows that $(S^k X)_{r,c,\ell} = X_{(c \bmod 2^k)2^{m-k} + \lfloor r/2^k \rfloor,\, (r \bmod 2^k)2^{m-k} + \lfloor c/2^k \rfloor,\, \ell}$, giving $S^m = \tau$ at $k = m$. □

In the shared library `lean-machine` (`LeanMachine.ShuffleShear`), we construct explicit transition tables for one Stone shuffle pass, one Stone unshuffle pass, and within-row cyclic shear, and transport them into `Transpose.TM` in `ShuffleShearBridge.lean`:

**Proposition 8.4** (Verified $O(L)$-time multitape Turing machines for shuffle, unshuffle, and row shear)**.** Lean: `stoneShuffleTM_threeTape_linear`, `stoneUnshuffleTM_twoTape_linear`, `rowShearTM_twoTape_linear` [`ShuffleShearBridge.lean`]

*With $b = \mathrm{entryBits}(d)$ (*`Transpose.entryBits`*) and $L = \mathrm{inputLength}(d) = d^2 b$ (*`Transpose.inputLength`*):*

*(a)* `stoneShuffleTransposeTM` *has 3 work tapes ($k = 3$) and, for every $m \geq 1$ ($d = 2^m$) and every matrix $A$ (*`Transpose.Matrix`*), computes* `stoneShuffleTMMatrix` *on $(m, A)$ within $150L$ steps.*

*(b)* `stoneUnshuffleTransposeTM` *has 2 work tapes ($k = 2$) and, for every $m \geq 1$ ($d = 2^m$) and every matrix $A$ (*`Transpose.Matrix`*), computes* `stoneUnshuffleTMMatrix` *on $(m, A)$ within $150L$ steps.*

*(c)* `rowShearTransposeTM` *has 2 work tapes ($k = 2$) and, for every $d \geq 1$ and every matrix $A$ (*`Transpose.Matrix`*), computes within-row cyclic shear* `rowShearTMMatrix` *on $(d, A)$ (sending entry $(r, c, \ell)$ to $A_{r,(c+r) \bmod d,\ell}$) within $200L$ steps.*

*Outline of the formal proof.* Because $d$ and $b$ are not hardcoded in the finite control, the machines in `LeanMachine.ShuffleShear` first determine unary counters for $b$ and $d$ (or $H = d^2/2$) from the input length $L = d^2 b$ in $O(L)$ steps. The 3-work-tape shuffle machine stores unary counters for $b$ and $H$ on two work tapes, buffers the first $H$ words on a third work tape, and interleaves $b$-bit words from the buffer and the second half of the input tape. The 2-work-tape unshuffle machine emits even-indexed $b$-bit words directly from the input tape while buffering odd-indexed words on tape 0 using a unary $b$-counter on tape 1, then replays tape 0. The 2-work-tape row-shear machine keeps a save/emit row

template on tape 0: for row $r$, it buffers the first $rb$ bits on tape 1, streams the remaining $(d-r)b$ bits directly from the input tape to the output, and then emits the $rb$ buffered bits from tape 1 (advancing the save region on tape 0 by $b$ cells for the next row). `computesWithin_toTransposeTM` transports each proof identically to `Transpose.ComputesWithin`. □

Two controls clarify the boundary of $\mathrm{rk}_\otimes$ and of single-cell information measures:

**Proposition 8.5** (Within-row shear $\mathrm{rk}_\otimes = d$ and row-diagonal source-arity preservation)**.** Lean: `control_RowShearAndRowDiagonalAlternation` [`OperatorDualityProofs.lean`] Lean: `rowDiagonal_SourceArityGaugeInvariance` [`DualityAndSwapConsequences.lean`]

*Let $F$ be any field, $d \geq 1$, and $b \geq 1$.*

*(a) Within-row cyclic shear* `rowShearLinear` *on $(F, d, b)$ lies in the row-diagonal subalgebra* `IsRowDiagonalOp` *($(KX)_r = B_r(X_r)$) yet has operator Schmidt rank $d$.*

*(b) For any $s \in \mathbb{N}$, the output-row source-arity bound $\leq s$ (*`HasOutputRowSourceArityLe`*) is preserved under two-sided composition $D_1 \circ K \circ D_0$ with arbitrary row-diagonal operators $D_0, D_1$ (over any commutative semiring $F$), whereas $\mathrm{rk}_\otimes(\mathrm{id}) \leq 1 < d = \mathrm{rk}_\otimes(\mathcal{S}_{d,b} \circ \mathrm{id})$ separates $\mathrm{rk}_\otimes$ from row-diagonal gauge invariance for all $d \geq 2$ (writing $\mathcal{S}_{d,b} :=$ `rowShearLinear`$(F, d, b)$).*

*(c) On $d = 2^m$ ($m \geq 0$), every alternating product $D_t U_t D_{t-1} \cdots U_1 D_0$ (*`alternatingRowDiagProduct`*) of $t+1$ row-diagonal operators $D_i$ and $t$ shuffle or unshuffle passes $U_j \in \{S, S^{-1}\}$ has output-row source arity at most $2^t$; if it equals $\tau$, then $2^m \leq 2^t$ and $t \geq m$.*

*Outline of the formal proof.* In (a), `rowShearLinear` $= \sum_{r=0}^{d-1} E_{r,r} \otimes \sigma^r$ (where $(\sigma v)_{c,\ell} = v_{c+1,\ell}$), and linear independence of the $d$ cyclic column shifts $\sigma^0, \ldots, \sigma^{d-1}$ forces $\mathrm{rk}_\otimes = d$. In (b) and (c), row-diagonal operators act within each row without mixing source rows, while each $S$ or $S^{-1}$ combines at most two source rows into each output row; since row 0 of $\tau(X)$ depends on all $2^m$ input rows when $b \geq 1$, equality with $\tau$ forces $2^m \leq 2^t$. □

Over finite fields, Proposition 8.5(c) is a linear special case of [4, Proposition 6.25(a)] (proved in text), which bounds arbitrary (not necessarily linear) pipelines over a $q$-symbol alphabet on $d = 2^m$ whose stages read input rows inside affine cosets of $\mathbb{F}_2^m$: row-diagonal maps have coset dimension 0 and $S, S^{-1}$ have coset dimension 1, so [4, Proposition 6.25(a)] bounds transposed sets by $q^{(d^2+2^t d)/2}$ and subsumes the conclusion $2^m \leq 2^t$ of Proposition 8.5(c). By contrast, Theorem 8.3(a) allows dense column-uniform linear row operators $A \otimes I_{db}$ (which read all $2^m$ rows and have coset dimension $m$ in [4, Proposition 6.25(a)]) while excluding row-varying column operators such as `rowShearLinear`.

**Proposition 8.6** (Skew one-time-pad zero-mutual-information control)**.** Lean: `control_SkewOneTimePadInformationTrap` [`OperatorDualityProofs.lean`] Lean: `uniformShannonMI_eq_zero_of_indep`, `skewOneTimePad_ShannonMutualInfoZero` [`DualityAndSwapConsequences.lean`]

*For $d \geq 1$ and $b \geq 1$, define $W(X) := X^\mathsf{T} \oplus X$ on $\{0,1\}^{d \times d \times b}$ (*`skewXorPad`*). Then $W(X) \oplus X = X^\mathsf{T}$ for every $X$, and under uniformly random $X$, every single entry $W_{r,c}$ is statistically independent of every source column $X_{:,c'}$, so its discrete Shannon mutual information (*`uniformShannonMI`*) satisfies*

$$I(W_{r,c}\,;\, X_{:,c'}) \;=\; 0 \qquad \text{for all } r, c, c' \in \mathbb{Z}_d.$$

*Outline of the formal proof.* When $r = c$, $W_{r,r} = 0$ is constant. When $r \neq c$, the two entries $X_{c,r}$ and $X_{r,c}$ lie in distinct columns $r \neq c$, so at least one lies outside column $c'$ and acts as an independent uniform pad on the other, giving exact product-count independence and zero Shannon mutual information. □

## 8.2 Anti-diagonal untwisting of shift-collision equations

We close this section by recording a `TensorProduct` packaging of the shift-covariance subspace of [10, Lemma 3.1] (and the shift-link covariance spaces of [4, Section 6]) together with the compatibility characterization of [10]'s Lean library. In [10], Lemma 3.1 (`CapacityAmbient.lean`) establishes the necessity inclusion $C_E \subseteq H_E$ (`SharedDecoders.IsSharedCode.commonPreimage_le_covariance`), the anti-diagonal row normalization (`SharedDecoders.covarianceSpace`, `SharedDecoders.covarianceEquiv`), its dimension $bd\chi(E)$ (`SharedDecoders.finrank_covarianceSpace`), and its rotation invariance (`SharedDecoders.covarianceSpace_rot`), while the two-way compatibility equivalence (`SharedDecoders.isSharedCode_iff_compatible`) is proved in `CapacityGluing.lean` of [10]'s Lean library. Theorem 8.7(a),(c) and the upper bound of Proposition 8.8(a) below restate those properties; what is new here is the packaging of the anti-diagonal coordinate change $a = c + j \in \mathbb{Z}_d$ (`antiDiagLinear`, `fromAntiDiagLinear`),

$$Z_{c,a} \;:=\; X_{c,\,a-c}, \qquad X_{c,j} \;=\; Z_{c,\,j+c} \qquad (c, a, j \in \mathbb{Z}_d),$$

for $d \geq 1$ as the explicit linear tensor isomorphism $\Psi =$ `antiDiagTensorEquiv` $: (\mathbb{Z}_d \to F) \otimes_F (\mathbb{Z}_d \to A) \xrightarrow{\sim} \mathrm{Mat}_{d\times d}(A)$ sending $u \otimes v$ to $(c, j) \mapsto u_c\, v_{j+c}$, together with the converse construction of Proposition 8.8(a) and Proposition 8.8(b).

**Theorem 8.7** (Pairwise shift-collision characterization and anti-diagonal tensor separation; cf. [10, Lemma 3.1] and `SharedDecoders.isSharedCode_iff_compatible`)**.** Lean: `pairwiseShiftCollisionCharacterization`, `antiDiagonalTensorSeparation` [`OperatorDualityProofs.lean`]

Lean: `antiDiagTensorEquiv_intertwines_rot`, `shiftSetCovariance_tensor_finrank` [`DualityAndSwapConsequences.lean`]

*Let $F$ be a field, $d \geq 1$, $A$ an $F$-vector space, and $p : \mathbb{Z}_d \to \kappa$.*

*(a) A linear subspace $C \leq \mathrm{Mat}_{d\times d}(A)$ satisfies* `SharedDecoders.IsSharedCode` *for $p$ if and only if for every shift $s \in \mathbb{Z}_d$ and every $Y \in C$ with $\mathrm{rot}_s(Y) \in C$, $Y$ lies in* `InPairwiseShiftCovariance`*: whenever $p(c) = p(c + s)$, one has $Y_{c+s,\,j} = Y_{c,\,j+s}$ for all $j \in \mathbb{Z}_d$ (cf.* `SharedDecoders.isSharedCode_iff_compatible`*).*

*(b) Under $\Psi : (\mathbb{Z}_d \to F) \otimes_F (\mathbb{Z}_d \to A) \xrightarrow{\sim} \mathrm{Mat}_{d\times d}(A)$, column rotation $\mathrm{rot}_t$ intertwines with the pure right-factor coordinate shift $\mathrm{id} \otimes \sigma^t$ (*`coordShiftLinear`*, $(\sigma^t v)_a = v_{a+t}$).*

*(c) For any shift set $E \subseteq \mathbb{Z}_d$, a matrix $X \in \mathrm{Mat}_{d\times d}(A)$ lies in the shift-set covariance subspace $H_E(p)$ (*`shiftSetCovarianceSubmodule`*, cf.* `SharedDecoders.covarianceSpace`*) if and only if its anti-diagonal coordinates $Z_{c,a} = X_{c,a-c}$ are constant in $c$ on every connected component of* `SwapRigidity.Link` *$p$ $E$ (*`antiDiagonalTensorSeparation`*). When $\dim_F A < \infty$, writing $U_E(p) \leq (\mathbb{Z}_d \to F)$ for the subspace of vectors constant on those components (*`leftComponentSubmodule`*), $\Psi$ maps $U_E(p) \otimes_F (\mathbb{Z}_d \to A)$ onto $H_E(p)$ and*

$$\dim_F H_E(p) \;=\; \dim_F U_E(p) \cdot d \cdot \dim_F A$$

*(cf.* `SharedDecoders.finrank_covarianceSpace`*).*

*Outline of the formal proof.* In (a) (where linearity of $C$ is not used), determinism of $f_{p(c)}$ on $C$ means that whenever $X_1, X_2 \in C$ satisfy $p(c_1) = p(c_2)$ and $\mathrm{rot}_{c_1}(X_1) = \mathrm{rot}_{c_2}(X_2)$, row $c_1$ of $X_1$ equals row $c_2$ of $X_2$: given $Y \in C$ with $\mathrm{rot}_s(Y) \in C$ and $p(c) = p(c+s)$, applying this to $X_1 = \mathrm{rot}_s(Y) \in C$ at $c_1 = c$ and $X_2 = Y \in C$ at $c_2 = c + s$ (so $\mathrm{rot}_c(X_1) = \mathrm{rot}_{c+s}(Y) = \mathrm{rot}_{c_2}(X_2)$) yields $(\mathrm{rot}_s Y)_{c,j} =$

$Y_{c+s,j}$, i.e. $Y_{c,j+s} = Y_{c+s,j}$; conversely, if $\mathrm{rot}_{c_1}(X_1) = \mathrm{rot}_{c_2}(X_2)$ for $X_1, X_2 \in C$ with $p(c_1) = p(c_2)$, setting $s = c_2 - c_1$ gives $X_1 = \mathrm{rot}_s(X_2)$, so taking $Y = X_2 \in C$ gives $\mathrm{rot}_s(Y) = X_1 \in C$ and hence $(X_2)_{c_1+s,j} = (X_2)_{c_1,j+s} = (X_1)_{c_1,j}$. In (b), $\Psi(\mathrm{id} \otimes \sigma^t(u \otimes v))_{c,j} = u_c v_{j+c+t} = \mathrm{rot}_t(\Psi(u \otimes v))_{c,j}$. In (c), substituting $X_{c,j} = Z_{c,j+c}$ turns $X_{c',j+\delta'} = X_{c,j+\delta}$ under $c + \delta = c' + \delta'$ into $Z_{c',a} = Z_{c,a}$ independently for each anti-diagonal $a = c + \delta + j$. □

**Proposition 8.8** (Two-subspace Grassmann relaxation and alternating-mask shift separation; cf. [10, Lemma 3.1])**.** Lean:
`maskFloor_GenericSubspaceRealization` [`DualityAndSwapConsequences.lean`]

*Let $F$ be a field, $d \geq 1$, and $A$ a finite-dimensional $F$-vector space.*

*(a) For any $p : \mathbb{Z}_d \to \kappa$, any shift $s \in \mathbb{Z}_d$, and any subspaces $C_0, C_s \leq \mathrm{Mat}_{d\times d}(A)$ with $\dim_F C_0 = \dim_F C_s$ and $C_0 \cap C_s \leq H_{\{0,s\}}(p)$, Grassmann modularity gives $2\dim_F C_0 \leq d^2 \dim_F A + \dim_F H_{\{0,s\}}(p)$. Conversely, once the right-factor shift coupling $C_s = \mathrm{rot}_{-s}(C_0)$ is dropped, choosing two independent subspaces of dimension $\lfloor (d - \dim_F U_{\{0,s\}}(p))/2 \rfloor$ inside a complement of $U_{\{0,s\}}(p)$ in $(\mathbb{Z}_d \to F)$ constructs $C_0 = \Psi(V_0 \otimes \top)$ and $C_s = \Psi(V_s \otimes \top)$ with $C_0 \cap C_s = H_{\{0,s\}}(p)$, $\dim_F C_0 = \dim_F C_s$, and*

$$d^2 \dim_F A + \dim_F H_{\{0,s\}}(p) \;\leq\; 2\dim_F C_0 + d \dim_F A.$$

*(b) On even $d = 2m$ ($m \geq 2$) with the alternating parity partition $p : \mathbb{Z}_{2m} \to \mathrm{Bool}$ ($p(c) =$ `true` $\iff c$ is odd, for $c \in \{0, \ldots, 2m-1\}$), shift $s = 1$ satisfies $p(c) \neq p(c+1)$ everywhere so $U_{\{0,1\}}(p) = \top$ and $H_{\{0,1\}}(p) = \top$, whereas shift $s = 2$ has $\dim_F U_{\{0,2\}}(p) = 2$ and forces $2\dim_F C \leq ((2m)^2 + 4m) \dim_F A$ for every shared code $C$.*

*Outline of the formal proof.* Part (a) applies $\dim(C_0 + C_s) + \dim(C_0 \cap C_s) = 2\dim C_0$ for the upper bound, and chooses disjoint sets of $\lfloor (d - \dim U_{\{0,s\}}(p))/2 \rfloor$ non-representative indices on the left tensor factor for the converse construction. Part (b) checks that shift 1 has no monochromatic links on the alternating coloring while shift 2 connects all even indices and all odd indices into two components. □

# 9 Cut-cone evaluation, random-order capacity, and negative-bracket cut examples

## 9.1 Cut-causal cone evaluation on frozen `Transpose.TM` fibres

Let $M$ be a deterministic multitape Turing machine (`Transpose.TM`) with $1 \leq M.k$ (available without loss of generality for any transposer by Lemma 2.3), and let $C \subseteq \Sigma_d^{d\times d}$ be a frozen-trace $\Gamma$-cut fibre (`IsFrozenTraceCutFibre`) over a step horizon $T$: all $X \in C$ halt at step $T$ with output $X^{\mathsf{T}}$, follow a common input/work head trajectory (inPos, workPos) and output emission schedule emitStep, and agree on the post-step control state and written work-tape symbols at every cut step $u \in \Gamma \subseteq \{0, \ldots, T-1\}$. Along workPos, a direct causal step $u \to v$ (`TMDepStep`) holds for $u < v < T$ when either $v = u + 1$ (the control-state spine edge) or $u$ is the most recent step visiting the cell scanned by one of the $M.k$ work heads at step $v$ (`IsLastWorkVisit`). A descending path is $\Gamma$-free (`IsCutFreeCausalPath`) if none of its strict predecessors lies in $\Gamma$; the resulting $\Gamma$-truncated causal cone of step $v$ is `trueConeSteps` (`InCutCausalCone`), and its syntactic input-cell support is `cutConeInputCells`.

The causal-cone determination principle and out-degree cone-size bound go back in text to [5, Lemma 4.4 and Corollary 4.5], with earlier Lean formalizations for abstract functional DAGs in [7,

Lemma 3.3(b)] (`MergeDepth/FieldDag.lean`, `MergeDepth.FunDag.cone`), for cone-size bounds on skeletons and block machines of `Transpose.TM` in [7, Lemma 4.7(a)] (`MergeDepth/SkeletonCones.lean`, `MergeDepth.card_coneV_le` giving $\sum_{h\le D}(k+1)^h$; and `MergeDepth/BlockMachine.lean`, `MergeDepth.blockMachine_card_cone_le`), and for native cut replay in [12] (`AdaptiveTransfer/FrozenCutInvoice.lean`, `AdaptiveTransfer.stepValue_eq_of_uncut_native_cut`, `AdaptiveTransfer.native_cut_recovers`). What is new in Lemma 9.1 below are the $(M.k+1)^D$ cone-size instance directly on the native step DAG (`TMDepStep`: control-state spine plus last-work-visit edges) of frozen trace/cut fibres and the rooted $(1+M.k)$-ary symbolic tree representation (`CausalConeTree` in `MachineSweepConsequences.lean`), extracted relative to column $c$ (`traceConeTree`) and evaluated bottom-up on $\mathrm{rot}_c(X)$ directly through $M.\delta$ (`evalConeTree`, syntactic support `treeCellSupport`):

**Lemma 9.1** (Syntactic cut-cone support and symbolic tree evaluation on frozen fibres; cf. [5, Lemma 4.4 and Corollary 4.5], [7, Lemmas 3.3(b) and 4.7(a)], [12])**.** Lean: `tmTraceCutConeDetermination` [`NativeSweepInterfaceProofs.lean`]
Lean: `evalConeTree_cellSupport`, `evalConeTree_entryCellSupport_card_le`, `evalConeTree_traceConeTree_eq` [`MachineSweepConsequences.lean`]
*Let $M$ be a* `Transpose.TM` *with $1 \le M.k$, let $d \ge 1$ and $D \ge 1$, and let $C \subseteq \Sigma_d^{d\times d}$ satisfy* `IsFrozenTraceCutFibre` *such that every $\Gamma$-free causal path $\pi$ has at most $D$ vertices ($\pi$.length $\le D$).*

*(a) For every step $v < T$, $|\texttt{trueConeSteps}(\mathrm{workPos}, T, \Gamma, v)| \le (M.k+1)^D$. For every output bit $(r, c, \ell)$ emitted at step $v = \mathrm{emitStep}(\texttt{outputBitIdx}(d, r, c, \ell))$, the explicit syntactic cone support $Q_{\mathrm{bit}} := \texttt{cutConeInputCells}(d, T, \mathrm{inPos}, \mathrm{workPos}, \Gamma, v)$ consists of those cells visited by* inPos *on* `trueConeSteps`*, satisfies $|Q_{\mathrm{bit}}| \le (M.k+1)^D$, and uniquely determines $X_{c,r,\ell}$ across $C$: whenever $X_1, X_2 \in C$ agree on $Q_{\mathrm{bit}}$, one has $X_1(c,r,\ell) = X_2(c,r,\ell)$.*

*(b) Every depth-$D$* `CausalConeTree` *has $|\texttt{treeCellSupport}(\mathrm{tree})| \le (M.k+1)^D$ and depends on $\mathrm{rot}_c(X)$ only through* `treeCellSupport`(tree)*. Given any cut-value assignment* cutVal *agreeing with the frozen $\Gamma$-cut states and work symbols on $C$, evaluating* `traceConeTree` *at depth $D$ on $\mathrm{rot}_c(X)$ ($X \in C$) via* `evalConeTree` *reproduces the post-step control state, written work symbols, and emitted output bit $X_{c,r,\ell}$ of the live machine run. Across all $b = \texttt{Transpose.entryBits}(d)$ bit-lane trees of an output entry, the union of* `treeCellSupport` *has size at most $\Delta = b(M.k+1)^D$.*

*Outline of the formal proof.* Each step $w < T$ has at most $1 + M.k$ direct causal predecessors ($w-1$ for the control state and at most one last-visit step per work tape), so inducting on the depth budget $m \le D$ gives the geometric sum bound $1 + (M.k+1)\big((M.k+1)^{m-1} - 1\big) < (M.k+1)^m$. Defining $Q_{\mathrm{bit}} =$ `cutConeInputCells` explicitly as the union of input cells read along `trueConeSteps` avoids self-referential singleton footprints: strong induction on $w < T$ shows that inputs in $C$ agreeing on $Q_{\mathrm{bit}}$ induce identical transitions $M.\delta$ throughout the $\Gamma$-stopped cone of $v$, and `evalConeTree` replays those exact transitions bottom-up from initial blank leaves and frozen $\Gamma$-cut leaves on $\mathrm{rot}_c(X)$. □

## 9.2 Lean formalization of the $1/(\Delta+1)$ random-order bound for shared set codes

For any recovery relation of maximum footprint size $\Delta$, the acyclic-subgraph bound of Bar-Yossef, Birk, Jayram, and Kol [2, Theorem 3] combined with a uniformly random vertex ordering gives a maximum acyclic induced subgraph of size at least $(\#\mathrm{clean})/(\Delta+1)$ and hence rate at most $1 - 1/(\Delta+1)$ up to self-referential entries. This elementary bound has been in the series since [5, Proposition 4.14(a)–(b)] (proved in text there, with the $1 - 1/(\Delta+1)$ and acyclic-subgraph forms recalled in [4, Sections 4.1 and 6.7], and ordered-recovery injectivity formalized in `LeanMachine.Or`

deredRecovery.eq_of_rank). We record the Lean 4 formalization of its column-class storage-code analogue in the notation of Definition 2.2.

Given $p : \mathbb{Z}_d \to \operatorname{Fin} M$ and $Q : \operatorname{Fin} M \to \mathbb{Z}_d \to \operatorname{Finset}(\mathbb{Z}_d \times \mathbb{Z}_d)$, evaluating $f_{p(c)}(\mathrm{rot}_c(X))_r$ reads the shifted footprint (shiftedFootprint)

$$F_{c,r} \;:=\; \{(r',\, c+u) \;:\; (r',u) \in Q_{p(c),r}\} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d,$$

and an input entry $(c,r)$ is *clean* (cleanEntries) if $(r,c) \notin \mathcal{E}$ and $(c,r) \notin F_{c,r}$. Note that decoder sharing enters the following bound *only* through the $dM\Delta$ count of self-referential entries $(c,r) \in F_{c,r}$; once clean entries are isolated, the random-order bound uses only the individual footprints $F_{c,r}$.

**Lemma 9.2** (Self-referential entries and clean-entry count)**.** Lean: mem_shiftedFootprint_self_iff, selfReferentialAndCleanCount [BoundedAritySetCapacityProofs.lean]

*For all $d \geq 1$, $M, \Delta \in \mathbb{N}$, $p : \mathbb{Z}_d \to \operatorname{Fin} M$, $Q : \operatorname{Fin} M \to \mathbb{Z}_d \to \operatorname{Finset}(\mathbb{Z}_d \times \mathbb{Z}_d)$, and $\mathcal{E} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$:*

*(a) $(c,r) \in F_{c,r}$ if and only if $(c, r-c) \in Q_{p(c),r}$.*

*(b) If $|Q_{m,r}| \leq \Delta$ for all $(m,r)$, then $|F_{c,r}| \leq \Delta$ for every clean entry $(c,r)$ and*

$$|\texttt{cleanEntries}(p,Q,\mathcal{E})| \;\geq\; d^2 - |\mathcal{E}| - dM\Delta.$$

*Outline of the formal proof.* For each fixed $(m,r)$, the map $c \mapsto (c, r-c)$ is injective on $\mathbb{Z}_d$, so at most $|Q_{m,r}| \leq \Delta$ columns $c \in p^{-1}(m)$ satisfy $(c,r) \in F_{c,r}$. Summing over $m \in \operatorname{Fin} M$ and $r \in \mathbb{Z}_d$ bounds self-referential entries by $dM\Delta$. □

**Theorem 9.3** (Random-order capacity bound for bounded-arity shared set codes; after [5, Proposition 4.14(a)–(b)] and [2, Theorem 3])**.** Lean: orderedRecoverable_injective_and_card_le, exists_orderedRecoverableCells_bound, randomOrder_BoundedAritySetCapacity, corollary_OneClassArityOneHalf, corollary_OneClassArityTwoTwoThirds, asymptotic_SublinearClassesBrokenFractionRefutesRateOne [SetCapacityConsequences.lean]

*Let $d \geq 1$, let $M, \Delta \in \mathbb{N}$, let $\Sigma$ be a finite alphabet with $|\Sigma| \geq 2$, and let $C \subseteq \Sigma^{d \times d}$ satisfy* IsBoundedAritySetCode$(\Delta, p, \mathcal{E}, f, Q, C)$.

*(a) For every ranking $\mathrm{rk} : \mathbb{Z}_d \times \mathbb{Z}_d \to \mathbb{N}$, let $A(\mathrm{rk}) :=$ orderedRecoverableCells$(p,Q,\mathcal{E},\mathrm{rk})$ be the set of clean entries $(c,r)$ satisfying $\mathrm{rk}(y) < \mathrm{rk}(c,r)$ for all $y \in F_{c,r}$. Then restriction to $(\mathbb{Z}_d \times \mathbb{Z}_d) \setminus A(\mathrm{rk})$ is injective on $C$ (cf. LeanMachine.OrderedRecovery.eq_of_rank), so $|C| \leq |\Sigma|^{d^2 - |A(\mathrm{rk})|}$, and averaging over all permutations of $\mathbb{Z}_d \times \mathbb{Z}_d$ produces an injective ranking $\mathrm{rk}_*$ with $(\Delta+1)|A(\mathrm{rk}_*)| \geq |$cleanEntries$(p,Q,\mathcal{E})|$.*

*(b) Consequently, for every (empty or nonempty) code $C$,*

$$\frac{\log|C|}{\log|\Sigma|} \;\leq\; d^2 \;-\; \frac{d^2 - |\mathcal{E}| - dM\Delta}{\Delta+1}.$$

*At $M = 1, \mathcal{E} = \emptyset$, this gives rate at most $\frac{1}{2} + \frac{1}{2d}$ at $\Delta = 1$ and $\frac{2}{3} + \frac{2}{3d}$ at $\Delta = 2$; and for any fixed $\Delta \geq 1$, $\delta \in (0,1]$, any sequence of finite alphabets $(\Sigma_d)$ with $|\Sigma_d| \geq 2$, $|\mathcal{E}_d| \leq (1-\delta)d^2$, and $M_d = o(d/\Delta)$, the rate is eventually at most $1 - \frac{\delta}{2(\Delta+1)}$.*

*Outline of the formal proof.* In (a), strong induction on $\mathrm{rk}(c,r)$ shows that two codewords agreeing outside $A(\mathrm{rk})$ agree on $A(\mathrm{rk})$ as well, because every $y \in F_{c,r}$ strictly precedes $(c,r)$. For each clean entry $x = (c,r)$, $|F_{c,r} \cup \{x\}| \leq \Delta + 1$; for a uniformly random permutation $\pi$ of $\mathbb{Z}_d \times \mathbb{Z}_d$, composing

$\pi$ with the transposition swapping $x$ and the $\pi$-maximum element of $F_{c,r} \cup \{x\}$ partitions the symmetric group so that $x$ comes last in $F_{c,r} \cup \{x\}$ on at least a $1/(\Delta+1)$ fraction of permutations, and double counting selects $\mathrm{rk}_*$. Combining (a) with Lemma 9.2(b) and $\log|\Sigma| > 0$ gives (b) (with $\log 0 = 0$ in Lean handling $C = \emptyset$). □

**Remark 9.4** (Comparison with small-window and unrestricted-arity bounds)**.** In `BoundedAritySetCapacityProofs.lean` and `SetCapacityConsequences.lean` (`oneHopRecoverableInjective`, `existsRecoverableSubsetBound`, `theorem1_BoundedAritySetCapacity`, `growingClassesPositiveDeficit`, and `pMin_two_sided_theta_inv_arity`) we also formalized the weaker one-hop product-Bernoulli variant with factor $\Delta^\Delta/(\Delta+1)^{\Delta+1} \in (1/(e(\Delta+1)),\, 1/(2(\Delta+1))]$ for $\Delta \geq 1$, together with explicit $M = 1, \mathcal{E} = \emptyset$ nonlinear lower bounds for all $d \geq 2$ and $|\Sigma| \geq 2$ (`boundaryControl_DeficitVanishesAtDeltaEqD`, `unrestrictedArity_RateLowerBoundAndTendstoOne`, and `unrestrictedArity_ExactLogarithmicAsymptoticAllDimensions`): $|C| = \lfloor |\Sigma|^d/d \rfloor^d$ at cell arity $\Delta = d$ (where the clean-entry coefficient $d^2 - dM\Delta$ vanishes), $|C| \geq |\Sigma|^{d(d-1)}$ at cell arity $\Delta = 2d-1$, and $|C| \geq |\Sigma|^{d^2}/d$ for prime $d$ and $|C| \geq |\Sigma|^{d^2}/(2d)$ for all $d \geq 2$ at $\Delta = d^2$ (wrapping the full-view orbit construction of [10, Theorem A.1 and Corollary A.2], which reaches rate $1 - o(1)$). At $\Delta \leq 2$, stronger results (proved in text, allowing a separate recovery function for each position) are already in [4]: at window $w = 1$, Proposition 6.21(b) of [4] gives the same exponent $|C| \leq q^{(d^2+Md)/2}$ (rate $\frac{1}{2} + \frac{1}{2d}$ at $M = 1$) without requiring shared functions, and at window $w \leq 2$, Proposition 6.22(b) of [4] improves on the $\frac{2}{3} + \frac{2}{3d}$ bound of Theorem 9.3(b) to $|C| \leq q^{d(d+1)/2+d-1}$ (rate $\frac{1}{2} + O(1/d)$). Meanwhile, Problem 9.4(b) of [4] asks whether rate $\frac{1}{2} + o(1)$ holds at fixed windows $w \geq 3$ over a *fixed* alphabet of $q$ symbols, rather than merely rate $< 1$.

### 9.3 Negative brackets for two column-class cut reductions

**Remark 9.5** (Why subspace bounds and column-class set capacity stop short of Statement (T))**.** It is natural to ask whether the linear shared-decoder bounds of Sections 3 to 6 or the set-capacity infrastructure of Lemma 9.1 and Theorem 9.3 can bound a periodic stretch of a multitape Turing machine by conditioning on genie advice or a bounded-depth trace cut $\Gamma$. Three distinct barriers prevent a direct reduction:

(1) *Nonlinear set fibres versus linear subspaces ([4, Problem 9.4(b)]; [10, Theorem A.1 and Corollary A.2]).* Freezing genie advice or a cut transcript on a finite-alphabet machine selects a *nonlinear subset* $C \subseteq \Sigma_d^{d \times d}$ rather than a linear subspace, so the linear decoder and Fourier-filtration bounds of Sections 3 to 6 do not apply directly. Moreover, Theorem A.1 and Corollary A.2 of [10] prove that shared set decoders with unbounded window reach rate $1 - o(1)$ ($|C| \geq |\Sigma|^{d^2}/(2d)$), so any set-level bridge must enforce a bounded read window.

(2) *Depth-$D$ cut cost exceeds the $1/(\Delta+1)$ deficit ([5, Propositions 4.13 and 4.14(c)], proved in text).* In [5, Propositions 4.13 and 4.14(c)], path depth counts edges (stated for oblivious machines, with the same spine-window counting argument applying to the common trajectory of a frozen-trace fibre), so a $\Gamma$-free path of at most $D$ vertices (Lemma 9.1) has edge-depth at most $D-1$: cutting all causal paths down to $D$ vertices over a run of length $T \geq L$ at cut bit-cost $\varepsilon L = \Omega(L/D)$ forces $(M.k+1)^D > 1/\varepsilon$ (evaluating [5, Proposition 4.14(c)] at edge-depth $D-1$), while the entry-level cone arity $\Delta = b(M.k+1)^D$ grows exponentially in $D$ and includes $b = \max(1, \lceil \log_2 d \rceil)$. Thus an $O(1/\Delta)$ rate deficit is always smaller than the cut-advice rate and, as proved in [5, Proposition 4.14(c)], cannot by itself contradict a transposer.

(3) *Per-entry cut labels along spine edges, and separated high-rate storage codes ([4, commentary on Problem 9.4(b)]; [8, 11]).* Proposition 9.6 below illustrates that two naive column-class reductions of a depth-$D$ $\Gamma$-cut fibre—discarding all cut-touching outputs, or positing a worst-case full-product class count $M_{\mathrm{cls}} := M_0\big((M.q+1)(M.g+1)^{M.k}\big)^{L_{\mathrm{cut}}}$ across $L_{\mathrm{cut}}$ per-column cut slots—make the numerator $d^2 - |\mathcal{E}| - dM\Delta$ in Theorem 9.3(b) strictly negative. (Note that Proposition 9.6(b) uses only $M_{\mathrm{cls}} \geq L_{\mathrm{cut}} + 1$ and does not bound how many cut patterns actually occur: on periodic stretches few cut patterns may occur, where the bracket need not be negative and the obstruction is instead item (2) above.) In Proposition 9.6, this occurs because `TMDepStep` includes the control-state spine edge $u \to u+1$, attaching cut labels to individual output steps rather than whole columns; this reinforces the observation in the commentary on [4, Problem 9.4(b)] that cut patterns divide outputs into boundedly many per-entry classes, motivating the per-position fixed-alphabet window formulation of [4, Problem 9.4(b)]. Meanwhile, linear-time transposers must use curved, head-dependent, non-piecewise-affine motion on interacting work tapes rather than uniform home-based passes ([5, Theorems 3.10 and 3.12 and Corollary 3.24]), and [8, 11] construct bounded-degree paired-distance-separated storage codes above rate $\frac{1}{2}$ and approaching rate 1, so graph degree and separation alone cannot rule out linear-time transposition without exploiting one-dimensional head trajectory geometry.

**Proposition 9.6** (Negative brackets for two specific column-class cut reductions)**.** `Lean: perColumnCutAbsorption_bracket_neg` [`MachineSweepConsequences.lean`]

*Let $M$ be a* `Transpose.TM` *with $1 \leq M.k$, let $d \geq 1$ and $D \in \mathbb{N}$, and let $C \subseteq \Sigma_d^{d\times d}$ be a* nonempty *fibre satisfying* `IsFrozenTraceCutFibre` *whose $\Gamma$-free causal paths have at most $D$ vertices. Set $\Delta := $* `Transpose.entryBits`$(d) \cdot (M.k+1)^D$.

*(a)* Discarding cut-touching outputs*: let $\mathcal{E}_{\mathrm{cut}}$ (*`cutTouchingOutputCells` *for $d$, $T$,* workPos*,* emitStep*, and $\Gamma$) be the set of output coordinates $(r,c)$ having a bit cone that meets $\Gamma$ along a direct* `TMDepStep`*. Then $d^2 \leq |\mathcal{E}_{\mathrm{cut}}| + D$, and for every $M_0 \geq 1$ and every $\mathcal{E}_{\mathrm{sched}} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$, setting $\mathcal{E} := \mathcal{E}_{\mathrm{sched}} \cup \mathcal{E}_{\mathrm{cut}}$ gives*

$$d^2 - |\mathcal{E}| - dM_0\Delta \;<\; 0.$$

*(b)* Worst-case full-product class refinement across per-column cut slots*: suppose every cut step $u \in \Gamma$ touched by a bit cone of an unbroken output $(r,c) \notin \mathcal{E}_{\mathrm{sched}}$ lies among $L_{\mathrm{cut}}$ column-indexed cut slots* $\mathrm{colCutStep}(c) : \mathrm{Fin}\, L_{\mathrm{cut}} \to \mathbb{N}$*. Then $d^2 \leq |\mathcal{E}_{\mathrm{sched}}| + D + dL_{\mathrm{cut}}D$, and for every $M_0 \geq 1$, whenever $(M.q+1)(M.g+1)^{M.k} \geq 2$, the worst-case full-product class count $M_{\mathrm{cls}} := M_0\big((M.q+1)(M.g+1)^{M.k}\big)^{L_{\mathrm{cut}}}$ satisfies*

$$d^2 - |\mathcal{E}_{\mathrm{sched}}| - dM_{\mathrm{cls}}\Delta \;<\; 0.$$

*Outline of the formal proof.* Since $C \neq \emptyset$, the $d^2$ lane-0 emission steps $v(r,c) = \mathrm{emitStep}(\mathrm{idx}_0(r,c))$ (via `outputBitIdx` at lane 0) are pairwise distinct. For any $(r,c)$ with $v(r,c) \geq D$, the descending control-state spine from $v(r,c)$ cannot avoid $\Gamma$ for $D$ steps without creating a $\Gamma$-free path of $D+1$ vertices; taking the largest $u \in \Gamma \cap [v(r,c) - D, v(r,c))$ places $u+1$ in the $\Gamma$-cut cone of $v(r,c)$ with $u \to u+1$, so $(r,c) \in \mathcal{E}_{\mathrm{cut}}$. At most $D$ emission steps satisfy $v(r,c) < D$, giving $d^2 - |\mathcal{E}_{\mathrm{cut}}| \leq D < 2^D \leq \Delta \leq dM_0\Delta$ and proving (a). In (b), each slot $u = \mathrm{colCutStep}(c,s)$ can serve as that most recent cut step for at most $D$ emission steps $v(r,c) \in (u, u+D]$, leaving at most $D + dL_{\mathrm{cut}}D \leq d(L_{\mathrm{cut}}+1)D$ unbroken entries; meanwhile $(M.q+1)(M.g+1)^{M.k} \geq 2$ implies $M_{\mathrm{cls}} \geq 2^{L_{\mathrm{cut}}} \geq L_{\mathrm{cut}} + 1$, and $D < \Delta$ yields $d(L_{\mathrm{cut}}+1)D < dM_{\mathrm{cls}}\Delta$. □

### 9.4 Open problems

Statement (T) remains open. Within the models studied in this paper and in [4, 9, 10], four concrete algebraic and combinatorial problems remain unresolved:

1. **Macroscopic rank deficit for $M_1 \geq 2$ and positive characteristic in two-stage linear composition ([10, Sections 6 and 8]).** For fixed $M_1 \geq 2$ and $M_2 \geq 1$ (at $M_1 = 1$, [10, Theorem 6.6] already gives $\Omega(bd^2/M_2)$ for $d \geq 4M_2 + 1$), can the $\Omega(bd(\log d)^{1/(2M_2-1)})$ error-rank bound of Theorem 4.6 be strengthened to a positive normalized deficit $\Omega_{M_1,M_2}(bd^2)$ as asked in [10, Section 8], and extended to fields in which $\mathrm{char}(F) \mid d$ (such as $\mathbb{F}_2$ at $d = 2^m$)?

2. **Closing $\frac{19}{32}$ to $\frac{1}{2} + o(1)$ and positive characteristics in Problem 9.1(a)–(b) of [4].** Can the $\frac{19}{32}\,bd^2 + \frac{3}{2}\,bd^{3/2}$ two-class ceiling of Theorem 5.4 be lowered to $(\frac{1}{2} + o(1))bd^2$, can the averaged multi-class bound of Theorem 5.5 ($M \geq 3$) be extended from primes to composite $d$, and can Theorems 5.2, 5.4 and 5.5 be extended to fields with $\mathrm{char}(F) \mid d$ ([4, Problem 9.1(a)])? And for $\{0,1,3\} \mid \{2,4,5\}$ on $\mathbb{Z}_6$ (Theorem 6.2), what is $D_1(\{0,1,3\},\{2,4,5\})$ over fields of characteristic 2 ($25 \leq D_1 \leq 28$) and characteristic 3 ($21 \leq D_1 \leq 28$ from symmetric matrices and [4, Theorem 6.4]; whether 25 is attainable in characteristic 3 is [9, Problem 6.1])?

3. **Remaining dimensions in Remark 5.6 and two-sided multi-scale bounds in Problem 9.3 of [4], and periodic Walsh masks.** Does any three-term swap code exceed $d(d+1)/2$ at $d = 5$ or at any $d \geq 8$ outside $\{9, 12\}$ ([4, Remark 5.6])? Is $\lfloor \frac{1}{2}(d^2 + \sum_{i,j} \min(|I_i|, |J_j|)) \rfloor$ (Remark 7.6) an upper bound on swap-code dimension for all two-sided interval partitions, and can Theorem 7.4 be extended from two scales to general multi-scale dyadic interval profiles ([4, Problem 9.3])? On the periodic variant of [4, Problem 9.3] ($d = 2^m$), when both $A_a$ and $B_a$ lie in the at most $(m+1)2^m$-dimensional span of periodic Walsh digit masks composed with rotations (Proposition 7.7), is $\dim_F \texttt{SwapRigidity.swapCode}(A, B) \leq (1 - c)d^2$ for a constant $c > 0$?

4. **Fixed-alphabet windows, oblivious/adaptive interfaces, and multi-scale invariants ([4, Problems 9.4 and 9.5]).** Over a fixed finite alphabet of $q$ symbols and a fixed window $w \geq 3$ on a spread one-class relation, does the $\frac{1}{2} + o(1)$ rate bound hold with a separate recovery function for each position or a single shared function ([4, Problem 9.4(b)]), or in its whole-row $\rho$-code form ([4, Problem 9.4(c)])? On the machine side, [4, Problem 9.4(a)] asks for a cone-system analogue of Theorem TP on *oblivious* schedules (while [12, Theorem H.5 and Corollary H.6] transfer entry-relation storage bounds to cut runs of *adaptive* transposers), and [4, Problem 9.5] asks for a quantity subadditive over machine steps, large for the swap at every scale separately, and *"not raised at all scales by a single linear-time operation"*.

## A Formalization architecture

The Lean 4 formalization is split between the shared repository package `lean-machine` (`../lean-machine`) and this paper's package `BoundedArityDecoders` (`lean/`), both pinned to Lean 4 and Mathlib version `v4.34.1` [22]. Through Lake path dependencies in `lean/lakefile.toml`, `BoundedArityDecoders` requires `SharedDecoders` ([10], which transitively imports `SwapRigidity` from [4] and `Transpose` from [6]), `ClassUniformMemo` ([9]), and `LeanMachine`. None of the earlier papers is modified, and for this paper `lean-machine` is extended only by registering four reusable modules under `LeanMachine/ShuffleShear/` in the package root, axiom audit, and `README`:

- `Model.lean`, `StoneShuffle.lean`, `StoneUnshuffle.lean`, and `RowShear.lean`: standalone deterministic multitape Turing machine semantics and verified linear-time machines for Stone's shuffle (3 work tapes, $\leq 150L$ steps), Stone's unshuffle (2 work tapes, $\leq 150L$ steps), and within-row cyclic shear (2 work tapes, $\leq 200L$ steps).

Inside `lean/BoundedArityDecoders/`, twenty-seven public modules are organized into three layers:

(1) *Nine definition and specification modules* (`AffineCubeComposition.lean`, `SumsetRankDeficit.lean`, `ExtensionLinearRun.lean`, `ExtensionAperiodicSix.lean`, `ExtensionTwoSidedProfile.lean`, `SwapCodeAnswers.lean`, `OperatorDuality.lean`, `NativeSweepInterface.lean`, and `BoundedAritySetCapacity.lean`), defining the algebraic operators, one-stage linear run-extension invariants, coarse–fine interval decompositions, syntactic causal cones, and named proposition types;

(2) *Eleven primary proof modules* (`AffineCubeCompositionProofs.lean`, `SumsetRankDeficitProofs.lean`, `ExtensionLinearRunSingleShift.lean`, `ExtensionLinearRunSharpness.lean`, `ExtensionLinearRunProofs.lean`, `ExtensionAperiodicSixProofs.lean`, `ExtensionTwoSidedProfileProofs.lean`, `SwapCodeAnswersProofs.lean`, `OperatorDualityProofs.lean`, `NativeSweepInterfaceProofs.lean`, and `BoundedAritySetCapacityProofs.lean`), proving the two-stage composition theorems and finite controls, the one-stage linear run-extension and $\mathbb{Z}_6$ bounds, the two-sided coarse–fine and three-term/interval swap-code theorems, the Operator Schmidt Rank bounds and row-shear / mutual-information controls, the frozen-fibre cut-causal cone determination theorem, and the one-hop bounded-arity set-capacity and clean-entry bounds; and

(3) *Seven consequence, transfer, and bridge modules* (`ExtensionFieldTransfer.lean`, `AffineCubeConsequences.lean`, `SumsetRankConsequences.lean`, `DualityAndSwapConsequences.lean`, `ShuffleShearBridge.lean`, `MachineSweepConsequences.lean`, and `SetCapacityConsequences.lean`), proving the algebraic-closure primitive-root and scalar-extension transfer lemmas, the explicit constants $c(M_1, M_2)$ and $D_0(M_1, M_2)$, the general-field finite controls at $d = 13$ and $d = 4$, and the complex witness and bridge at $d = 12$, Stone's iteration $S^m = \tau$ and the `TensorProduct` anti-diagonal equivalences, the transport of `LeanMachine.ShuffleShear` to `Transpose.TM`, the symbolic `CausalConeTree` evaluator, the $1/(\Delta+1)$ random-order bound, and the negative column-class cut-reduction bounds.

Every one of the 41 numbered theorems, lemmas, propositions, corollaries, and examples in the manuscript carries a proved `theorem` declaration in these modules (38 marked `\formal` and 3 marked `\partformal`). The parts proved in the text only are:

- in Theorem 3.3, the asymptotic $o(d^{1/2^{M_2}})$ and $(1-\varepsilon)\log_2\log_2 d$ specializations of the Lean pointwise inequality and the greedy translate-covering step turning Behrend's 3-AP-free set [3] into a $2^{O(\sqrt{\log d})}$-class partition of $\mathbb{Z}_d$;
- in Example 4.9, the alternative $(b = 1, M_2 = 2)$ parameter reading of $q = bM_2k = 4$ (Lean proves the shift-equivariant `SharedDecoders.bandRowSpace` bridge for $b = 2, M_2 = 1, k = 2$);
- in Corollary 7.5, applying [4, Theorem 6.4] to `InContiguousCoarseFineSpan` for the one-sided baseline $(d^2 + (\min(s_A, s_B) + 1)d)/2$.

Outside numbered theorem statements, remarks and prose comparisons with other papers (including the van der Waerden regime comparisons in Remark 3.4 and Remark 4.7(a), the finite crossover and $(u, v)$-minimization comparisons in Section 5 checked in exact rational arithmetic in `tools/verify_linear_capacity_crossovers.py`, the general $(m, s)$ padded dimension formula $4m(m+1) + \frac{s(s+1)}{2} + 2ms$ in Section 7.1 verified in Lean at $(m, d) \in \{(1, 4), (1, 5), (2, 7)\}$, the positive-characteristic Mahler binomial observation in Remark 5.6, and the numerical search in Remark 7.6) are proved in the text (or checked by script) only unless a Lean declaration is named.

# B Index of Lean declarations

The following declarations are cited by formal-status lines. Names are relative to `BoundedArityDecoders`. A partial status retains the gap stated at its theorem.

**`AffineCubeCompositionProofs.lean`** `cauchySchwarzAffineCubeExtraction`, `control_D13TwoCubeAPFreeWitness`, `control_D4SameIntervalCodedFactorization`, `lemma_AffineCubeCommonMessageGrowth`, `theorem2A_TwoStageRamseyFreeCutoff`.

**`AffineCubeConsequences.lean`** `control_D13_GeneralFieldAndProgressionBridge`, `control_D4_AllWidthsAndBinaryFields`, `denseSubset_AffineCubeCommonMessageBound`, `singleSubset_CauchySchwarzAffineCubeExtraction`, `theorem2A_GrowingClassesNonFactorization`.

**`BoundedAritySetCapacityProofs.lean`** `mem_shiftedFootprint_self_iff`, `selfReferentialAndCleanCount`.

**`DualityAndSwapConsequences.lean`** `antiDiagTensorEquiv_intertwines_rot`, `bonus1_UnifiedThreeTermExceedsPairingBound`, `bonus2_SublinearProfileAndTendstoHalf`, `bonus2_SublinearProfileExplicitThreshold`, `bonus2_WalshDichotomyStructuralProperties`, `boundedOSRPipelinePassLowerBound`, `isCyclicInterval_cyclicBoundaryCount_le_two`, `isPeriodicScaleDyadicMask_cyclicBoundaryCount`, `isRowOrColUniform_rankLe_one`, `maskFloor_GenericSubspaceRealization`, `rowDiagonal_SourceArityGaugeInvariance`, `shiftSetCovariance_tensor_finrank`, `skewOneTimePad_ShannonMutualInfoZero`, `stoneShuffleGridLinear_iterate_eq_swap`, `uniformShannonMI_eq_zero_of_indep`, `walshBitSet_cyclicBoundaryCount`, `walshBitSet_joint_separates_points`, `walshDigitMaskRotSpan_finrank_le`.

**`ExtensionAperiodicSixProofs.lean`** `aperiodicSix_finrank_le_25`, `aperiodicSix_maxDim_eq_25`, `aperiodicSix_shift_inf_finrank_le_14`, `extension_Theorem2_AperiodicSixExact`, `extension_Theorem2_Preliminaries`.

**`ExtensionFieldTransfer.lean`** `coprimeToChar_iff_natCast_ne_zero`, `exists_isPrimitiveRoot_algebraicClosure`, `exists_isPrimitiveRoot_of_isAlgClosed`, `extendLinearMapPi_comp_map`, `finrank_le_of_pi_ringHom`, `linearIndependent_fin_pi_ringHom`.

**`ExtensionLinearRun.lean`** `laneRot_apply`.

**`ExtensionLinearRunProofs.lean`** `extension_Theorem1B_MultiClassPrime`, `extension_Theorem1B_TwoClass`, `linearClassUniform_multiClass_prime_finrank_le`, `linearClassUniform_multiClass_prime_finrank_le_sharp`, `linearClassUniform_multiClass_prime_maxDim_le`, `linearClassUniform_multiClass_prime_maxDim_le_sharp`,

linearClassUniform_twoClass_finrank_le, linearClassUniform_twoClass_prime_finrank_le, multiClassLeadingCoeff_sub_flag, multiClass_simplex_max_iff.

**ExtensionLinearRunSharpness.lean** alternatingSharpCode_finrank_and_shift_inf, extension_LinearModelAndCharBridges, extension_Theorem1A_AlternatingSharpness, laneOneMatrixEquiv_laneRot, matrixToSharedStateEquiv_laneRot.

**ExtensionLinearRunSingleShift.lean** extension_Theorem1A_SingleShift, finrank_vanishCoeffs_le, linearClassUniform_singleShift_finrank_le.

**ExtensionTwoSidedProfileProofs.lean** extension_Theorem3_ConcreteWinOverTP, extension_Theorem3_TwoSidedCoarseFine, swapCode_finrank_le_contiguousCoarseFine, swapCode_finrank_le_twoSidedCoarseFine.

**MachineSweepConsequences.lean** evalConeTree_cellSupport, evalConeTree_entryCellSupport_card_le, evalConeTree_traceConeTree_eq, padDummyWorkTape_transposesWithin, perColumnCutAbsorption_bracket_neg.

**NativeSweepInterfaceProofs.lean** entryBits_eq_clog, tmTraceCutConeDetermination.

**OperatorDualityProofs.lean** antiDiagonalTensorSeparation, control_RowShearAndRowDiagonalAlternation, control_SkewOneTimePadInformationTrap, hasOperatorSchmidtRankLe_comp, operatorSchmidtRankStoneShuffle, operatorSchmidtRankSubmultiplicative, operatorSchmidtRankSwap, pairwiseShiftCollisionCharacterization, stoneShufflePipelinePassLowerBound.

**SetCapacityConsequences.lean** asymptotic_SublinearClassesBrokenFractionRefutesRateOne, canonicalWordAlphabet_card_and_ge_two, corollary_OneClassArityOneHalf, corollary_OneClassArityTwoTwoThirds, exists_orderedRecoverableCells_bound, orderedRecoverable_injective_and_card_le, randomOrder_BoundedAritySetCapacity.

**ShuffleShearBridge.lean** rowShearTM_twoTape_linear, stoneShuffleTM_threeTape_linear, stoneUnshuffleTM_twoTape_linear, tmMatrix_gridLinear_bridges.

**SumsetRankConsequences.lean** control_D12_BandRowSpaceChannelEquiv, control_D12_UnconditionalExists, d12AddChar_injective_and_primitive, exists_primitive_twelfth_root_complex, finsetNatDiam_shift_min, sumsetAnnihilatingPolynomial_bandRowSpace, theorem2BExplicitConst_pos, theorem2BExplicitThreshold_eq_pow, theorem2B_ExplicitConstant, theorem2B_OneSecondStage_OmegaLog, unshiftedToShiftedSumset_explicit, unshiftedToShiftedSumset_intrinsicDiam.

**SumsetRankDeficitProofs.lean** control_D12SingleClassAbstractBand, cyclicChannelShift_aeval_ker_finrank_le, iteratedKovariSosTuranSumset, stackedBandPullbackInjective, sumsetAnnihilatingPolynomial, sumsetTwoHopRankBound, theorem2B_TwoStageLogPowerRankDeficit, unshiftedToShiftedSumset, vandermondeArbitrarySubsetInjective.

**SwapCodeAnswersProofs.lean** bonus1_ThreeTermBlockSwapCodeDim, bonus1_ThreeTermPaddedSevenAndFive, bonus2_MultiscaleFullProfileAttainsFullSwap, bonus2_MultiscaleIntervalBothSides, walshDigitMask_in_periodicDyadicSpan.

**Other declarations named in the text** `CanonicalWordAlphabet`, `CausalConeTree`, `CoprimeToChar`, `GridState`, `HasCoarseFineIntervalDecomposition`, `HasExactOperatorSchmidtRank`, `HasOperatorSchmidtRankLe`, `HasOutputRowSourceArityLe`, `InAlignedDyadicIntervalRotSpan`, `InContiguousCoarseFineSpan`, `InCutCausalCone`, `InMultiscaleIntervalRotSpan`, `InMultiscalePeriodicDyadicRotSpan`, `InPairwiseShiftCovariance`, `IsAlignedDyadicInterval`, `IsAperiodicSixPartition`, `IsBoundedAritySetCode`, `IsColUniformRowOp`, `IsCutFreeCausalPath`, `IsCyclicInterval`, `IsFrozenTraceCutFibre`, `IsLastWorkVisit`, `IsLinearClassUniformCode`, `IsPeriodicScaleDyadicMask`, `IsRowDiagonalOp`, `IsRowUniformColOp`, `IsThreeByThreeSymmetricZeroTraceBlocks`, `ShiftLinkCovarianceSpace`, `SyntacticCellSupport`, `SyntacticColumnSupport`, `TMDepStep`, `ThreeAPFreeMod`, `altTwoColoring`, `alternatingRowDiagProduct`, `alternatingSharpCode`, `alternatingSharpDecoder`, `antiDiagLinear`, `antiDiagTensorEquiv`, `boundaryControl_DeficitVanishesAtDeltaEqD`, `charBandEval`, `cleanEntries`, `composeGridOps`, `computesWithin_toTransposeTM`, `contiguousCoarseComplement`, `contiguousFineSet`, `coordShiftLinear`, `cutConeInputCells`, `cutTouchingOutputCells`, `cyclicBoundaryCount`, `cyclicChannelShift`, `cyclicRunCount`, `cyclicRunEndpointCount`, `encodeTMMatrixToGrid`, `evalConeTree`, `existsRecoverableSubsetBound`, `extendLinearMapPi`, `finsetNatDiam`, `fourierCoeffs`, `fromAntiDiagLinear`, `gridSwapLinear`, `growingClassesPositiveDeficit`, `halfIntervalPartition4`, `intervalMaskRotMatrix`, `laneOneMatrixEquiv`, `laneRot`, `leftComponentSubmodule`, `linearClassUniformMaxDim`, `matrixToSharedStateEquiv`, `modeLift`, `multiClassLeadingCoeff`, `oneHopRecoverableInjective`, `orbitClassCard`, `orbitRunCount`, `orderedCubeMap`, `orderedRecoverableCells`, `outputBitIdx`, `pMin_two_sided_theta_inv_arity`, `rowShearLinear`, `rowShearTMMatrix`, `rowShearTransposeTM`, `shiftGcd`, `shiftOrbit`, `shiftOrbitLen`, `shiftSetCovarianceSubmodule`, `shiftTransitionCount`, `shiftedFootprint`, `skewXorPad`, `stepRunBound`, `stoneShuffleGridLinear`, `stoneShuffleTMMatrix`, `stoneShuffleTransposeTM`, `stoneUnshuffleGridLinear`, `stoneUnshuffleTMMatrix`, `stoneUnshuffleTransposeTM`, `theorem1ASingleShiftBound`, `theorem1_BoundedAritySetCapacity`, `theorem2BExplicitConst`, `theorem2BExplicitThreshold`, `theorem3UnsymmetrizedBound`, `threeTermBlockA`, `threeTermBlockB`, `threeTermPaddedA`, `threeTermPaddedB`, `traceConeTree`, `treeCellSupport`, `trueConeSteps`, `uniformShannonMI`, `unrestrictedArity_ExactLogarithmicAsymptoticAllDimensions`, `unrestrictedArity_RateLowerBoundAndTendstoOne`, `vanishCoeffs`, `walshBitSet`, `walshDigitMaskRotMatrix`.

# C Building and verifying

The repository must be cloned together with its sibling directories (`lean-machine` and the earlier papers of the series) so that relative Lake path dependencies resolve. Both `lean-machine` and `BoundedArityDecoders` pin Lean 4 and Mathlib to version `v4.34.1`.

To verify the shared shuffle, unshuffle, and row-shear machines in `../lean-machine`, run from this paper's directory:

```
cd ../lean-machine
lake exe cache get
lake build
lake env lean Axioms.lean
```

Then return to this paper's directory, regenerate the declaration index and axiom audit, verify all proofs in `lean/`, run the optional crossover and numerical scripts for Section 5 and Remark 7.6, and build the manuscript PDF:

```
cd ../Ramsey-Free-Composition-and-One-Stage-Capacity-Bounds-for-Linear-Shared-Decoders
python3 tools/lean_index.py
cd lean
lake exe cache get
```

```
lake build
lake env lean PaperAxioms.lean
cd ..
python3 tools/verify_linear_capacity_crossovers.py
python3 tools/verify_two_sided_svd.py
cd paper
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

Running `python3 tools/lean_index.py` regenerates `lean/BoundedArityDecoders.lean` from the module list under `lean/BoundedArityDecoders/`, and regenerates `lean/PaperAxioms.lean` and Appendix B (`paper/sections/leanindex.tex`) from the `\formal`, `\partformal`, and `\lean` commands in `paper/sections/*.tex`. Every proved declaration audited in `Axioms.lean` and `PaperAxioms.lean` depends at most on Lean's three standard axioms `propext`, `Classical.choice`, and `Quot.sound`, with no `sorry`, `admit`, `native_decide`, `unsafe`, `implemented_by`, `extern`, or custom `axiom` declarations.

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